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# STINGRAY
## REV 1.0
## FAB C
## RETAIL

www.mundoyakara.com

Rules: (when possible)
* MSB-->LSB: top-->bottom
* When possible: inputs on left, outputs on right
* Page order: chip interfaces, termination, power, decoupling
* Avoid off-page connectors for on-page connections
* Bussed signals are grouped on symbols
* Unnamed nets are displayed with half-sized text
* Transmitter name used as prefix with RX and TX connections
* V_ prefix for voltage rail signal names
* _DP/_DN suffix for differential pairs
* _N suffix for active low or n junction
* _P suffix for p junction
* _EN suffix for enable
* CLK for clock, RST for reset
* PWRGD for power good
* Rev and fab set with custom variables.  Tools->options->variables

[PAGE_TITLE=COVER PAGE]

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| PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|
| | 1/74 | 1/74 | C | 1.0 |

# GCPU SETUP

25 IN CPU_RST_N

FT3T10 FTP 1

FT3T11 FTP 1

65 25 IN CPU_PWRGD

1 R3E10 10 KOHM 5% CH 402 2

1 C5R66 360 PF 5% 50 V EMPTY 603 2

1 C5R11 360 PF 5% 50 V EMPTY 603 2

R4P5 2 1 1.27 KOHM 1% 402 EMPTY

1 R4P4 2 KOHM 1% 2 EMPTY 402

CPU_RST_V1P1_N OUT 65

GCPU_VEJLE_BASE

U5E1 14 OF 17 IC

GCPU VERSION 1

D5 HARD_RESET_B

T5 POWER_GOOD

CHECKSTOP_B E5 CPU_CHECKSTOP_N OUT 65

22 IN CPU_CLK_DP R4D1 2 1 0 OHM 5% 402 CH

IN CPU_CLK_DP_R2 P2 CPU_CLK_DP

22 IN CPU_CLK_DN R4D2 2 1 0 OHM 5% 402 CH

IN CPU_CLK_DN_R2 P1 CPU_CLK_DN

V_CPUCORE

1 R4R22 100 OHM 5% 2 EMPTY 402

1 R4R23 100 OHM 5% 2 EMPTY 402

DB4R3 1

DB4R2 1

| CPU_DBG_RST_EN | INTERNAL PULLDN |
|---|---|
| GPU_DBG_RST_EN | INTERNAL PULLDN |

CPU_DBG_RST_EN P5 CPU_DBG_RST_EN<DN>

GPU_DBG_RST_EN R5 GPU_DBG_RST_EN<DN>

OUT EDRAM_PSRO_DOUT N6 PSRO_DOUT

OUT CPU_PSRO0_OUT G7 PSRO0_OUT

VID6 C24 CPU_VREG_APS6 OUT 43

VID5 B24 CPU_VREG_APS5 OUT 43

VID4 A24 CPU_VREG_APS4 OUT 43

VID3 C23 CPU_VREG_APS3 OUT 43

VID2 B23 CPU_VREG_APS2 OUT 43

VID1 A23 CPU_VREG_APS1 OUT 43

1 FTP FT7P8

1 FTP FT7P7

1 FTP FT7P6

1 FTP FT7P5

1 FTP FT7P4

1 FTP FT7P3

V_CPUPLL

WHEN V_CPUPLL=1.83V
SET VGATE=1.20V
ACTUAL=1.202V

N: IF V_CPUPLL CHANGES
VGATE RESISTORS SHOULD BE ADJUSTED

1 R5T8 562 OHM 1% 2 CH 402

CPU_VGATE V6 V_GATE

1 R5T9 1.07 KOHM 1% 2 CH 402

<DN>PULSE_LIMIT_BYPASS J6 CPU_LIMIT_BYPASS OUT

<DN>PLL_BYPASS F8 CPU_PLL_BYPASS OUT

OUT RESISTOR0_DP L7 RESISTOR0_DP

OUT RESISTOR0_DN M7 RESISTOR0_DN

SRVID A3 CPU_SRVID OUT 51

EFU_POWERON C1 VREG_EFUSE_EN OUT 52

OUT CPU_EXT_CLK_EN F7 EXT_CLK_EN<DN>

CPU_TINIT J7 TI_39_TINIT

GND E7

TE CORE_HF_BGR_PLL M6 CORE_HF_BGR_PLL OUT

1 R4R21 200 OHM 5% 2 CH 402

X818336-001 BGA_2

6 LAYER ONLY SIGNALS

| | |
|---|---|
| CORE_HF_BGR_PLL | 6 LAYER ONLY; TP ONLY |
| CPU_LIMIT_BYPASS | 6 LAYER ONLY; TP ONLY, INTERNAL PULLDN |
| CPU_PLL_BYPASS | 6 LAYER ONLY; TP ONLY, INTERNAL PULLDN |
| CPU_CORE_HF_CLKOUT_DN | 6 LAYER ONLY; TP ONLY |
| CPU_CORE_HF_CLKOUT_DP | 6 LAYER ONLY; TP ONLY |
| CPU_EXT_CLK_EN | 6 LAYER ONLY; TP ONLY, INTERNAL PULLDN |
| CPU_DLL_SNIF_OUT | 6 LAYER ONLY; TP ONLY |
| CPU_VDDS0_DP | 6 LAYER ONLY |
| CPU_VDDS0_DN | 6 LAYER ONLY |
| CPU_VDDS1_DP | 6 LAYER ONLY |
| CPU_VDDS1_DN | 6 LAYER ONLY |
| RESISTOR0_DP | 6 LAYER ONLY |
| RESISTOR0_DN | 6 LAYER ONLY |
| EDRAM_PSRO_DOUT | 6 LAYER ONLY |

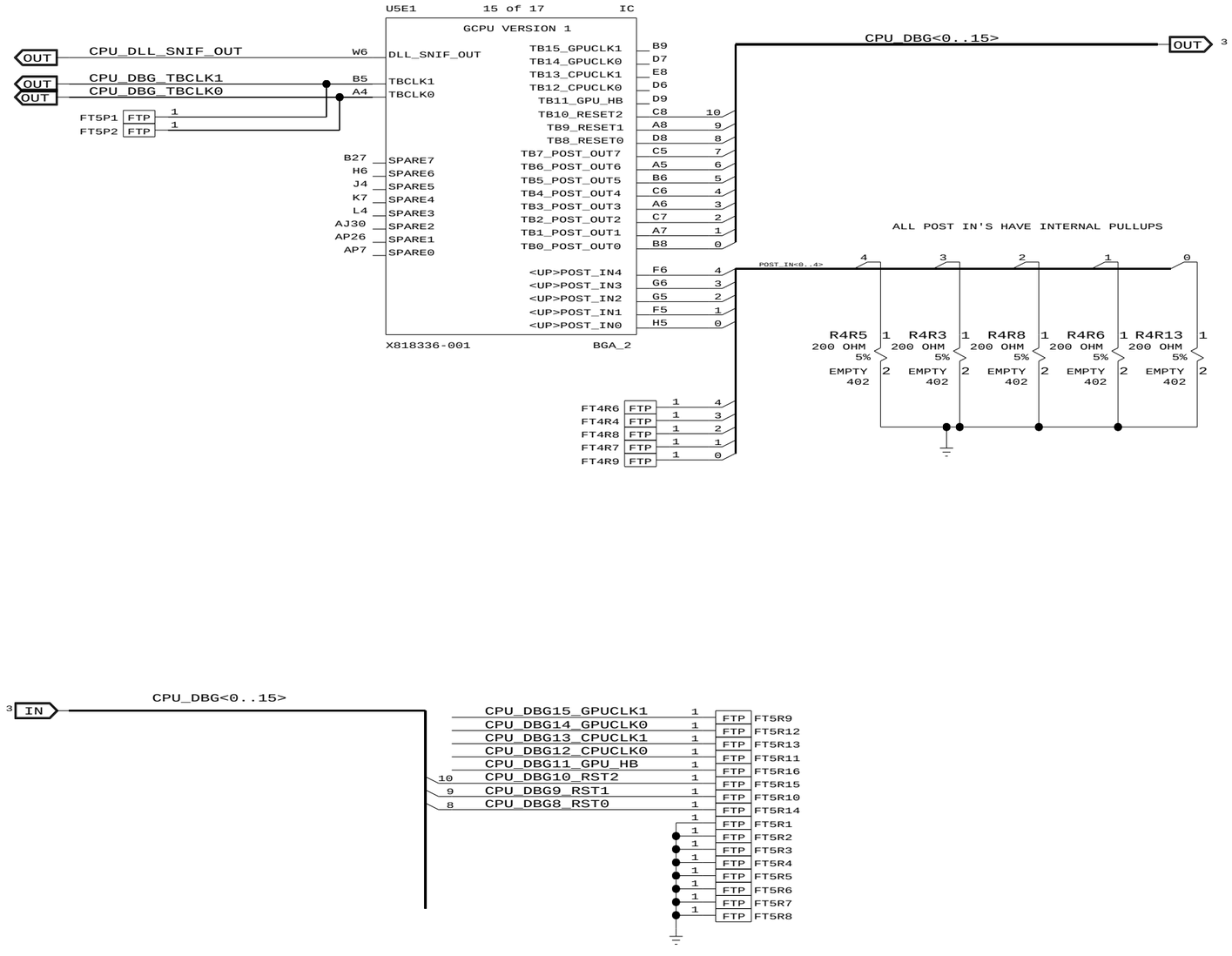
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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 2/74 | 2/74 | C | 1.0 |

# GCPU, DEBUG BUS





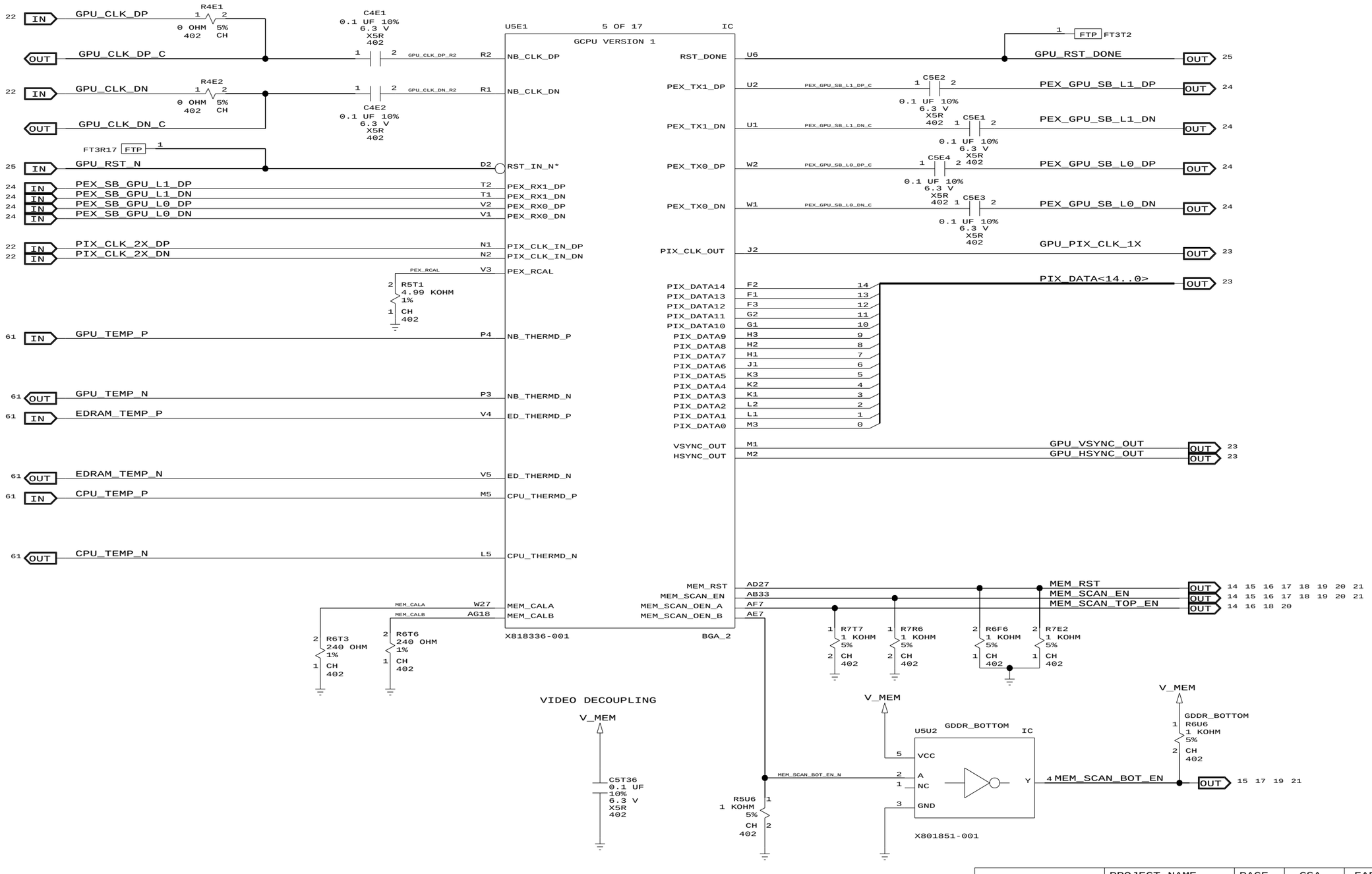


[PAGE_TITLE=GCPU, DEBUG BUS]

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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 3/74 | 3/74 | C | 1.0 |

# GCPU, VIDEO + PCIEX

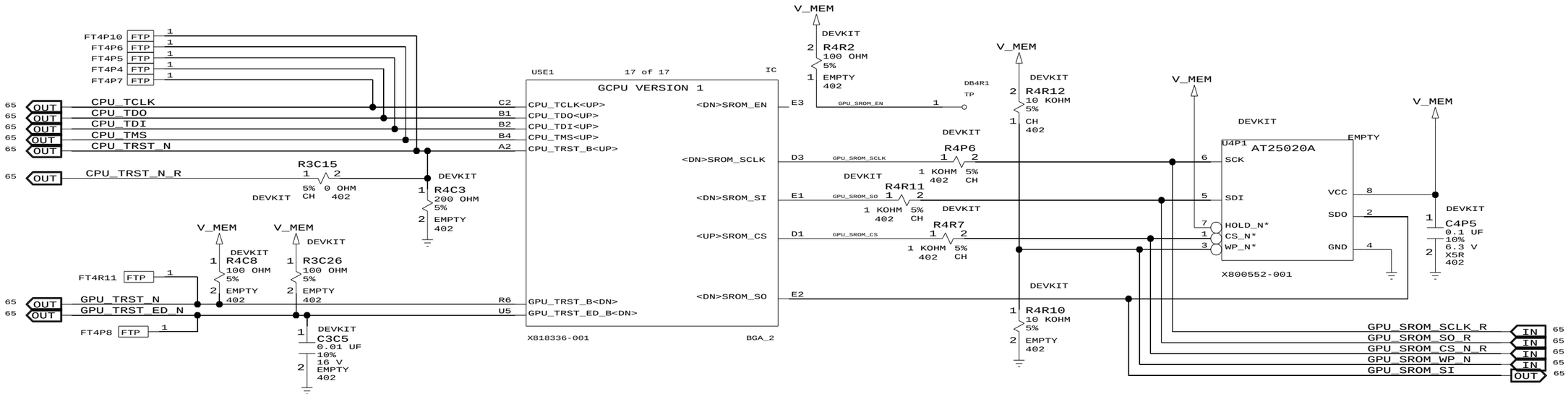



[PAGE_TITLE=GCPU, VIDEO + PCIEX]

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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 4/74 | 4/74 | C | 1.0 |

# GCPU, EEPROM + JTAG

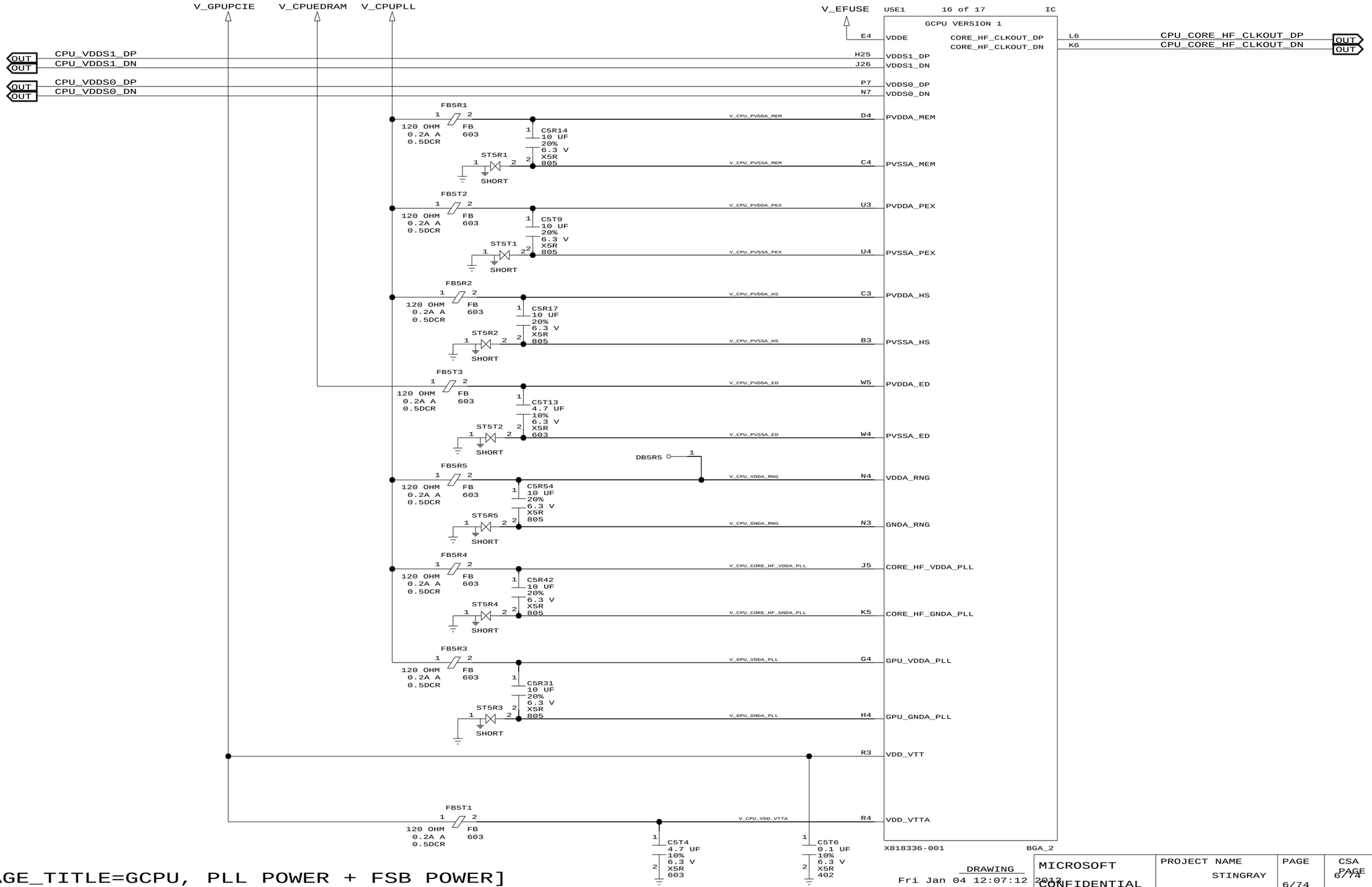



[PAGE_TITLE=GCPU, EEPROM + JTAG]

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|---|---|---|---|---|---|
| | STINGRAY | 5/74 | 5/74 | C | 1.0 |

# GCPU, PLL POWER + FSB POWER

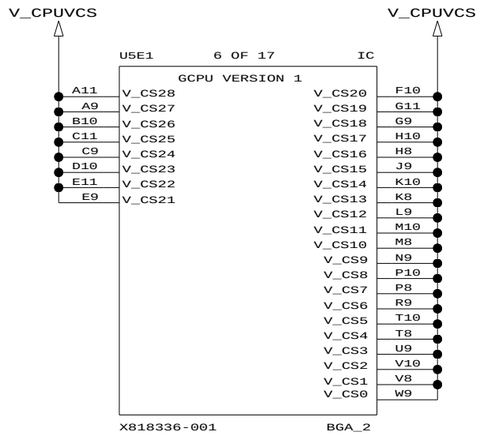


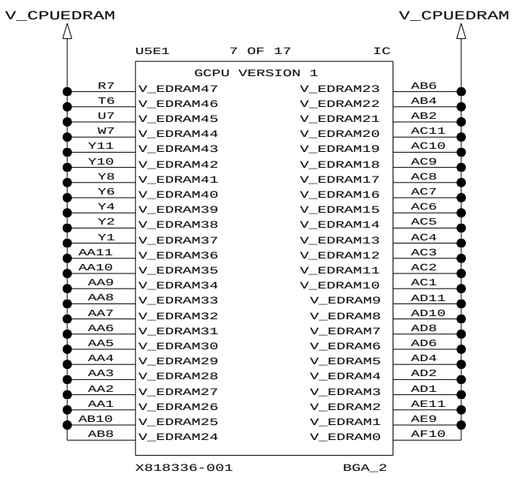

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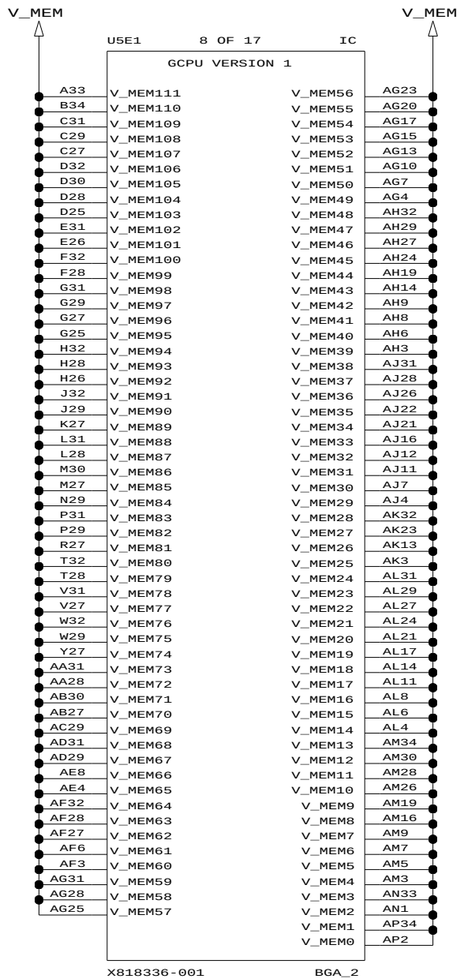
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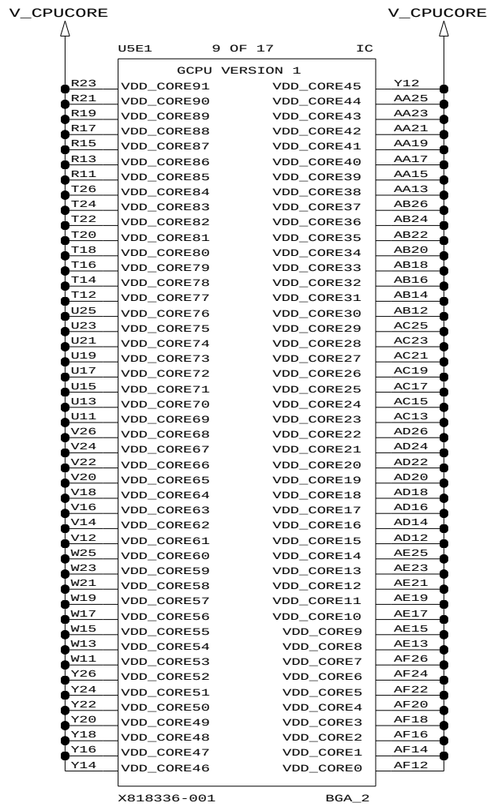
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|---|---|---|---|---|---|

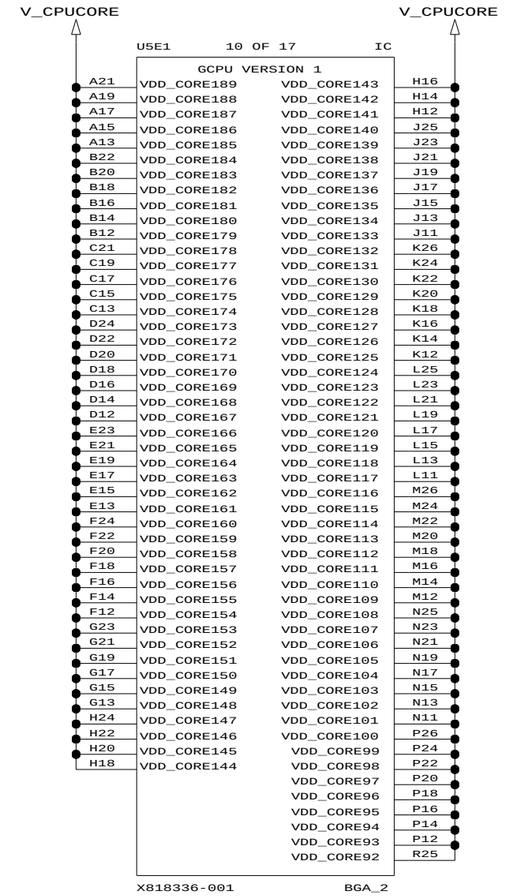
# GCPU, POWER



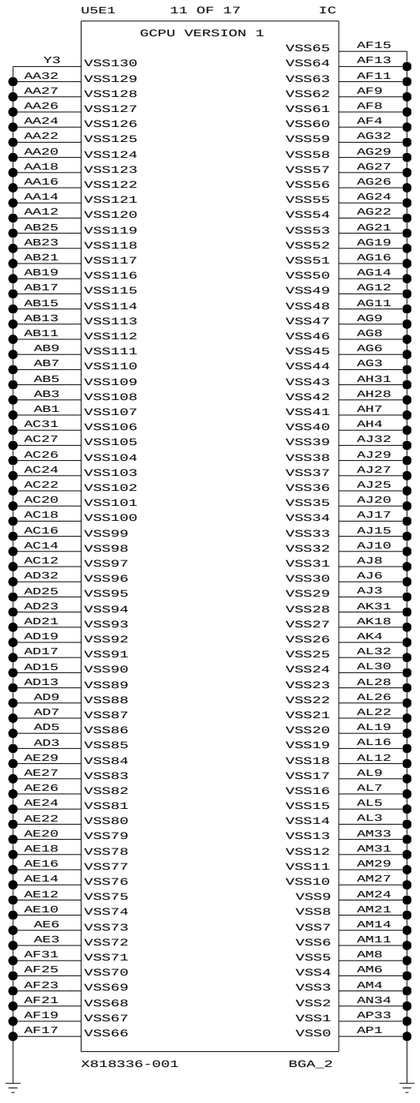

[PAGE_TITLE=GCPU, POWER]

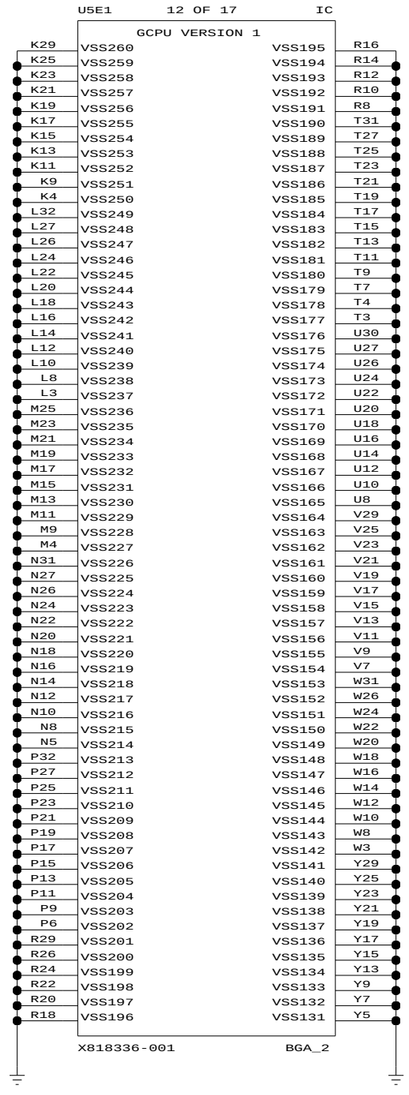
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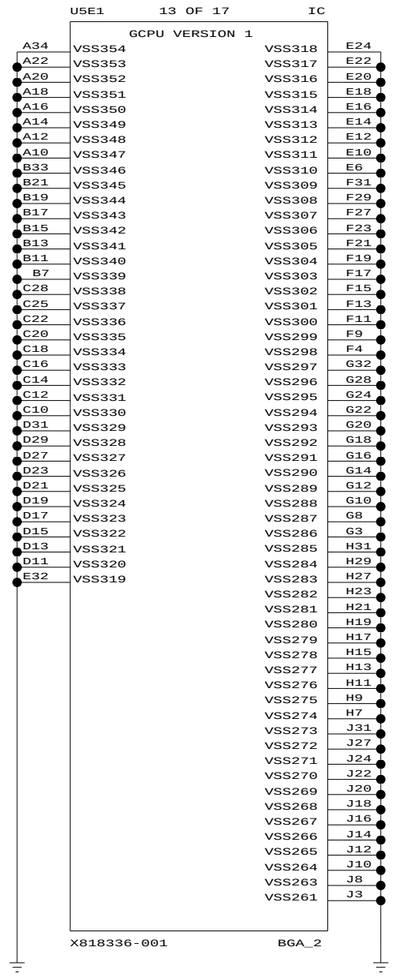
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| MICROSOFT | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 7/74 | 7/74 | C | 1.0 |

# GCPU, POWER



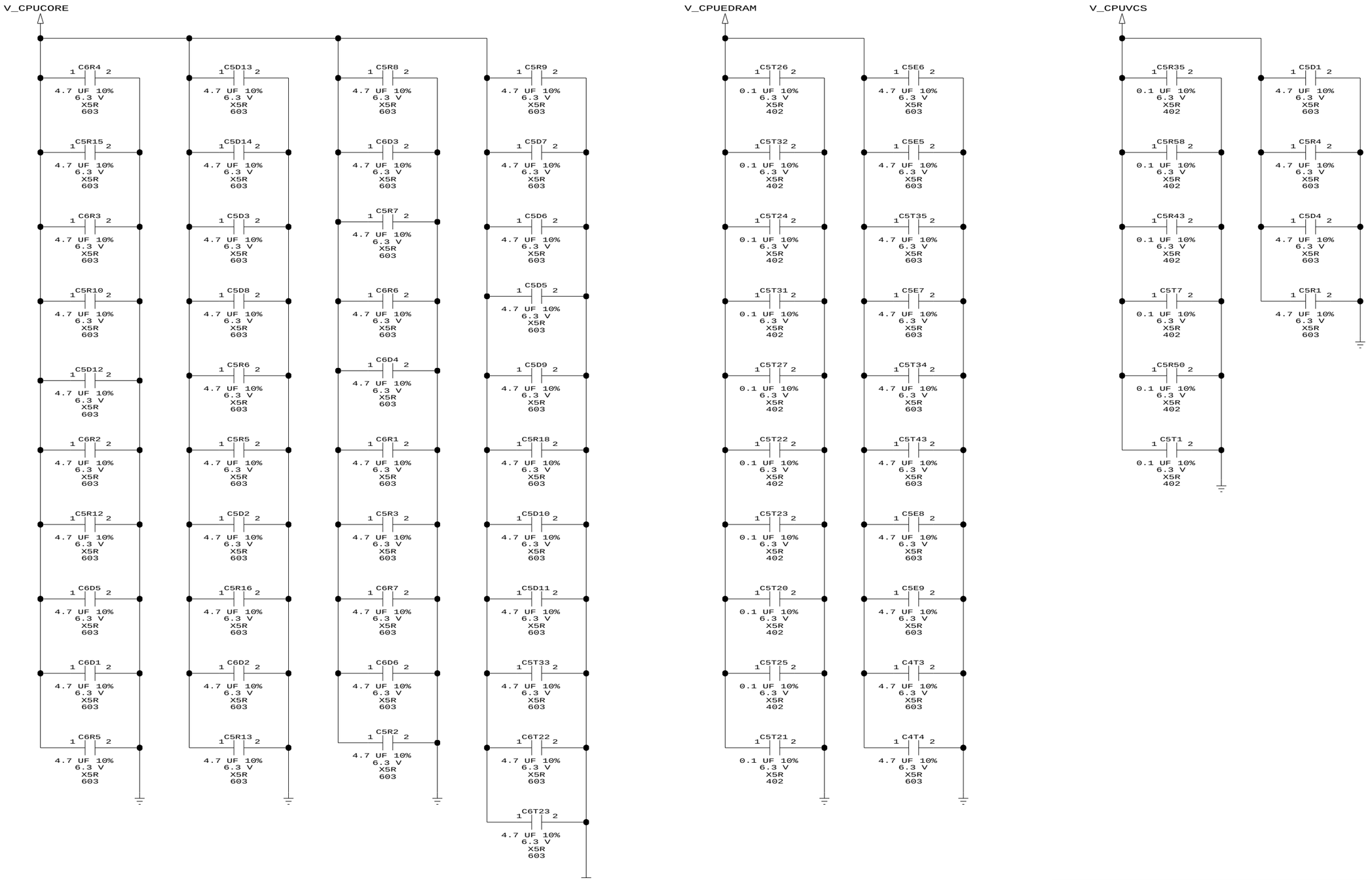




[PAGE_TITLE=GCPU, POWER]

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| MICROSOFT CONFIDENTIAL | PROJECT NAME STINGRAY | PAGE 8/74 | CSA PAGE 8/74 | FAB C | REV 1.0 |
|---|---|---|---|---|---|

# GCPU, DECOUPLING

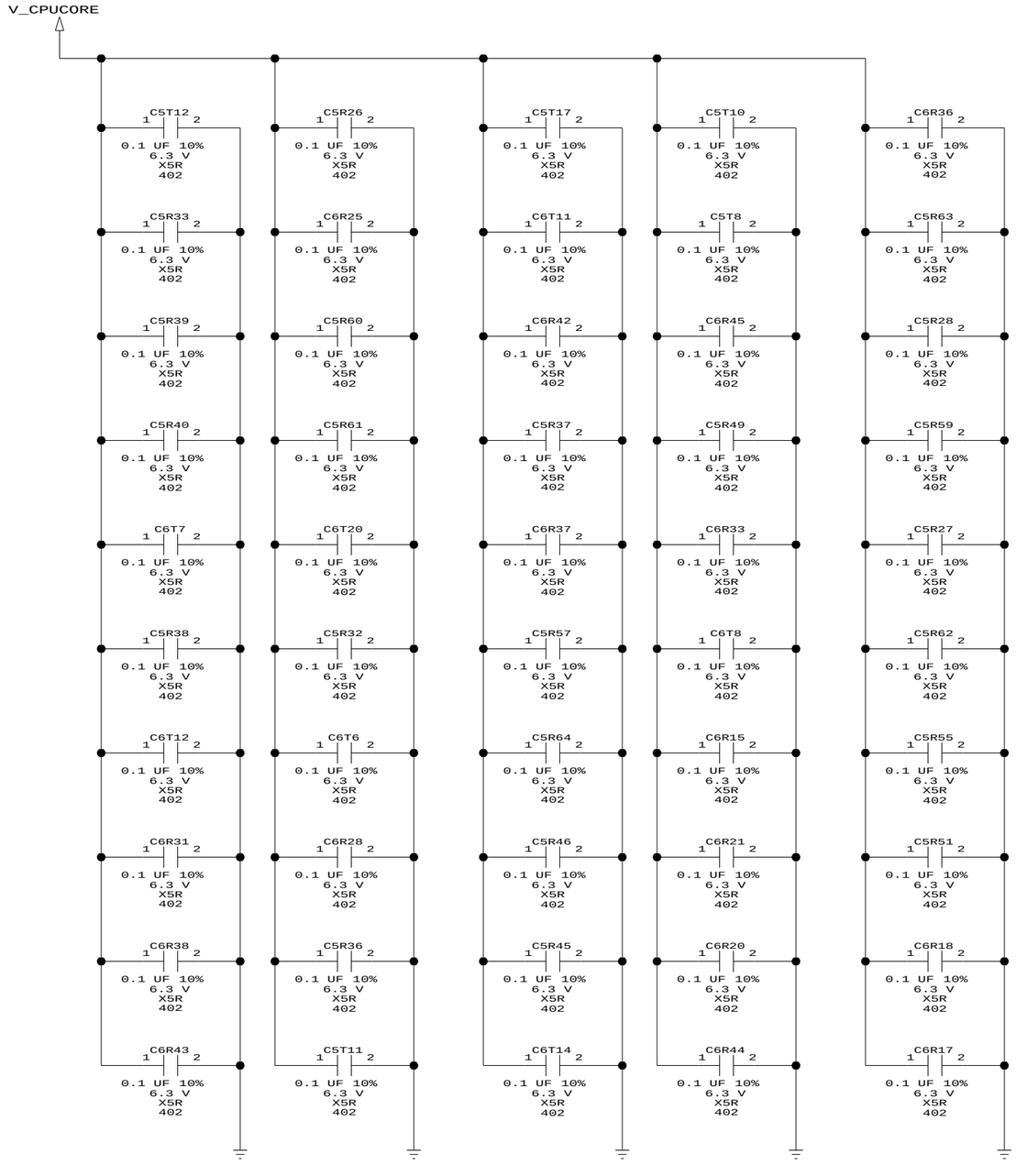



[PAGE_TITLE=GCPU, DECOUPLING]

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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 9/74 | 9/74 | C | 1.0 |

# GCPU, DECOUPLING

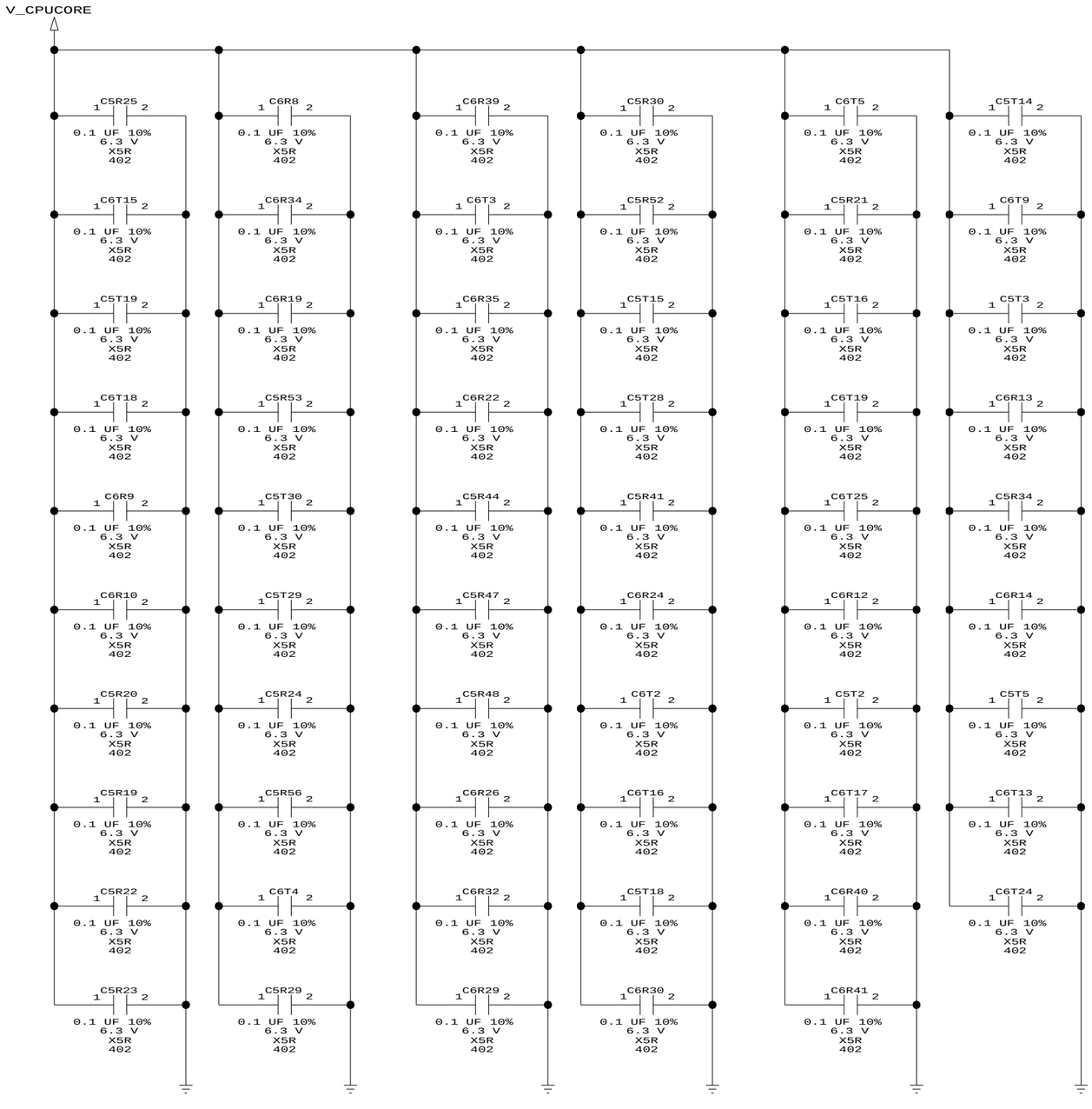



[PAGE_TITLE=GCPU, DECOUPLING]

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| MICROSOFT CONFIDENTIAL | PROJECT NAME STINGRAY | PAGE 10/74 | CSA PAGE 10/74 | FAB C | REV 1.0 |
|---|---|---|---|---|---|

# GCPU, DECOUPLING




[PAGE_TITLE=GCPU, DECOUPLING]

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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 11/74 | 11/74 | C | 1.0 |

# GPU, MEMORY CONTROLLER 0 PARTITION A & B

U5E1 4 OF 17 IC

GCPU VERSION 1

| | | | | |
|---|---|---|---|---|
| MA_DQ31 | T34 | MA_DQ31 | | |
| MA_DQ30 | T33 | MA_DQ30 | | |
| MA_DQ29 | U31 | MA_DQ29 | | |
| MA_DQ28 | R33 | MA_DQ28 | | |
| MA_DQ27 | U33 | MA_DQ27 | | |
| MA_DQ26 | R32 | MA_DQ26 | | |
| MA_DQ25 | V33 | MA_DQ25 | | |
| MA_DQ24 | R31 | MA_DQ24 | | |
| MA_WDQS3 | R34 | MA_WDQS3 | | |
| MA_RDQS3 | U32 | MA_RDQS3 | | |
| MA_DM3 | U34 | MA_DM3 | | |
| MA_DQ23 | Y34 | MA_DQ23 | | |
| MA_DQ22 | Y33 | MA_DQ22 | | |
| MA_DQ21 | Y32 | MA_DQ21 | | |
| MA_DQ20 | AA33 | MA_DQ20 | | |
| MA_DQ19 | W33 | MA_DQ19 | | |
| MA_DQ18 | AB31 | MA_DQ18 | | |
| MA_DQ17 | V34 | MA_DQ17 | | |
| MA_DQ16 | AB32 | MA_DQ16 | | |
| MA_WDQS2 | AA34 | MA_WDQS2 | | |
| MA_RDQS2 | Y31 | MA_RDQS2 | | |
| MA_DM2 | W34 | MA_DM2 | | |
| MA_DQ15 | W28 | MA_DQ15 | | |
| MA_DQ14 | R30 | MA_DQ14 | | |
| MA_DQ13 | W30 | MA_DQ13 | | |
| MA_DQ12 | R28 | MA_DQ12 | | |
| MA_DQ11 | V28 | MA_DQ11 | | |
| MA_DQ10 | T29 | MA_DQ10 | | |
| MA_DQ9 | U29 | MA_DQ9 | | |
| MA_DQ8 | U28 | MA_DQ8 | | |
| MA_WDQS1 | T30 | MA_WDQS1 | | |
| MA_RDQS1 | V30 | MA_RDQS1 | | |
| MA_DM1 | V32 | MA_DM1 | | |
| MA_DQ7 | Y30 | MA_DQ7 | | |
| MA_DQ6 | AD28 | MA_DQ6 | | |
| MA_DQ5 | Y28 | MA_DQ5 | | |
| MA_DQ4 | AD30 | MA_DQ4 | | |
| MA_DQ3 | AB28 | MA_DQ3 | | |
| MA_DQ2 | AC30 | MA_DQ2 | | |
| MA_DQ1 | AB29 | MA_DQ1 | | |
| MA_DQ0 | AC28 | MA_DQ0 | | |
| MA_WDQS0 | AC32 | MA_WDQS0 | | |
| MA_RDQS0 | AA29 | MA_RDQS0 | | |
| MA_DM0 | AA30 | MA_DM0 | | |
| | AB34 | MA_VREF0 | | |
| MA_CLK1_DP | AE28 | MA_CLK1_DP | OUT | 15 |
| MA_CLK1_DN | AE30 | MA_CLK1_DN | OUT | 15 |
| MA_CLK0_DP | AC33 | MA_CLK0_DP | OUT | 14 |
| MA_CLK0_DN | AC34 | MA_CLK0_DN | OUT | 14 |
| MA_A12 | AL33 | | | |
| MA_A11 | AH34 | MA_A<11..0> | OUT | 14 15 |
| MA_A10 | AL34 | | | |
| MA_A9 | AG34 | | | |
| MA_A8 | AD33 | | | |
| MA_A7 | AF34 | | | |
| MA_A6 | AE33 | | | |
| MA_A5 | AD34 | | | |
| MA_A4 | AF33 | | | |
| MA_A3 | AG33 | | | |
| MA_A2 | AH33 | | | |
| MA_A1 | AK33 | | | |
| MA_A0 | AJ34 | | | |
| MA_BA2 | AE34 | MA_BA<2..0> | OUT | 14 15 |
| MA_BA1 | AE32 | | | |
| MA_BA0 | AK34 | | | |
| MA_CKE | AJ33 | MA_CKE | OUT | 14 15 |
| MA_WE_N* | AF29 | MA_WE_N | OUT | 14 15 |
| MA_CAS_N* | AG30 | MA_CAS_N | OUT | 14 15 |
| MA_RAS_N* | AH30 | MA_RAS_N | OUT | 14 15 |
| MA_CS1_N* | AF30 | MA_CS1_N | OUT | 14 15 |
| MA_CS0_N* | AE31 | MA_CS0_N | OUT | 14 |

X818336-001 BGA_2

R6T8 4.75 KOHM 1% 402 CH

MA_WDQS1 OUT 15 14 12

MA_DQ12 BI 15 14 12

R6T7 4.75 KOHM 1% 402 CH

MA_DQ4 BI 15 14 12

MA_WDQS0 OUT 15 14 12

V_MEM

R6T5 549 OHM 1% EMPTY 402

MA_VREF0

R6T4 1.27 KOHM 1% EMPTY 402

NEED NEW VREF RESISTORS

MEMORY CONTROLLER A, DECOUPLING

V_MEM

C6T27 0.1 UF 10% 6.3 V X5R 402

C6T10 0.1 UF 10% 6.3 V X5R 402

C6T32 0.1 UF 10% 6.3 V X5R 402

C6T29 0.1 UF 10% 6.3 V X5R 402

TO CHANGE GPU VREF, CHANGE THESE RESISTORS TO MATCH THE TABLE
R6T4, R6T1, R5T3, R5T4

| MEM VREF | RESISTOR VALUE |
|---|---|
| 70% | 1.27KOHM |
| 72% | 1.40KOHM |
| 73% | 1.47KOHM |
| 74% | 1.54KOHM |

THESE ARE THE GPU VREFS NEEDED FOR VARIOUS MEMORIES. CONSULT WITH MEM TEAM FOR USAGE.

U5E1 3 OF 17 IC

GCPU VERSION 1

| | | | | |
|---|---|---|---|---|
| MB_DQ31 | H33 | MB_DQ31 | | |
| MB_DQ30 | G34 | MB_DQ30 | | |
| MB_DQ29 | H34 | MB_DQ29 | | |
| MB_DQ28 | F34 | MB_DQ28 | | |
| MB_DQ27 | K31 | MB_DQ27 | | |
| MB_DQ26 | D34 | MB_DQ26 | | |
| MB_DQ25 | K32 | MB_DQ25 | | |
| MB_DQ24 | C34 | MB_DQ24 | | |
| MB_WDQS3 | E34 | MB_WDQS3 | | |
| MB_RDQS3 | J33 | MB_RDQS3 | | |
| MB_DM3 | J34 | MB_DM3 | | |
| MB_DQ23 | M32 | MB_DQ23 | | |
| MB_DQ22 | M34 | MB_DQ22 | | |
| MB_DQ21 | M31 | MB_DQ21 | | |
| MB_DQ20 | M33 | MB_DQ20 | | |
| MB_DQ19 | K33 | MB_DQ19 | | |
| MB_DQ18 | N34 | MB_DQ18 | | |
| MB_DQ17 | K34 | MB_DQ17 | | |
| MB_DQ16 | P33 | MB_DQ16 | | |
| MB_WDQS2 | N33 | MB_WDQS2 | | |
| MB_RDQS2 | L34 | MB_RDQS2 | | |
| MB_DM2 | L33 | MB_DM2 | | |
| MB_DQ15 | J28 | MB_DQ15 | | |
| MB_DQ14 | C33 | MB_DQ14 | | |
| MB_DQ13 | J30 | MB_DQ13 | | |
| MB_DQ12 | D33 | MB_DQ12 | | |
| MB_DQ11 | F33 | MB_DQ11 | | |
| MB_DQ10 | E33 | MB_DQ10 | | |
| MB_DQ9 | G30 | MB_DQ9 | | |
| MB_DQ8 | F30 | MB_DQ8 | | |
| MB_WDQS1 | E30 | MB_WDQS1 | | |
| MB_RDQS1 | G33 | MB_RDQS1 | | |
| MB_DM1 | H30 | MB_DM1 | | |
| MB_DQ7 | K30 | MB_DQ7 | | |
| MB_DQ6 | P28 | MB_DQ6 | | |
| MB_DQ5 | K28 | MB_DQ5 | | |
| MB_DQ4 | P30 | MB_DQ4 | | |
| MB_DQ3 | M28 | MB_DQ3 | | |
| MB_DQ2 | N30 | MB_DQ2 | | |
| MB_DQ1 | M29 | MB_DQ1 | | |
| MB_DQ0 | N28 | MB_DQ0 | | |
| MB_WDQS0 | N32 | MB_WDQS0 | | |
| MB_RDQS0 | L29 | MB_RDQS0 | | |
| MB_DM0 | L30 | MB_DM0 | | |
| | P34 | MB_VREF0 | | |
| MB_CLK1_DP | B32 | MB_CLK1_DP | OUT | 17 |
| MB_CLK1_DN | A32 | MB_CLK1_DN | OUT | 17 |
| MB_CLK0_DP | B31 | MB_CLK0_DP | OUT | 16 |
| MB_CLK0_DN | A31 | MB_CLK0_DN | OUT | 16 |
| MB_A12 | C30 | | | |
| MB_A11 | F25 | MB_A<11..0> | OUT | 16 17 |
| MB_A10 | A25 | | | |
| MB_A9 | E29 | | | |
| MB_A8 | B30 | | | |
| MB_A7 | B29 | | | |
| MB_A6 | A30 | | | |
| MB_A5 | C32 | | | |
| MB_A4 | A29 | | | |
| MB_A3 | E25 | | | |
| MB_A2 | G26 | | | |
| MB_A1 | B25 | | | |
| MB_A0 | F26 | | | |
| MB_BA2 | E28 | MB_BA<2..0> | OUT | 16 17 |
| MB_BA1 | B28 | | | |
| MB_BA0 | E27 | | | |
| MB_CKE | A27 | MB_CKE | OUT | 16 17 |
| MB_WE_N* | B26 | MB_WE_N | OUT | 16 17 |
| MB_CAS_N* | C26 | MB_CAS_N | OUT | 16 17 |
| MB_RAS_N* | D26 | MB_RAS_N | OUT | 16 17 |
| MB_CS1_N* | A28 | MB_CS1_N | OUT | 16 17 |
| MB_CS0_N* | A26 | MB_CS0_N | OUT | 16 |

X818336-001 BGA_2

R6R2 4.75 KOHM 1% 402 CH

MB_WDQS1 OUT 17 16 12

MB_DQ12 BI 17 16 12

R6R1 4.75 KOHM 1% 402 CH

MB_DQ4 BI 17 16 12

MB_WDQS0 OUT 17 16 12

V_MEM

R6T2 549 OHM 1% EMPTY 402

MB_VREF0

R6T1 1.27 KOHM 1% EMPTY 402

MEMORY CONTROLLER B, DECOUPLING

V_MEM

C6R27 0.1 UF 10% 6.3 V X5R 402

C6R11 0.1 UF 10% 6.3 V X5R 402

C6R16 0.1 UF 10% 6.3 V X5R 402

C6T30 4.7 UF 10% 6.3 V X5R 603

C6R23 0.1 UF 10% 6.3 V X5R 402

C6R46 0.1 UF 10% 6.3 V X5R 402

N: GPU VREF SET INTERNALLY BY DEFAULT. EXTERNAL RESISTOR DIVIDER USED TO MANUALLY SET GPU VREF VOLTAGE.

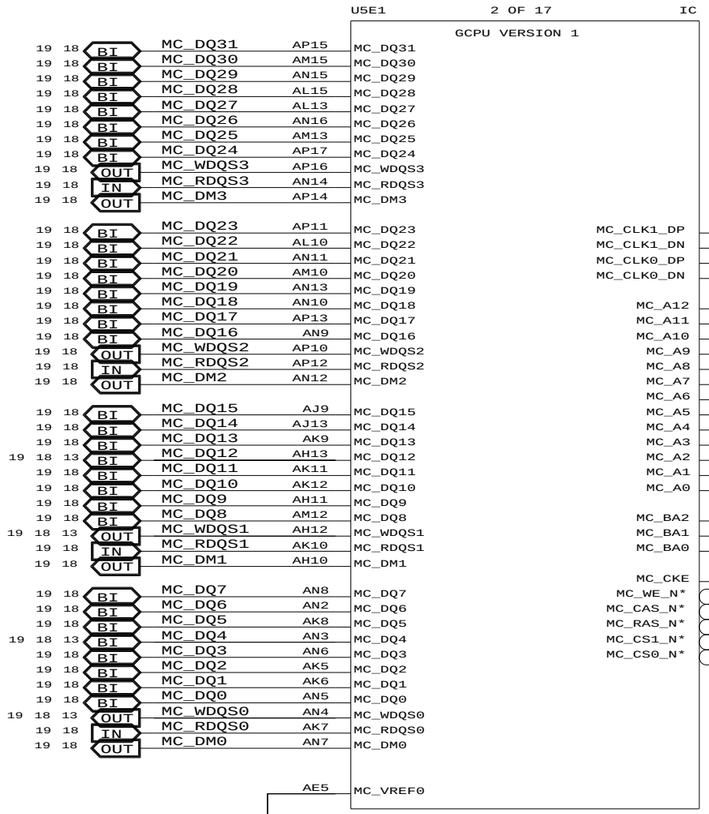
[PAGE_TITLE=GPU, MEMORY CONTROLLER A + B]

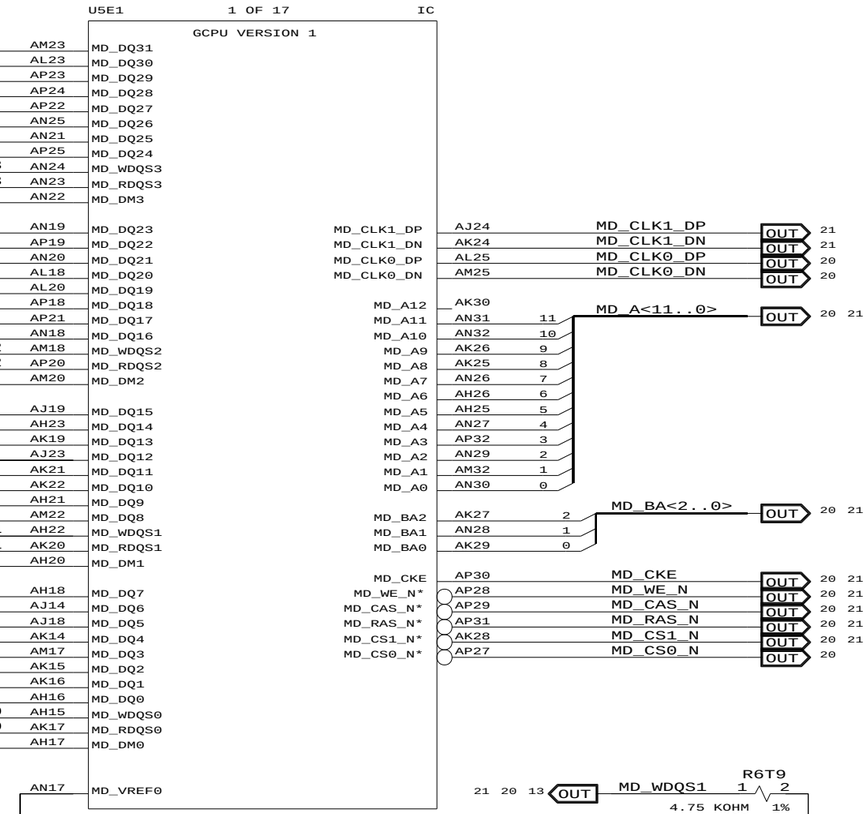
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| MICROSOFT | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| CONFIDENTIAL | STINGRAY | 12/74 | 12/74 | C | 1.0 |

# GPU, MEMORY CONTROLLER 1 PARTITION C & D




TO CHANGE GPU VREF, CHANGE THESE RESISTORS TO MATCH THE TABLE
R6T4, R6T1, R5T3, R5T4

| GPU MEM VREF | RESISTOR VALUE |
|---|---|
| 70% | 1.27KOHM |
| 72% | 1.40KOHM |
| 73% | 1.47KOHM |
| 74% | 1.54KOHM |

THESE ARE THE GPU VREFS NEEDED FOR VARIOUS MEMORIES. CONSULT WITH MEM TEAM FOR USAGE.

N: GPU VREF SET INTERNALLY BY DEFAULT. EXTERNAL RESISTOR DIVIDER USED TO MANUALLY SET GPU VREF VOLTAGE.

[PAGE_TITLE=[GPU, MEMORY CONTROLLER C + D]

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|---|---|---|---|---|---|

# MEMORY PARTITION A, TOP

CHIP SELECT = 0, MIRROR FUNCTION = 0

| MS_PART# | MATL | REF_DES | | DESCR. | BOM PROPERTY |
|---|---|---|---|---|---|
| U7E1 | X801996-024 | IC | IC,MEM,HYNIX | GDDR_TOP_HYNIX | PACK_IGNORE=TRUE |
| U7E1 | X801995-018 | IC | IC,MEM,SAMSUNG | GDDR_TOP_SAMSUNG | PACK_IGNORE=TRUE |
| U7E1 | X857474-001 | IC | IC,MEM,WINBOND | GDDR_TOP_WINBOND | PACK_IGNORE=TRUE |

V_MEM

R7E5 60.4 OHM 1% CH 402 (1 / 2)

R7E4 60.4 OHM 1% CH 402 (1 / 2)

U7E1 — BOM_IGNORE=TRUE — EMPTY

GDDR136 (1Gbit) MF=1

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 12 | IN | MA_CLK0_DP | J11 | CLK_DP | | | |
| 12 | IN | MA_CLK0_DN | J10 | CLK_DN | | | |
| 19 18 17 16 15 4 21 20 | IN | MEM_RST | V9 | RESET | | | |
| | | | | | DQ31 | T3 | MA_DQ31 BI 12 15 |
| | | | | | DQ30 | T2 | MA_DQ30 BI 12 15 |
| | | | | | DQ29 | R3 | MA_DQ29 BI 12 15 |
| | | | | | DQ28 | R2 | MA_DQ28 BI 12 15 |
| | | | | | DQ27 | M3 | MA_DQ27 BI 12 15 |
| | | | | | DQ26 | N2 | MA_DQ26 BI 12 15 |
| | | | | | DQ25 | L3 | MA_DQ25 BI 12 15 |
| | | | | | DQ24 | M2 | MA_DQ24 BI 12 15 |
| | | | | | WDQS3 | P2 | MA_WDQS3 IN 12 15 |
| | | | | | RDQS3 | P3 | MA_RDQS3 OUT 12 15 |
| | | | | | DM3 | N3 | MA_DM3 IN 12 15 |
| | | | J2 | A12 (1Gbit only, dual-load) | | | |
| 15 12 | IN | MA_A<11..0> | 11 L4 | A11/A7 | DQ23 | T10 | MA_DQ23 BI 12 15 |
| | | | 10 K2 | A10/A8 | DQ22 | T11 | MA_DQ22 BI 12 15 |
| | | | 9 M9 | A9/A3 | DQ21 | R10 | MA_DQ21 BI 12 15 |
| | | | 8 K11 | A8/A10 | DQ20 | R11 | MA_DQ20 BI 12 15 |
| | | | 7 L9 | A7/A11 | DQ19 | M10 | MA_DQ19 BI 12 15 |
| | | | 6 K10 | A6/A2 | DQ18 | N11 | MA_DQ18 BI 12 15 |
| | | | 5 H11 | A5/A1 | DQ17 | L10 | MA_DQ17 BI 12 15 |
| | | | 4 K9 | A4/A0 | DQ16 | M11 | MA_DQ16 BI 12 15 |
| | | | 3 M4 | A3/A9 | WDQS2 | P11 | MA_WDQS2 IN 12 15 |
| | | | 2 K3 | A2/A6 | RDQS2 | P10 | MA_RDQS2 OUT 12 15 |
| | | | 1 H2 | A1/A5 | DM2 | N10 | MA_DM2 IN 12 15 |
| | | | 0 K4 | A0/A4 | | | |
| | | | | | DQ15 | G10 | MA_DQ15 BI 12 15 |
| | | | | | DQ14 | F11 | MA_DQ14 BI 12 15 |
| | | | | | DQ13 | F10 | MA_DQ13 BI 12 15 |
| | | | | | DQ12 | E11 | MA_DQ12 BI 12 15 |
| | | | | | DQ11 | C10 | MA_DQ11 BI 12 15 |
| | | | | | DQ10 | C11 | MA_DQ10 BI 12 15 |
| 15 12 | IN | MA_BA<2..0> | 2 H10 | BA2/RAS_N | | | |
| | | | 1 G9 | BA1/BA0 | | | |
| | | | 0 G4 | BA0/BA1 | | | |
| 15 12 | IN | MA_CKE | H4 | CKE/WE_N | DQ9 | B10 | MA_DQ9 BI 12 15 |
| 15 12 | IN | MA_WE_N | H9 | WE_N/CKE | DQ8 | B11 | MA_DQ8 BI 12 15 |
| 15 12 | IN | MA_CAS_N | F4 | CAS_N/CS_N | WDQS1 | D11 | MA_WDQS1 IN 12 15 |
| 15 12 | IN | MA_RAS_N | H3 | RAS_N/BA2 | RDQS1 | D10 | MA_RDQS1 OUT 12 15 |
| 12 | IN | MA_CS0_N | F9 | CS_N/CAS_N | DM1 | E10 | MA_DM1 IN 12 15 |
| 15 12 | IN | MA_CS1_N | J3 | CS1_N (1Gbit only, single-load) | | | |
| 20 18 16 4 | IN | MEM_SCAN_TOP_EN | A9 | MF | | | |
| | | | | | DQ7 | G3 | MA_DQ7 BI 12 15 |
| | | | | | DQ6 | F2 | MA_DQ6 BI 12 15 |
| | | | | | DQ5 | F3 | MA_DQ5 BI 12 15 |
| | | | | | DQ4 | E2 | MA_DQ4 BI 12 15 |
| | | | | | DQ3 | C3 | MA_DQ3 BI 12 15 |
| | | | | | DQ2 | C2 | MA_DQ2 BI 12 15 |
| | | | | | DQ1 | B3 | MA_DQ1 BI 12 15 |
| | | | | | DQ0 | B2 | MA_DQ0 BI 12 15 |
| | | | | | WDQS0 | D2 | MA_WDQS0 IN 12 15 |
| | | | | | RDQS0 | D3 | MA_RDQS0 OUT 12 15 |
| | | | | | DM0 | E3 | MA_DM0 IN 12 15 |
| 19 18 17 16 15 4 21 20 | IN | MEM_SCAN_EN | V4 | SCAN_EN | | | |
| 15 14 | IN | MEM_A_VREF1 | H1 | VREF1 | | | |
| 15 | IN | MEM_A_VREF0 | H12 | VREF0 | | | |
| | | | | | ZQ | A4 | MA_ZQ_TOP |

X802980-019

R7E8 243 OHM 1% CH 402 (1 / 2)

V_MEM

U7E1 — EMPTY

GDDR136 (1Gbit) MF=0

| | | | |
|---|---|---|---|
| V1 | VDDQ<21> | VSSQ<19> | T12 |
| R12 | VDDQ<20> | VSSQ<18> | T9 |
| R9 | VDDQ<19> | VSSQ<17> | T4 |
| R4 | VDDQ<18> | VSSQ<16> | T1 |
| R1 | VDDQ<17> | VSSQ<15> | P12 |
| N12 | VDDQ<16> | VSSQ<14> | P9 |
| N9 | VDDQ<15> | VSSQ<13> | P4 |
| V12 | VDDQ<14> | VSSQ<12> | P1 |
| N4 | VDDQ<13> | VSSQ<11> | L11 |
| N1 | VDDQ<12> | VSSQ<10> | L2 |
| J9 | VDDQ<11> | VSSQ<9> | G11 |
| J4 | VDDQ<10> | VSSQ<8> | G2 |
| E12 | VDDQ<9> | VSSQ<7> | D12 |
| E9 | VDDQ<8> | VSSQ<6> | D9 |
| E4 | VDDQ<7> | VSSQ<5> | D4 |
| E1 | VDDQ<6> | VSSQ<4> | D1 |
| C12 | VDDQ<5> | VSSQ<3> | B12 |
| C9 | VDDQ<4> | VSSQ<2> | B9 |
| C4 | VDDQ<3> | VSSQ<1> | B4 |
| C1 | VDDQ<2> | VSSQ<0> | B1 |
| A12 | VDDQ<1> | | |
| A1 | VDDQ<0> | | |
| V2 | VDD<7> | VSS<7> | V3 |
| M12 | VDD<6> | VSS<6> | L12 |
| M1 | VDD<5> | VSS<5> | L1 |
| V11 | VDD<4> | VSS<4> | G12 |
| F12 | VDD<3> | VSS<3> | G1 |
| F1 | VDD<2> | VSS<2> | A10 |
| A11 | VDD<1> | VSS<1> | V10 |
| A2 | VDD<0> | VSS<0> | A3 |
| K12 | VDDA<1> | | |
| K1 | VDDA<0> | | |
| J12 | VSSA<1> | | |
| J1 | VSSA<0> | | |

X802980-019

MX_CS1_N CONNECTED TO J3 TO SUPPORT 1G RAM CONFIGS.

V_MEM

R7T5 549 OHM 1% CH 402 (1 / 2)

MEM_A_VREF1 — OUT 14 15

GDDR_TOP_SAMSUNG&GDDR_TOP_WINBOND

R7T4 1.27 KOHM 1% CH 402 (1 / 2)

C7T7 0.1 UF 10% 6.3 V X5R 402

TO CHANGE MEM VREF, CHANGE THESE RESISTORS TO MATCH THE TABLE
R7T4, R7E7, R7R4, R7D5, R5U4, R5F2, R6U4, R6F2

| MEM VREF | RESISTOR VALUE |
|---|---|
| 69% | 1.21KOHM |
| 70% | 1.27KOHM |
| 72% | 1.40KOHM |

THESE ARE THE MEM VREFS NEEDED FOR VARIOUS MEMORIES. CONSULT WITH MEM TEAM FOR USAGE.

| MS_PART# | MATL | REF_DES | | DESCR. | BOM PROPERTY |
|---|---|---|---|---|---|
| R7T4 | X801176-001 | CH | RES,1.47KOHM | GDDR_TOP_HYNIX | PACK_IGNORE=TRUE |

PARTITION A DECOUPLING

V_MEM

C7E8 4.7 UF 10% 6.3 V X5R 603 (1 / 2)

MEMORY A, TOP, DECOUPLING

V_MEM

| C7E14 | C7E11 | C7E6 | C7E5 | C7E13 | C7E10 | C7E4 | C7E7 |
|---|---|---|---|---|---|---|---|
| 0.1 UF 10% 6.3 V X5R 402 | 0.1 UF 10% 6.3 V X5R 402 | 0.1 UF 10% 6.3 V X5R 402 | 0.1 UF 10% 6.3 V X5R 402 | 0.1 UF 10% 6.3 V X5R 402 | 0.1 UF 10% 6.3 V X5R 402 | 0.1 UF 10% 6.3 V X5R 402 | 0.1 UF 10% 6.3 V X5R 402 |

[PAGE_TITLE=MEMORY PARTITION A, TOP]

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|---|---|---|---|---|---|

# MEMORY PARTITION A, BOTTOM

CHIP SELECT = 1, MIRROR FUNCTION = 1

| MS_PART# | MATL | REF_DES | | DESCR. | BOM PROPERTY |
|---|---|---|---|---|---|
| U7T1 | X801996-024 | IC | IC,MEM,HYNIX | GDDR_BOT_HYNIX | PACK_IGNORE=TRUE |
| U7T1 | X801995-018 | IC | IC,MEM,SAMSUNG | GDDR_BOT_SAMSUNG | PACK_IGNORE=TRUE |
| U7T1 | X857474-001 | IC | IC,MEM,WINBOND | GDDR_BOT_WINBOND | PACK_IGNORE=TRUE |

V_MEM

R7T3 60.4 OHM 1% CH 402

R7T2 60.4 OHM 1% CH 402

U7T1 BOM=GDDR_TOP EMPTY

GDDR136 (1Gbit) MF=1

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 12 | IN | MA_CLK1_DP | J11 | CLK_DP | DQ31 | T3 | MA_DQ23 | BI 12 14 |
| 21 12 | IN | MA_CLK1_DN | J10 | CLK_DN | DQ30 | T2 | MA_DQ22 | BI 12 14 |
| 17 16 14 4 20 19 18 | IN | MEM_RST | V9 | RESET | DQ29 | R3 | MA_DQ21 | BI 12 14 |
| | | | | | DQ28 | R2 | MA_DQ20 | BI 12 14 |
| | | | | | DQ27 | M3 | MA_DQ19 | BI 12 14 |
| | | | | | DQ26 | N2 | MA_DQ18 | BI 12 14 |
| | | | | | DQ25 | L3 | MA_DQ17 | BI 12 14 |
| | | | | | DQ24 | M2 | MA_DQ16 | BI 12 14 |
| | | | | | WDQS3 | P2 | MA_WDQS2 | BI 12 14 |
| | | | | | RDQS3 | P3 | MA_RDQS2 | IN/OUT 12 14 |
| | | | | | DM3 | N3 | MA_DM2 | IN 12 14 |
| 14 12 | IN | MA_A<11..0> | J2 | A12 (1Gbit only, dual-load) | DQ23 | T10 | MA_DQ31 | BI 12 14 |
| | | | L9 | A7/A11 | DQ22 | T11 | MA_DQ30 | BI 12 14 |
| | | | K11 | A8/A10 | DQ21 | R10 | MA_DQ29 | BI 12 14 |
| | | | M4 | A3/A9 | DQ20 | R11 | MA_DQ28 | BI 12 14 |
| | | | K2 | A10/A8 | DQ19 | M10 | MA_DQ27 | BI 12 14 |
| | | | L4 | A11/A7 | DQ18 | N11 | MA_DQ26 | BI 12 14 |
| | | | K3 | A2/A6 | DQ17 | L10 | MA_DQ25 | BI 12 14 |
| | | | H2 | A1/A5 | DQ16 | M11 | MA_DQ24 | BI 12 14 |
| | | | K4 | A0/A4 | WDQS2 | P11 | MA_WDQS3 | IN 12 14 |
| | | | M9 | A9/A3 | RDQS2 | P10 | MA_RDQS3 | OUT 12 14 |
| | | | K10 | A6/A2 | DM2 | N10 | MA_DM3 | IN 12 14 |
| | | | H11 | A5/A1 | | | | |
| | | | K9 | A4/A0 | DQ15 | G10 | MA_DQ7 | BI 12 14 |
| 14 12 | IN | MA_BA<2..0> | H3 | RAS_N/BA2 | DQ14 | F11 | MA_DQ6 | BI 12 14 |
| | | | G4 | BA0/BA1 | DQ13 | F10 | MA_DQ5 | BI 12 14 |
| | | | G9 | BA1/BA0 | DQ12 | E11 | MA_DQ4 | BI 12 14 |
| | | | | | DQ11 | C10 | MA_DQ3 | BI 12 14 |
| | | | | | DQ10 | C11 | MA_DQ2 | BI 12 14 |
| 14 12 | IN | MA_CKE | H9 | WE_N/CKE | DQ9 | B10 | MA_DQ1 | BI 12 14 |
| 14 12 | IN | MA_WE_N | H4 | CKE/WE_N | DQ8 | B11 | MA_DQ0 | BI 12 14 |
| 14 12 | IN | MA_CAS_N | F9 | CS_N/CAS_N | WDQS1 | D11 | MA_WDQS0 | IN 12 14 |
| 14 12 | IN | MA_RAS_N | H10 | BA2/RAS_N | RDQS1 | D10 | MA_RDQS0 | OUT 12 14 |
| 14 12 | IN | MA_CS1_N | F4 | CAS_N/CS_N | DM1 | E10 | MA_DM0 | IN 12 14 |
| | | | J3 | CS1_N (1Gbit only, single-load) | | | | |
| 21 19 17 4 | IN | MEM_SCAN_BOT_EN | A9 | MF | DQ7 | G3 | MA_DQ15 | BI 12 14 |
| | | | | | DQ6 | F2 | MA_DQ14 | BI 12 14 |
| 18 17 16 14 4 21 20 19 | IN | MEM_SCAN_EN | V4 | SCAN_EN | DQ5 | F3 | MA_DQ13 | BI 12 14 |
| | | | | | DQ4 | E2 | MA_DQ12 | BI 12 14 |
| | | | | | DQ3 | C3 | MA_DQ11 | BI 12 14 |
| | | | | | DQ2 | C2 | MA_DQ10 | BI 12 14 |
| | | | | | DQ1 | B3 | MA_DQ9 | BI 12 14 |
| | | | | | DQ0 | B2 | MA_DQ8 | BI 12 14 |
| | | | | | WDQS0 | D2 | MA_WDQS1 | IN 12 14 |
| | | | | | RDQS0 | D3 | MA_RDQS1 | OUT 12 14 |
| | | | | | DM0 | E3 | MA_DM1 | IN 12 14 |
| 15 14 | IN | MEM_A_VREF0 | H1 | VREF1 | | | | |
| 14 | IN | MEM_A_VREF1 | H12 | VREF0 | ZQ | A4 | MA_ZQ_BOT | |

X802980-019

GDDR_BOTTOM R7T6 243 OHM 1% CH 402

V_MEM

U7T1 EMPTY

GDDR136 (1Gbit) MF=0

| | | | |
|---|---|---|---|
| V1 | VDDQ<21> | VSSQ<19> | T12 |
| R12 | VDDQ<20> | VSSQ<18> | T9 |
| R9 | VDDQ<19> | VSSQ<17> | T4 |
| R4 | VDDQ<18> | VSSQ<16> | T1 |
| R1 | VDDQ<17> | VSSQ<15> | P12 |
| N12 | VDDQ<16> | VSSQ<14> | P9 |
| N9 | VDDQ<15> | VSSQ<13> | P4 |
| V12 | VDDQ<14> | VSSQ<12> | P1 |
| N4 | VDDQ<13> | VSSQ<11> | L11 |
| N1 | VDDQ<12> | VSSQ<10> | L2 |
| J9 | VDDQ<11> | VSSQ<9> | G11 |
| J4 | VDDQ<10> | VSSQ<8> | G2 |
| E12 | VDDQ<9> | VSSQ<7> | D12 |
| E9 | VDDQ<8> | VSSQ<6> | D9 |
| E4 | VDDQ<7> | VSSQ<5> | D4 |
| E1 | VDDQ<6> | VSSQ<4> | D1 |
| C12 | VDDQ<5> | VSSQ<3> | B12 |
| C9 | VDDQ<4> | VSSQ<2> | B9 |
| C4 | VDDQ<3> | VSSQ<1> | B4 |
| C1 | VDDQ<2> | VSSQ<0> | B1 |
| A12 | VDDQ<1> | | |
| A1 | VDDQ<0> | | |
| V2 | VDD<7> | VSS<7> | V3 |
| M12 | VDD<6> | VSS<6> | L12 |
| M1 | VDD<5> | VSS<5> | L1 |
| V11 | VDD<4> | VSS<4> | G12 |
| F12 | VDD<3> | VSS<3> | G1 |
| F1 | VDD<2> | VSS<2> | A10 |
| A11 | VDD<1> | VSS<1> | V10 |
| A2 | VDD<0> | VSS<0> | A3 |
| K12 | VDDA<1> | | |
| K1 | VDDA<0> | | |
| J12 | VSSA<1> | | |
| J1 | VSSA<0> | | |

X802980-019

V_MEM

R7E6 549 OHM 1% CH 402

MEM_A_VREF0 OUT 14 15

GDDR_TOP_SAMSUNG&GDDR_TOP_WINBOND

R7E7 1.27 KOHM 1% CH 402

C7E9 0.1 UF 10% 6.3 V X5R 402

V_MEM

MEMORY A, BOTTOM, DECOUPLING

C7T12 C7T9 C7T5 C7T4 C7T11 C7T8 C7T3 C7T6 — 0.1 UF 10% 6.3 V X5R 402

| MS_PART# | MATL | REF_DES | | DESCR. | BOM PROPERTY |
|---|---|---|---|---|---|
| R7E7 | X801176-001 | CH | RES,1.47KOHM | GDDR_TOP_HYNIX | PACK_IGNORE=TRUE |

[PAGE_TITLE=MEMORY PARTITION A, BOTTOM]

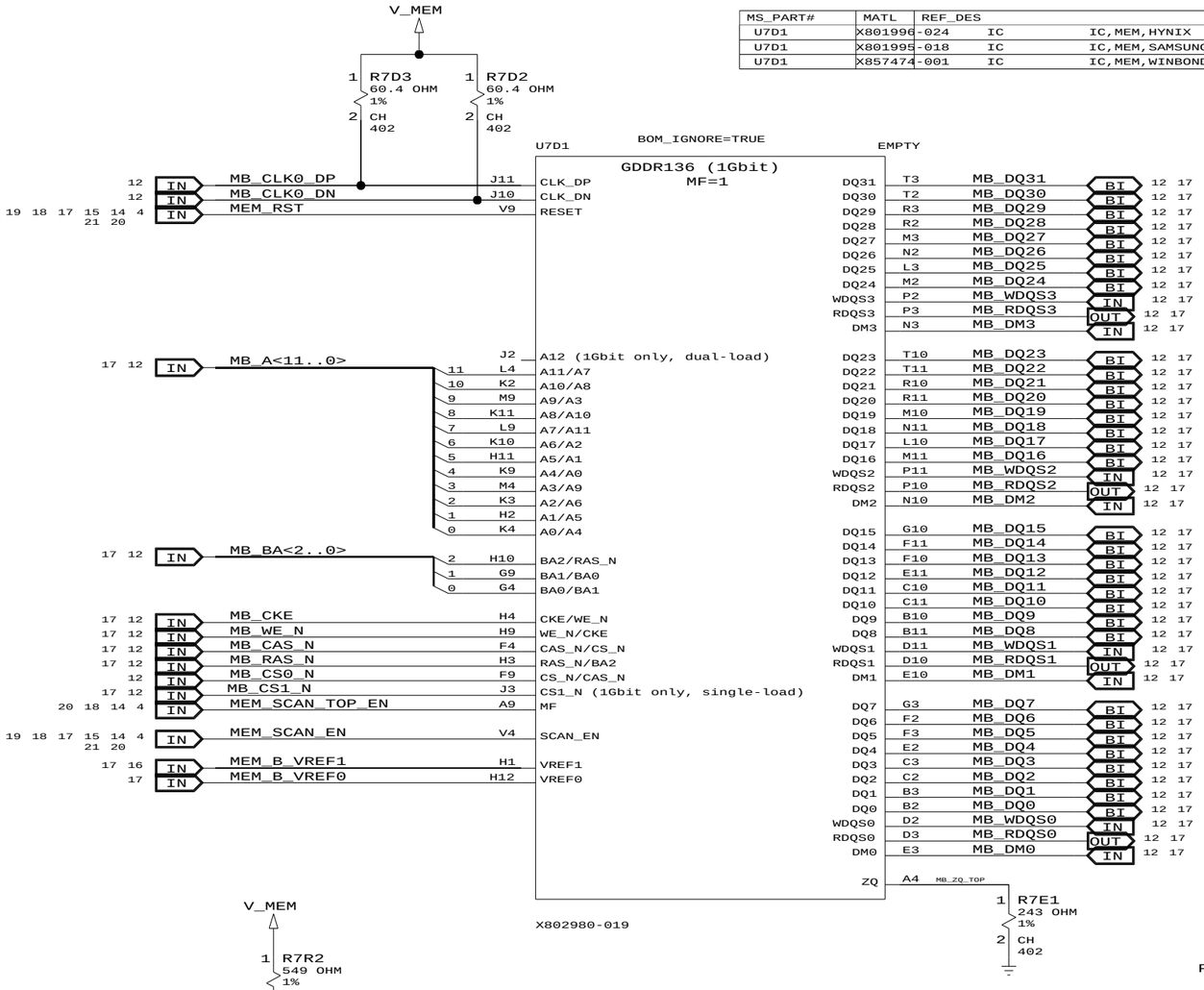
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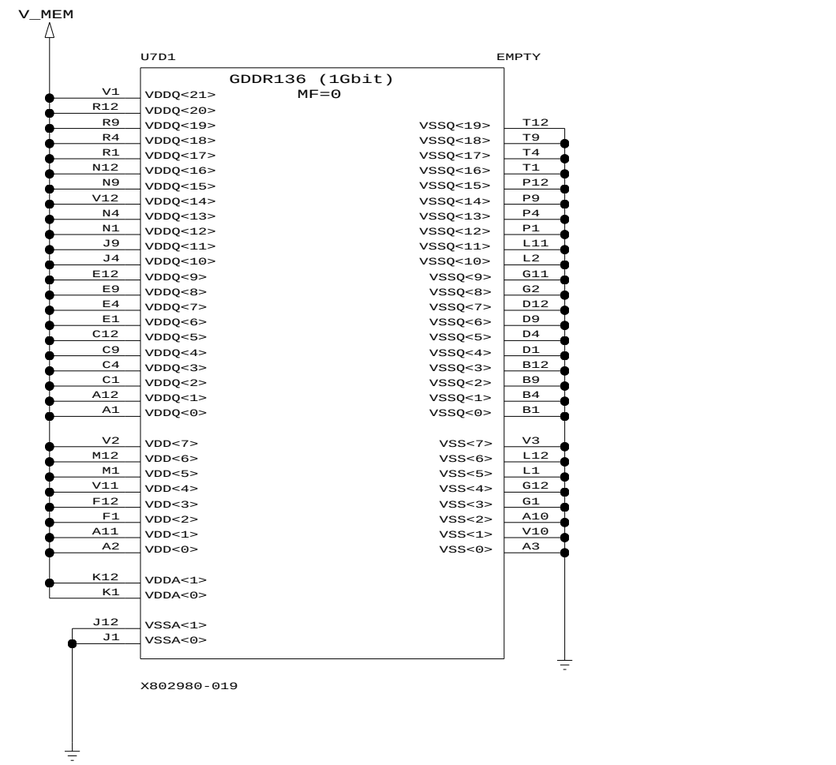
| MICROSOFT CONFIDENTIAL | PROJECT NAME STINGRAY | PAGE 15/74 | CSA PAGE 15/74 | FAB C | REV 1.0 |
|---|---|---|---|---|---|

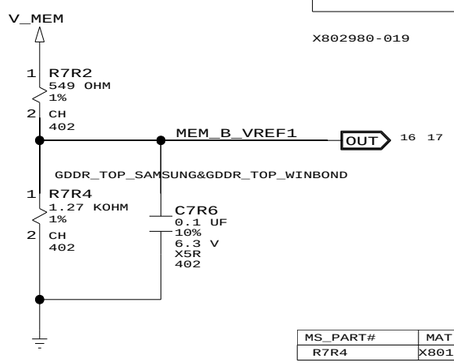
# MEMORY PARTITION B, TOP

CHIP SELECT = 0, MIRROR FUNCTION = 0

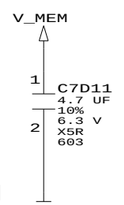
| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| U7D1 | X801996-024 | IC | IC,MEM,HYNIX | GDDR_TOP_HYNIX PACK_IGNORE=TRUE |
| U7D1 | X801995-018 | IC | IC,MEM,SAMSUNG | GDDR_TOP_SAMSUNG PACK_IGNORE=TRUE |
| U7D1 | X857474-001 | IC | IC,MEM,WINBOND | GDDR_TOP_WINBOND PACK_IGNORE=TRUE |

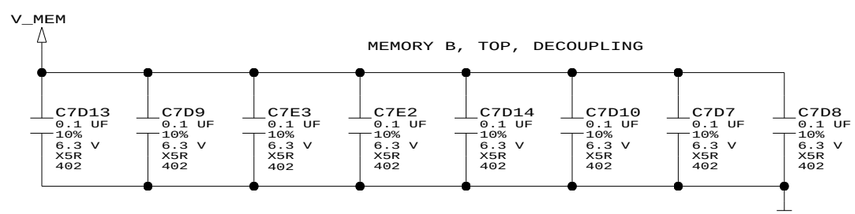



PARTITION B DECOUPLING




| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| R7R4 | X801176-001 | CH | RES,1.47KOHM | GDDR_TOP_HYNIX PACK_IGNORE=TRUE |

[PAGE_TITLE=MEMORY PARITION B, TOP]

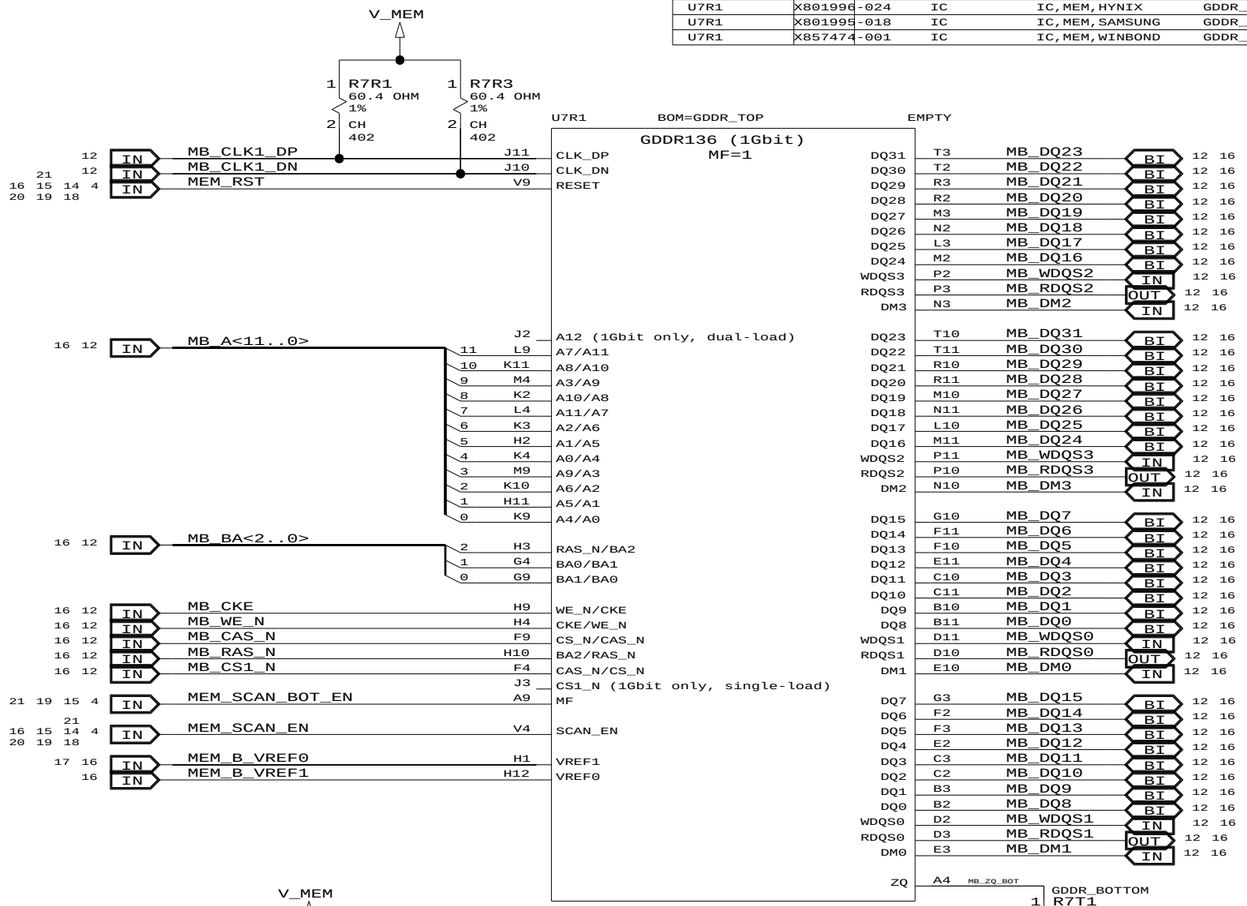
DRAWING
Fri Jan 04 12:07:15 2013

| MICROSOFT CONFIDENTIAL | PROJECT NAME STINGRAY | PAGE 16/74 | CSA PAGE 16/74 | FAB C | REV 1.0 |
|---|---|---|---|---|---|

# MEMORY PARTITION B, BOTTOM

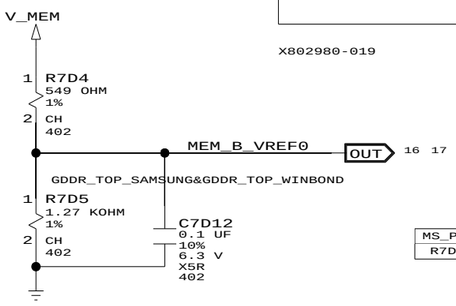
CHIP SELECT = 1, MIRROR FUNCTION = 1

| MS_PART# | MATL | REF_DES | | DESCR. | BOM PROPERTY |
|---|---|---|---|---|---|
| U7R1 | X801996-024 | IC | IC,MEM,HYNIX | GDDR_BOT_HYNIX | PACK_IGNORE=TRUE |
| U7R1 | X801995-018 | IC | IC,MEM,SAMSUNG | GDDR_BOT_SAMSUNG | PACK_IGNORE=TRUE |
| U7R1 | X857474-001 | IC | IC,MEM,WINBOND | GDDR_BOT_WINBOND | PACK_IGNORE=TRUE |

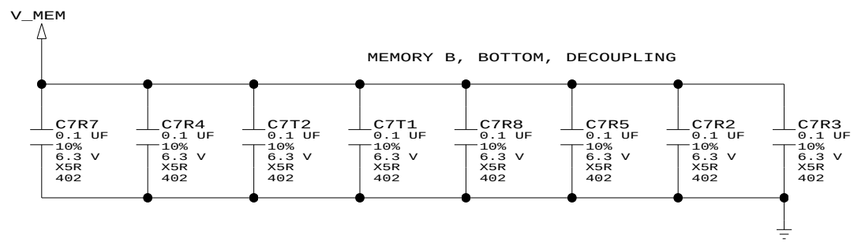









| MS_PART# | MATL | REF_DES | | DESCR. | BOM PROPERTY |
|---|---|---|---|---|---|
| R7D5 | X801176-001 | CH | RES,1.47KOHM | GDDR_TOP_HYNIX | PACK_IGNORE=TRUE |




[PAGE_TITLE=MEMORY PARITION B, BOTTOM]

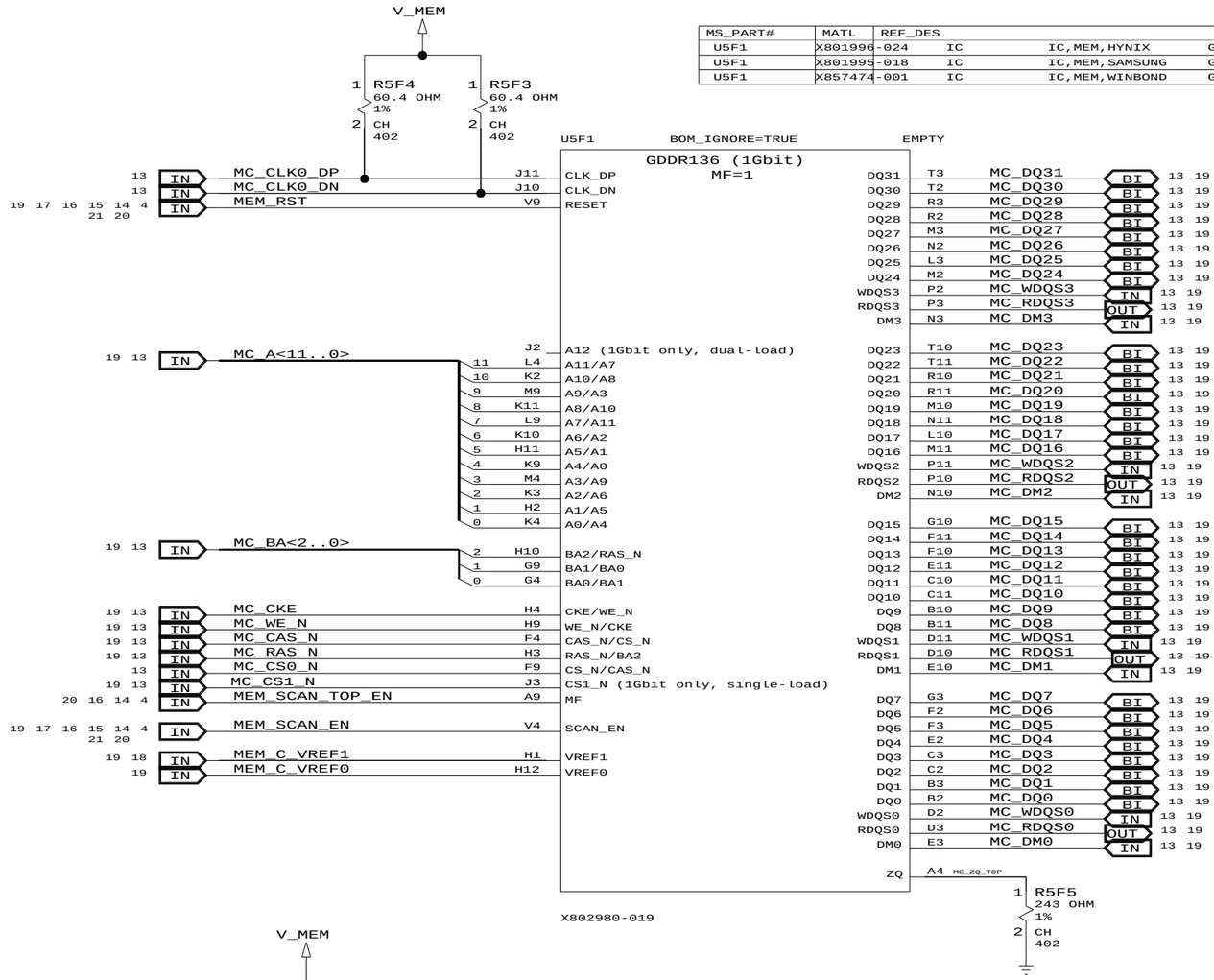
DRAWING
Fri Jan 04 12:07:15 2008

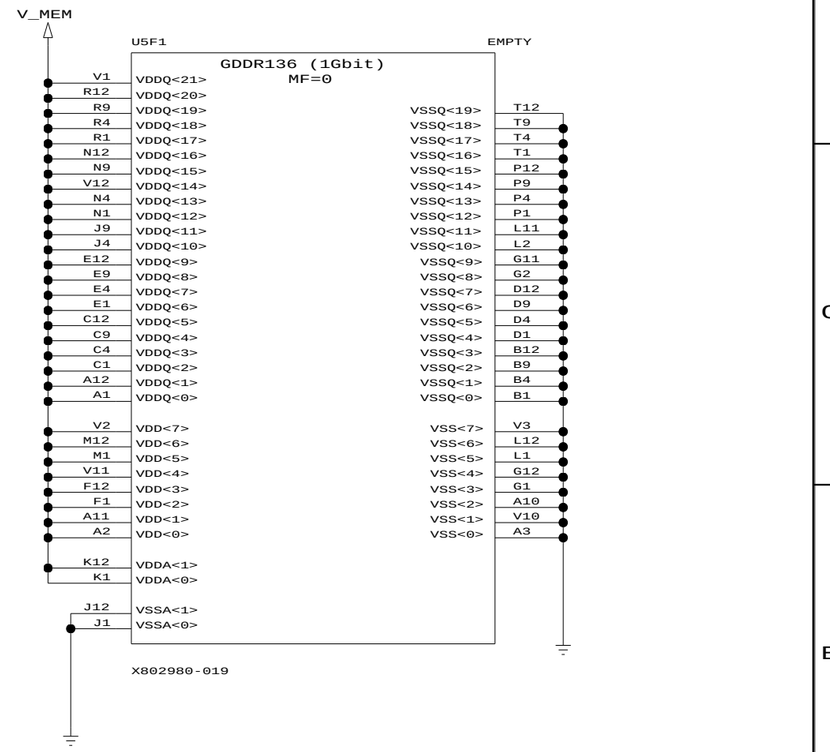
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 17/74 | 17/74 | C | 1.0 |

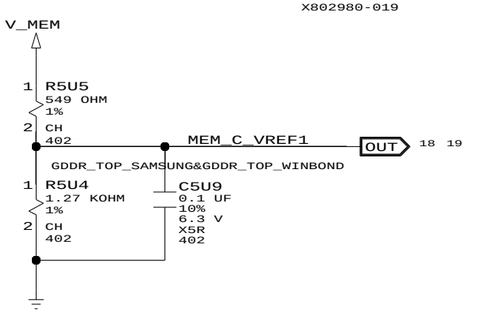
# MEMORY PARTITION C, TOP

CHIP SELECT = 0, MIRROR FUNCTION = 0

| MS_PART# | MATL | REF_DES | | DESCR. | BOM PROPERTY |
|---|---|---|---|---|---|
| U5F1 | X801996-024 | IC | IC,MEM,HYNIX | GDDR_TOP_HYNIX | PACK_IGNORE=TRUE |
| U5F1 | X801995-018 | IC | IC,MEM,SAMSUNG | GDDR_TOP_SAMSUNG | PACK_IGNORE=TRUE |
| U5F1 | X857474-001 | IC | IC,MEM,WINBOND | GDDR_TOP_WINBOND | PACK_IGNORE=TRUE |

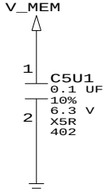


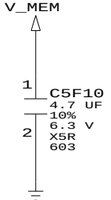



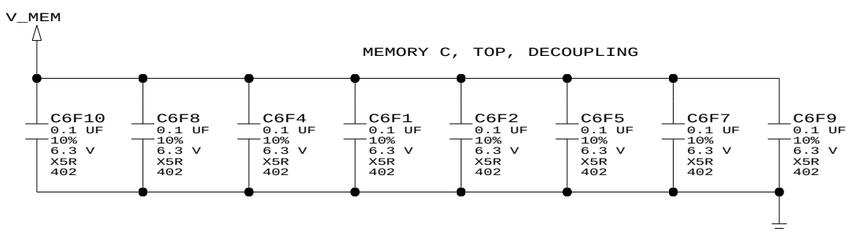

| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| R5U4 | X801176-001 | CH | RES,1.47KOHM | GDDR_TOP_HYNIX PACK_IGNORE=TRUE |

MC_CLK0
STITCHING CAP




PARTITION C DECOUPLING




[PAGE_TITLE=MEMORY PARTITION C, TOP]

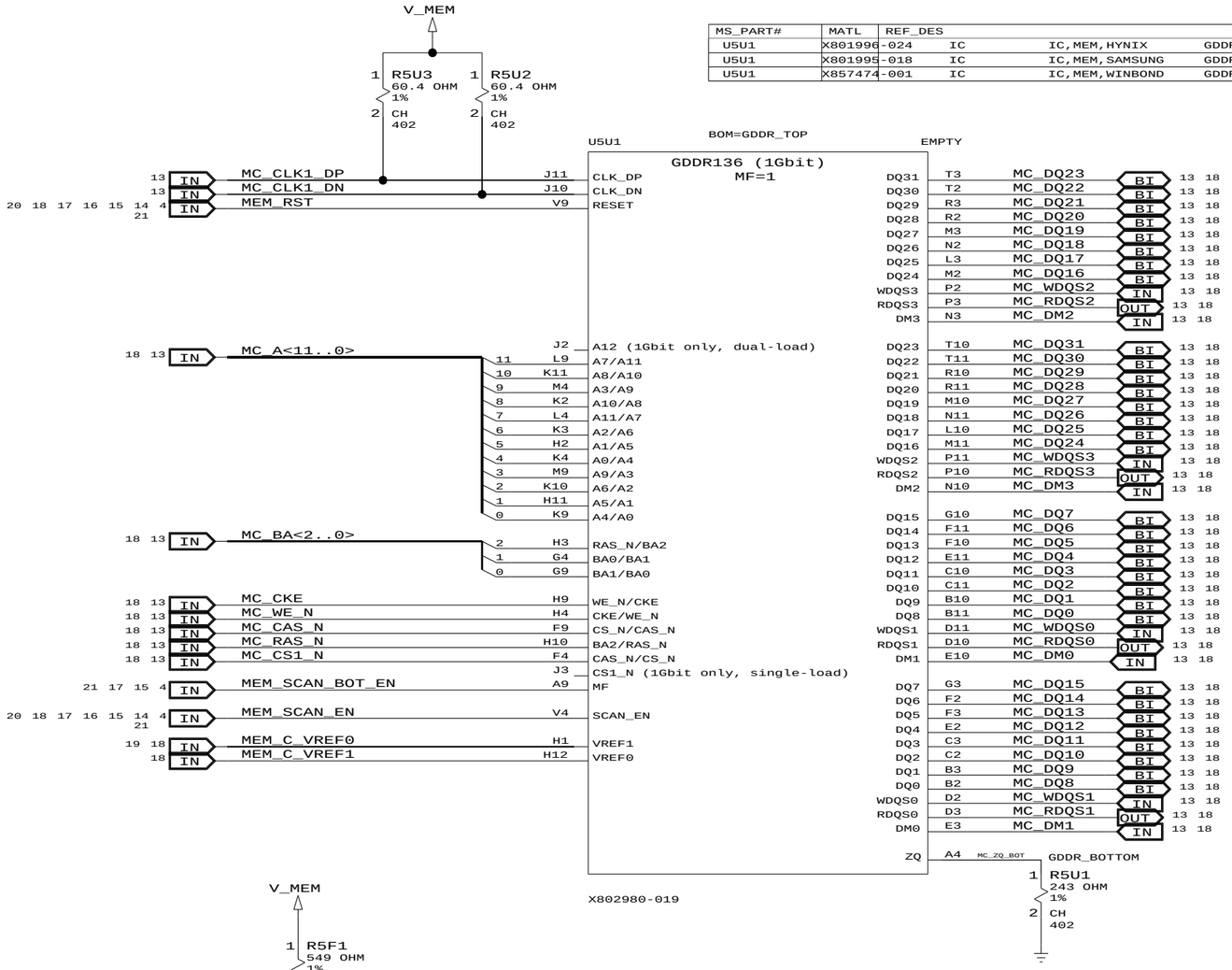
DRAWING
Fri Jan 04 12:07:15 2008

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 18/74 | 18/74 | C | 1.0 |

# MEMORY PARTITION C, BOTTOM

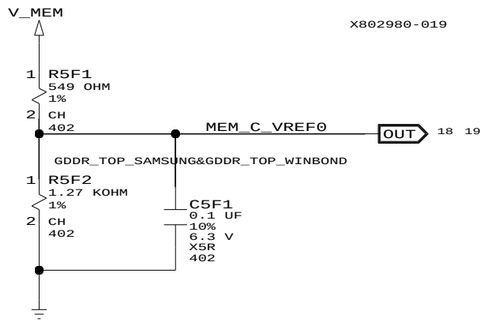
CHIP SELECT = 1, MIRROR FUNCTION = 1

| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| U5U1 | X801996-024 | IC | IC,MEM,HYNIX | GDDR_BOT_HYNIX PACK_IGNORE=TRUE |
| U5U1 | X801995-018 | IC | IC,MEM,SAMSUNG | GDDR_BOT_SAMSUNG PACK_IGNORE=TRUE |
| U5U1 | X857474-001 | IC | IC,MEM,WINBOND | GDDR_BOT_WINBOND PACK_IGNORE=TRUE |

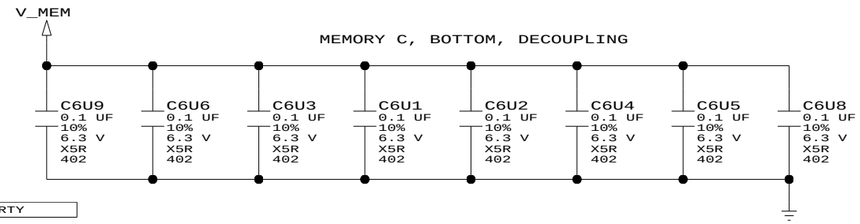









| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| R5F2 | X801176-001 | CH | RES,1.47KOHM | GDDR_TOP_HYNIX PACK_IGNORE=TRUE |



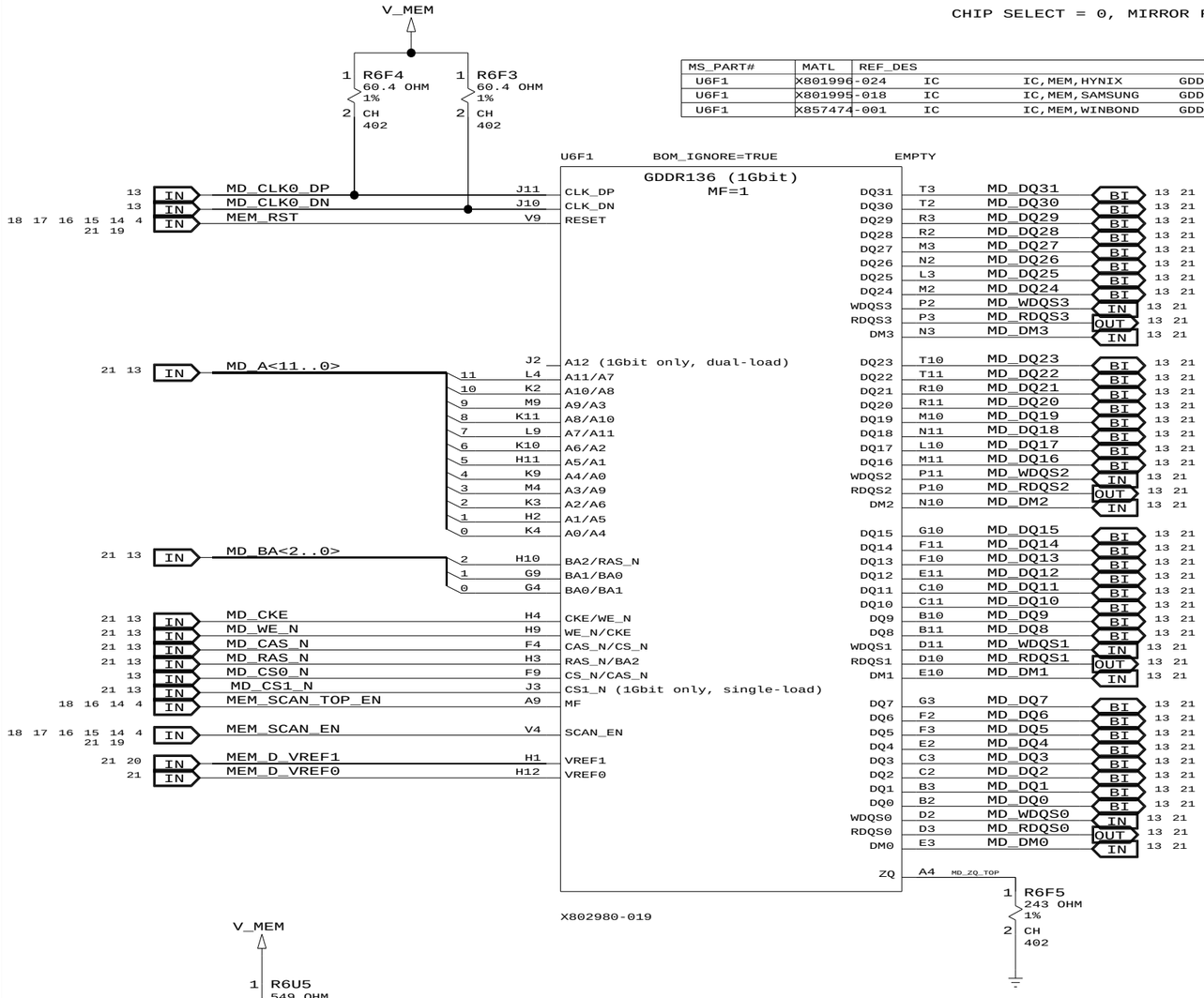

[PAGE_TITLE=MEMORY PARITION C, BOTTOM]

DRAWING
Fri Jan 04 12:07:15 2008

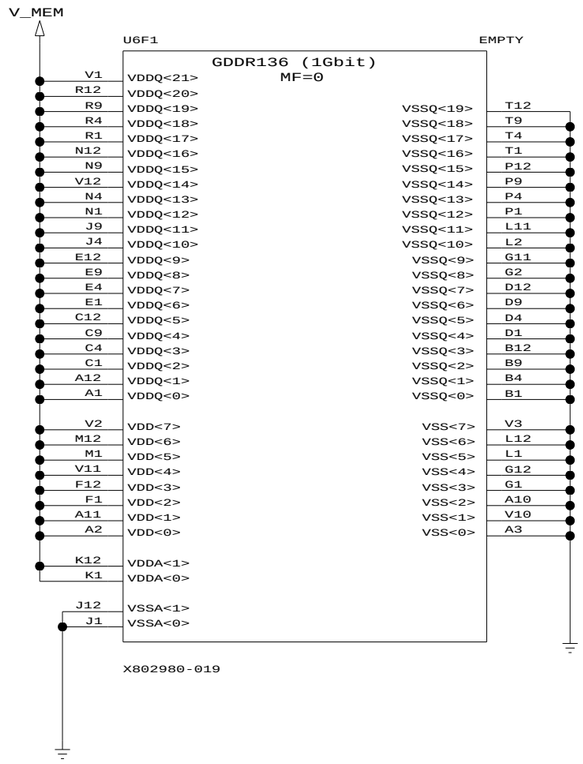
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 19/74 | 19/74 | C | 1.0 |

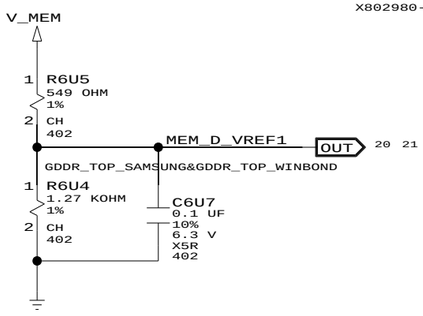
# MEMORY PARTITION D, TOP

CHIP SELECT = 0, MIRROR FUNCTION = 0



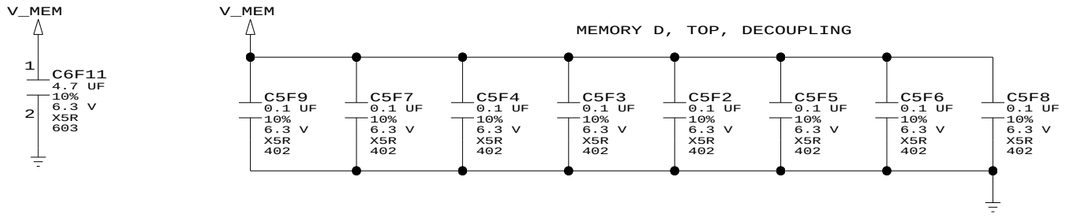

| MS_PART# | MATL | REF_DES | | DESCR. | BOM PROPERTY |
|---|---|---|---|---|---|
| U6F1 | X801996-024 | IC | IC,MEM,HYNIX | GDDR_TOP_HYNIX | PACK_IGNORE=TRUE |
| U6F1 | X801995-018 | IC | IC,MEM,SAMSUNG | GDDR_TOP_SAMSUNG | PACK_IGNORE=TRUE |
| U6F1 | X857474-001 | IC | IC,MEM,WINBOND | GDDR_TOP_WINBOND | PACK_IGNORE=TRUE |




| MS_PART# | MATL | REF_DES | | DESCR. | BOM PROPERTY |
|---|---|---|---|---|---|
| R6U4 | X801176-001 | CH | RES,1.47KOHM | GDDR_TOP_HYNIX | PACK_IGNORE=TRUE |

PARTITION D DECOUPLING




[PAGE_TITLE=MEMORY PARTITION D, TOP]

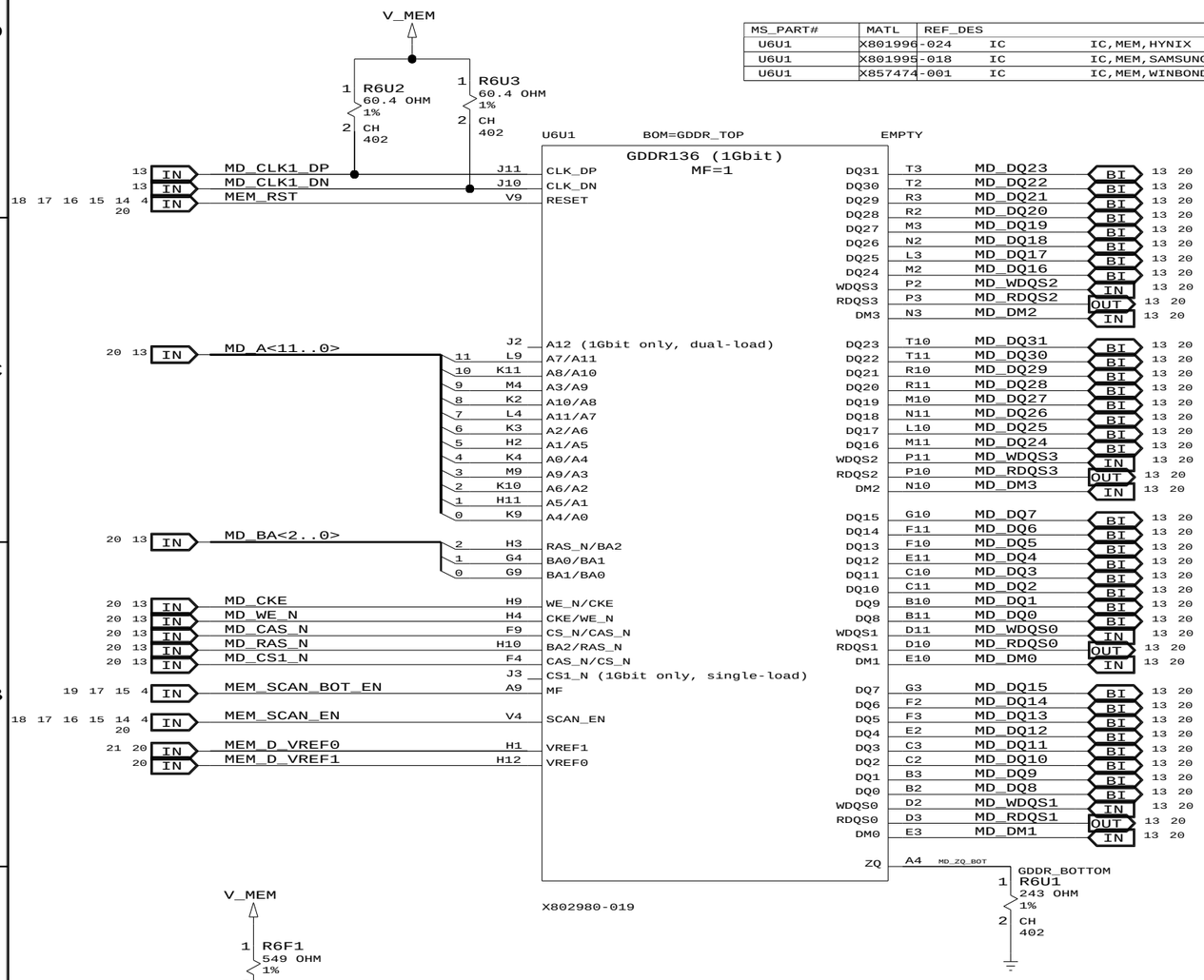
DRAWING
Fri Jan 04 12:07:16 2008

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 20/74 | 20/74 | C | 1.0 |

# MEMORY PARTITION D, BOTTOM

CHIP SELECT = 1, MIRROR FUNCTION = 1

| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| U6U1 | X801996-024 | IC | IC,MEM,HYNIX | GDDR_BOT_HYNIX PACK_IGNORE=TRUE |
| U6U1 | X801995-018 | IC | IC,MEM,SAMSUNG | GDDR_BOT_SAMSUNG PACK_IGNORE=TRUE |
| U6U1 | X857474-001 | IC | IC,MEM,WINBOND | GDDR_BOT_WINBOND PACK_IGNORE=TRUE |










| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| R6F2 | X801176-001 | CH | RES,1.47KOHM | GDDR_TOP_HYNIX PACK_IGNORE=TRUE |







[PAGE_TITLE=MEMORY PARTITION D, BOTTOM]

DRAWING

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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 21/74 | 21/74 | C | 1.0 |

# KSB, CLOCKS + STRAPPING + POR

XTAL BYPASS: STUFF R3, R4, R5, AND C153 AND
APPLY SQUARE WAVE TO XTAL_IN (VL=0V VH=1V)
KEEP R5 CLOSE TO KSB CHIP

RESISTOR VALUES CHOSEN TO GIVE V_12P0_DET 1P8V NOMINAL

V_12P0
R3R5 1% 84.5 KOHM CH 402
R3R17 1% 15 KOHM CH 402
C3R55 470 PF 5% 50 V EMPTY 402
DB3D2
Y3D1 25 MHZ SM XTAL
C3D7 27 PF 5% 50 V COG 402
C3D8 22 PF 5% 50 V NPO 402
R3D26 49.9 OHM 1% EMPTY 402
V_3P3STBY
R3D17 8.25 KOHM 1% EMPTY 402
R3D19 1.5 KOHM 1% EMPTY 402
C3D1 0.01 UF 10% 16 V EMPTY 402

U3D1 KSB IC
1 OF 9
X850744-001 PBGA404 I155

| | | |
|---|---|---|
| V_12P0_DET | W20 | V_12P0_DET |
| POR_BYPASS | B4 | POR_BYPASS<DN> |
| | | V_12P0_OK |
| | | V_RST_OK<DN> |
| | | MAIN_PWR_OK |
| | | SB_RST_N* |
| | | SMC_RST_N_OUT |
| XTAL_IN | AB21 | XTAL_IN |
| | | CPU_CLK_DP |
| | | CPU_CLK_DN |
| XTAL_OUT | AA21 | XTAL_OUT |
| | | NB_CLK_DP |
| | | NB_CLK_DN |
| | AA22 | XTAL_VSS |
| | | PCIEX_CLK_IN_DP |
| | | PCIEX_CLK_IN_DN |
| | | PIX_CLK_OUT_DP |
| | | PIX_CLK_OUT_DN |
| EXT_CLK48_IN | Y20 | CLK48<DN> |
| EXT_CLK48_SEL | W19 | CLK48_BYPASS<DN> |
| SMB_DATA | A4 | SMB_DATA |
| SMB_CLK | A5 | SMB_CLK |

DB2C2 POR_BYPASS
C5 V19 — SB_KSB_BASE
C4 — MAIN_PWR_OK
C9 SB_RST_N_R — D7 SMC_RST_N_R
U21 CPU_CLK_DP_R — V21 CPU_CLK_DN_R
V22 GPU_CLK_DP_R — W22 GPU_CLK_DN_R
Y15 EXT_PCIEX_CLK_DP — AA15 EXT_PCIEX_CLK_DN
U22 PIX_CLK_2X_DP_R — T22 PIX_CLK_2X_DN_R
DB3D3 EXT_CLK48_IN — DB3D1 EXT_CLK48_SEL
62 61 55 25 BI SMB_DATA — 62 61 55 25 BI SMB_CLK

V12P0_PWRGD — OUT 25 42 — FTP FT2P6
V_RST_OK — OUT — FTP FT3R3
SB_MAIN_PWRGD — IN 25 — FTP FT3P5
R3P2 1 KOHM 5% CH 402
SB_RST_N — IN 25
R2P3 10 KOHM 5% CH 402
R3P15 10 KOHM 5% CH 402
R3P16 1 KOHM 5% CH 402
SMC_RST_N — OUT 25 42 62
R3D6 33 OHM 5% 402 CH
R3D7 49.9 OHM 1% 402 CH
CPU_CLK_DP — OUT 2
CPU_CLK_DN — OUT 2
R3D8 33 OHM 5% 402 CH
R3D9 49.9 OHM 1% 402 CH
R3D10 33 OHM 5% 402 CH
R3D11 49.9 OHM 1% 402 CH
GPU_CLK_DP — OUT 4
GPU_CLK_DN — OUT 4
R3D12 33 OHM 5% 402 CH
R3D13 49.9 OHM 1% 402 CH
V_3P3
R3D28 10 KOHM 1% CH 402
R3D29 10 KOHM 1% CH 402
DB3R6
DB3R7
R3D5 33 OHM 5% 402 CH
R3D14 60.4 OHM 1% 402 CH
PIX_CLK_2X_DP — OUT 4
PIX_CLK_2X_DN — OUT 4
R3D3 33 OHM 5% 402 CH
R3D4 60.4 OHM 1% 402 CH
FTP FT3R7
FTP FT3R2

I2C ADDRESS

| | 0011 100 R/W HEX | 0111 000 R/W HEX |
|---|---|---|
| WRITE | 0011 100 0 0X38 | 0111 000 0 0X70 |
| READ | 0011 100 1 0X39 | 0111 000 1 0X71 |

| | 0011 110 R/W HEX |
|---|---|
| WRITE | 0011 110 0 0X3C |
| READ | 0011 110 1 0X3D |

[PAGE_TITLE=KSB, CLOCKS + STRAPING]

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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 22/74 | 22/74 | C | 1.0 |

# KSB, VIDEO + FAN

U3D1 KSB IC

2 OF 9

4 IN GPU_PIX_CLK_1X — J20 PIX_CLK_IN

4 IN PIX_DATA<14..0>

| | | |
|---|---|---|
| 14 | G20 | PIX_DATA14 |
| 13 | G21 | PIX_DATA13 |
| 12 | G22 | PIX_DATA12 |
| 11 | H20 | PIX_DATA11 |
| 10 | H21 | PIX_DATA10 |
| 9 | H22 | PIX_DATA9 |
| 8 | J21 | PIX_DATA8 |
| 7 | J22 | PIX_DATA7 |
| 6 | K21 | PIX_DATA6 |
| 5 | K22 | PIX_DATA5 |
| 4 | K20 | PIX_DATA4 |
| 3 | L21 | PIX_DATA3 |
| 2 | L22 | PIX_DATA2 |
| 1 | L20 | PIX_DATA1 |
| 0 | M20 | PIX_DATA0 |

4 IN GPU_HSYNC_OUT — M22 HSYNC_IN

4 IN GPU_VSYNC_OUT — M21 VSYNC_IN

DAC_RSET — D20 DAC_RSET

R3P11 5.11 KOHM 0.1% CH 402

DAC_RSET ROUTING SENSITIVE TO NOISE ROUTE MUST REMAIN LOCKED

61 38 IN BRD_TEMP_P — R4G2 0 OHM 5% 402 EMPTY — BRD_TEMP_P_DIODE — Q4G1 EMPTY

61 38 OUT BRD_TEMP_N — R4G1 0 OHM 5% 402 EMPTY — BRD_TEMP_N_DIODE

40 IN HDMI_HPD — D18 HDMI_HPD

KSB_RSET — H4 RSET

R2P4 2 KOHM 1% CH 402

FT3P1 FTP 1

25 IN SMC_PWM0 — R3P3 205 KOHM 1% 402 EMPTY — FAN_OP1_DP — F22 FAN_OP1_DP

C3P2 0.22 UF 10% 6.3 V EMPTY 402

36 IN FAN1_FDBK — F21 FAN_OP1_DN

P22 TEMP_N

66 61 IN CPU_TEMP_N_KSB — ST3C5 SHORT

66 61 IN GPU_TEMP_N_KSB — ST3C2 SHORT

66 61 IN EDRAM_TEMP_N_KSB — ST3C4 SHORT

61 IN BRD_TEMP_N_KSB — ST3C3 SHORT

23 IN CAL_TEMP_N — ST3C1 SHORT

| | | | |
|---|---|---|---|
| DAC_D_OUT_DP | A20 | VID_DACD_DP | OUT 37 |
| DAC_D_OUT_DN | F18 | | |
| DAC_B_OUT_DP | A21 | VID_DACB_DP | OUT 37 |
| DAC_B_OUT_DN | G18 | | |
| DAC_A_OUT_DP | C22 | VID_DACA_DP | OUT 37 |
| DAC_A_OUT_DN | G19 | | |
| DAC_C_OUT_DP | D22 | VID_DACC_DP | OUT 37 |
| DAC_C_OUT_DN | H18 | | |
| HSYNC_OUT | B19 | VID_HSYNC_OUT_R | OUT |
| VSYNC_OUT | C19 | VID_VSYNC_OUT_R | OUT |
| TMDS_TXC_DP | A16 | HDMI_TXC_DP | C3C4 0.1 UF 10% 6.3 V EMPTY 402 — HDMI_TXC_DP_R — R3C23 301 OHM 1% 603 EMPTY — OUT 40 |
| TMDS_TXC_DN | B16 | HDMI_TXC_DN | OUT 40 |
| TMDS_TX2_DP | B13 | HDMI_TX2_DP | C3C1 0.1 UF 10% 6.3 V EMPTY 402 — HDMI_TX2_DP_R — R3C20 301 OHM 1% 603 EMPTY — OUT 40 |
| TMDS_TX2_DN | C13 | HDMI_TX2_DN | OUT 40 |
| TMDS_TX1_DP | A14 | HDMI_TX1_DP | C3C2 0.1 UF 10% 6.3 V EMPTY 402 — HDMI_TX1_DP_R — R3C21 301 OHM 1% 603 EMPTY — OUT 40 |
| TMDS_TX1_DN | B14 | HDMI_TX1_DN | OUT |
| TMDS_TX0_DP | B15 | HDMI_TX0_DP | C3C3 0.1 UF 10% 6.3 V EMPTY 402 — HDMI_TX0_DP_R — R3C22 301 OHM 1% 603 EMPTY — OUT 40 |
| TMDS_TX0_DN | C15 | HDMI_TX0_DN | OUT 40 |
| DDC_SCK | A12 | HDMI_DDC_CLK | BI 40 55 |
| DDC_SDA | B12 | HDMI_DDC_DATA | BI 40 55 |
| FAN_OUT1 | E21 | FAN1_OUT | OUT 36 |
| TEMP3_P | N22 | CAL_TEMP_P | DB3P1 DB3R1 DB3P2 DB3R2 — C3R1 100 PF 5% 50 V EMPTY 402 |
| TEMP2_P | P20 | | GPU_TEMP_P_KSB — OUT 23 / 61 66 |
| TEMP1_P | P21 | | CPU_TEMP_P_KSB — OUT 61 66 |
| TEMP0_P | R21 | | EDRAM_TEMP_P_KSB — OUT 61 66 |
| TEMP4_P | N21 | | BRD_TEMP_P_KSB — OUT 61 |

CUSTOM THERMAL CALIBRATION PADS LOCATION MUST REMAIN LOCKED

X850744-001 PBGA404

[PAGE_TITLE=KSB, VIDEO + FAN]

DRAWING

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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 23/74 | 23/74 | C | 1.0 |

# KSB, PCIEX + SMM GPIO + JTAG

V_3P3

RETAIL

R2C14 10 KOHM 5% EMPTY 402
R2C12 10 KOHM 5% CH 402
R2C18 10 KOHM 5% EMPTY 402
R2C16 10 KOHM 5% EMPTY 402
R2C20 10 KOHM 5% EMPTY 402

DEVKIT

R2C13 1 KOHM 5% CH 402
R2C11 1 KOHM 5% CH 402
R2C17 1 KOHM 5% CH 402
R2C15 1 KOHM 5% CH 402
R2C19 1 KOHM 5% CH 402

DEBUG OUTPUT DISABLED = SB_GPIO<1> HIGH

SB_GPIO<0..15> BI 24 62

V_3P3STBY

R3G19 10 KOHM 5% CH 402

62 24 IN AUD_SPI_MOSI_R — R2C21 0 OHM 5% CH 402 — AUD_SPI_MOSI OUT 35

R2C1 10 KOHM 5% EMPTY 402

V_3P3STBY

R3G18 10 KOHM 5% CH 402

62 24 OUT AUD_SPI_MISO_R — R2C25 0 OHM 5% CH 402 — AUD_SPI_MISO IN 35

R2C24 10 KOHM 5% EMPTY 402

V_3P3STBY

R3G3 10 KOHM 5% CH 402

62 24 IN AUD_SPI_CLK_R — R2C23 0 OHM 5% CH 402 — AUD_SPI_CLK OUT 35

R2C22 10 KOHM 5% EMPTY 402

V_3P3STBY

R2G6 10 KOHM 5% CH 402

62 24 IN AUD_RDY_BSBY_R — R2G5 0 OHM 5% CH 402 — AUD_RDY_BSBY OUT 35

R2G7 10 KOHM 5% EMPTY 402

4 IN PEX_GPU_SB_L1_DP — Y18 PEX_RX1_DP
4 IN PEX_GPU_SB_L1_DN — AA18 PEX_RX1_DN
4 IN PEX_GPU_SB_L0_DP — Y16 PEX_RX0_DP
4 IN PEX_GPU_SB_L0_DN — AA16 PEX_RX0_DN

62 IN KER_DBG_RXD — A3 UART0_RXD<UP>

GPIO<1> = 0 ENABLE DEBUG OUTPUT
1 DISABLE DEBUG OUTPUT

GPIO<0,2,3> = 111 XENON
110 ZEPHYR A
101 ZEPHYR B
100 ZEPHYR C
011 FALCON
010 JASPER
001 TRINITY
000 CORONA

65 IN SB_TCLK — T19 TCK<DN>
65 OUT SB_TDO — R19 TDO
65 IN SB_TDI — R20 TDI<UP>
65 IN SB_TMS — U19 TMS<UP>
65 IN SB_TRST — N19 TRST<DN>

FT3T8 FTP
FT3T7 FTP
FT3T6 FTP
FT3T5 FTP
FT3T4 FTP

U3D1 KSB IC

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X850744-001 PBGA404

PEX_TX1_DP AA19 PEX_SB_GPU_L1_DP_C — C3D5 0.1 UF 10% 6.3 V X5R 402 — PEX_SB_GPU_L1_DP OUT 4
PEX_TX1_DN AB19 PEX_SB_GPU_L1_DN_C — C3D6 0.1 UF 10% 6.3 V X5R 402 — PEX_SB_GPU_L1_DN OUT 4
PEX_TX0_DP AA17 PEX_SB_GPU_L0_DP_C — C3D3 0.1 UF 10% 6.3 V X5R 402 — PEX_SB_GPU_L0_DP OUT 4
PEX_TX0_DN AB17 PEX_SB_GPU_L0_DN_C — C3D4 0.1 UF 10% 6.3 V X5R 402 — PEX_SB_GPU_L0_DN OUT 4

UART0_TXD A2 KER_DBG_TXD_R — R2C9 47 OHM 5% 402 — KER_DBG_TXD OUT 62

TST_PT E18 KSB_DEBUG — 1 DB3P3

GPIO15 F2 AUD_SPI_MISO_R IN 24 62
GPIO14 F3 AUD_SSB_R OUT 24 62
GPIO13 E1 SCART_RGB OUT 37 62
GPIO12 E4 AUD_RST_N OUT 33
GPIO11 E3 AUD_RDY_BSBY_R OUT 24 62
GPIO10 F4 62 FAN_TACH_SMM_R — R2P9 0 OHM 5% CH 402 — FAN_TACH IN 25 36
GPIO9 G4 WSS_CNTL0 OUT 37 62
GPIO8 G3 WSS_CNTL1 OUT 37 62
GPIO7 D3 AUD_SPI_MOSI_R OUT 24 62
GPIO6 D4 AUD_SPI_CLK_R OUT 24 62
GPIO5 E2 SB_GPIO<5> BI 24 62
GPIO4 C1 SB_GPIO<4> OUT 24 62
GPIO3 C2 SB_GPIO<3> OUT 24 62
GPIO2 D1 SB_GPIO<2> OUT 24 62
GPIO1 B1 SB_GPIO<1> BI 24 62
GPIO0 B2 SB_GPIO<0> OUT 24 62

V_3P3STBY

R3G20 10 KOHM 5% CH 402

62 24 IN AUD_SSB_R — R2P5 0 OHM 5% CH 402 — AUD_SSB OUT 35

R2C26 10 KOHM 5% EMPTY 402

## DEBUG BUS

| NET NAME | DEBUG BUS BIT |
|---|---|
| SB_GPIO<0> | 0 |
| SB_GPIO<2> | 1 |
| SB_GPIO<3> | 2 |
| SB_GPIO<4> | 3 |
| AUD_SPI_CLK_R | 4 |
| SB_GPIO<5> | 5 |
| WSS_CNTL1 | 6 |
| WSS_CNTL0 | 7 |
| FAN_TACH | 8 |
| AUD_RDY_BSBY_R | 9 |
| SCART_RGB | 10 |
| AUD_SSB_R | 11 |
| AUD_SPI_MISO_R | 12 |

[PAGE_TITLE=KSB, PCIEX + SMM GPIO + JTAG]

DRAWING
Fri Jan 04 12:07:17 2008

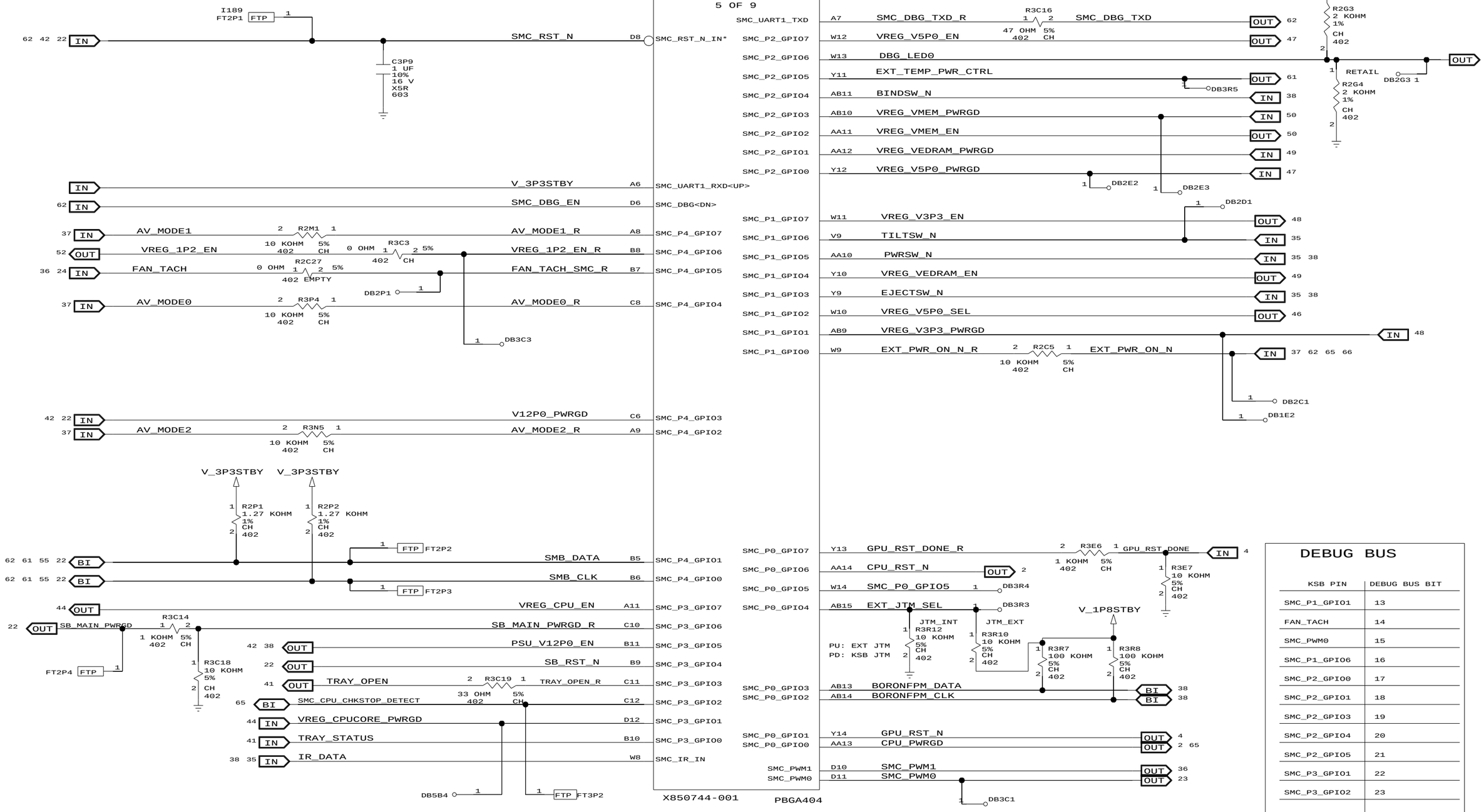
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 24/74 | 24/74 | C | 1.0 |

# KSB, SMC

N: DBG_LED0 PULLDOWN = SMC PRODUCTION MODE
DBG_LED0 PULLUP = SMC DEVELOPMENT MODE

U3D1 KSB IC

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X850744-001 PBGA404

SMC_P1,2,3,4 GPIOS ARE ON V3P3STBY. SMC_P0 GPIOS ARE ON 1P8VSTBY

## DEBUG BUS

| KSB PIN | DEBUG BUS BIT |
|---|---|
| SMC_P1_GPIO1 | 13 |
| FAN_TACH | 14 |
| SMC_PWM0 | 15 |
| SMC_P1_GPIO6 | 16 |
| SMC_P2_GPIO0 | 17 |
| SMC_P2_GPIO1 | 18 |
| SMC_P2_GPIO3 | 19 |
| SMC_P2_GPIO4 | 20 |
| SMC_P2_GPIO5 | 21 |
| SMC_P3_GPIO1 | 22 |
| SMC_P3_GPIO2 | 23 |

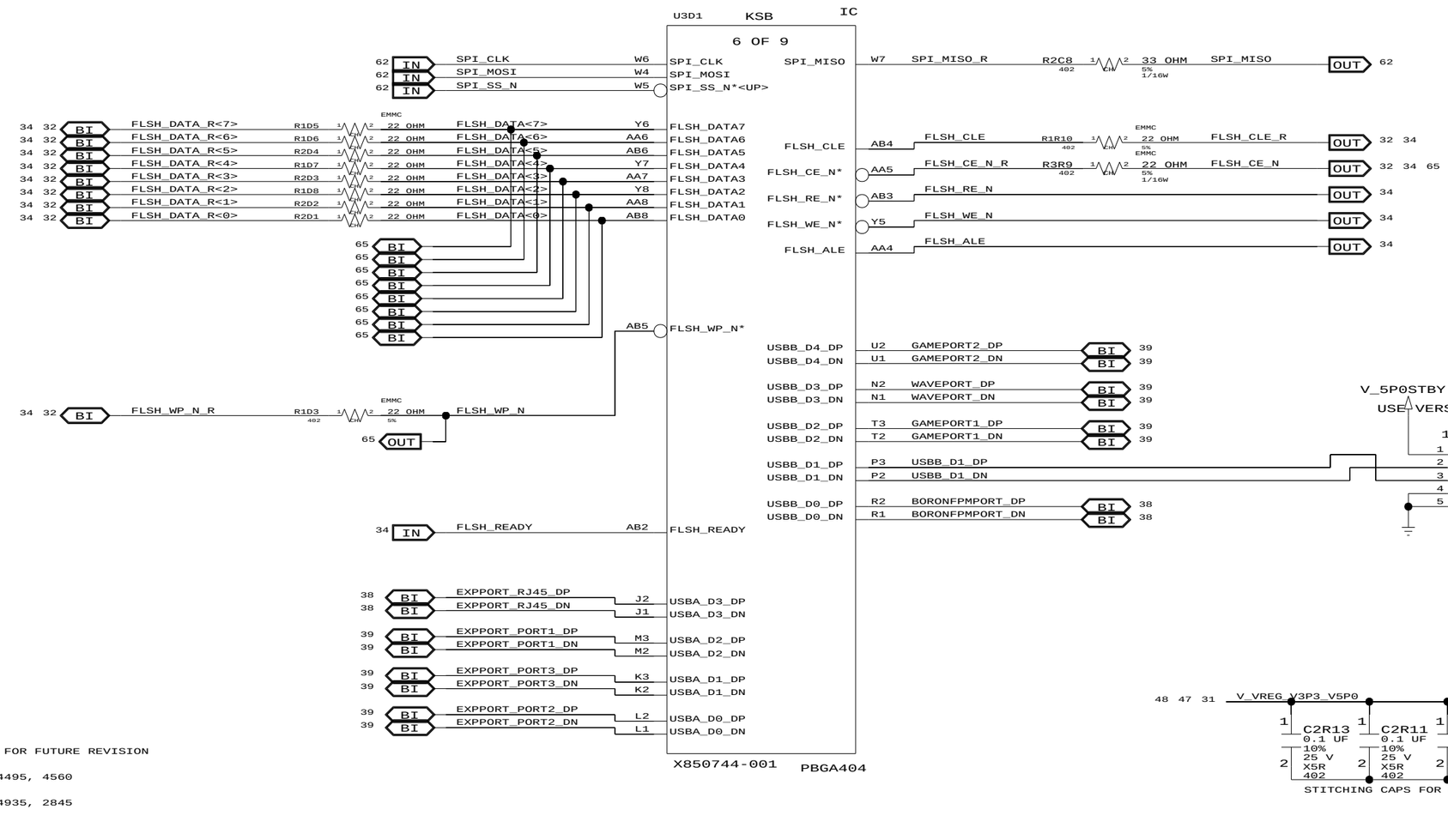
[PAGE_TITLE=KSB, SMC]

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Fri Jan 04 12:07:17 2008

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 25/74 | 25/74 | C | 1.0 |

# KSB, FLASH + USB + SPI

| MS_PART# | MATL | REF_DES | | DESCR. | BOM PROPERTY |
|---|---|---|---|---|---|
| R1D5 | X800967-001 | CH | RES,0OHM,0402 | NAND | PACK_IGNORE=TRUE |
| R1D6 | X800967-001 | CH | RES,0OHM,0402 | NAND | PACK_IGNORE=TRUE |
| R2D4 | X800967-001 | CH | RES,0OHM,0402 | NAND | PACK_IGNORE=TRUE |
| R1D7 | X800967-001 | CH | RES,0OHM,0402 | NAND | PACK_IGNORE=TRUE |
| R2D3 | X800967-001 | CH | RES,0OHM,0402 | NAND | PACK_IGNORE=TRUE |
| R1D8 | X800967-001 | CH | RES,0OHM,0402 | NAND | PACK_IGNORE=TRUE |
| R2D2 | X800967-001 | CH | RES,0OHM,0402 | NAND | PACK_IGNORE=TRUE |
| R2D1 | X800967-001 | CH | RES,0OHM,0402 | NAND | PACK_IGNORE=TRUE |
| R1D3 | X800967-001 | CH | RES,0OHM,0402 | NAND | PACK_IGNORE=TRUE |
| R1R10 | X800967-001 | CH | RES,0OHM,0402 | NAND | PACK_IGNORE=TRUE |
| R3R9 | X800451-001 | CH | RES,10OHM,0402 | NAND | PACK_IGNORE=TRUE |

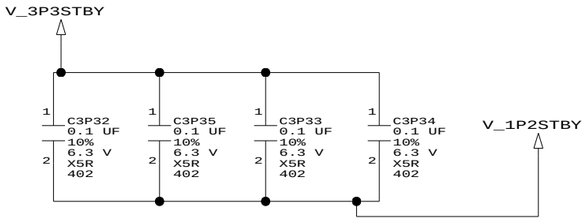



[PAGE_TITLE=KSB, FLASH + USB + SPI]

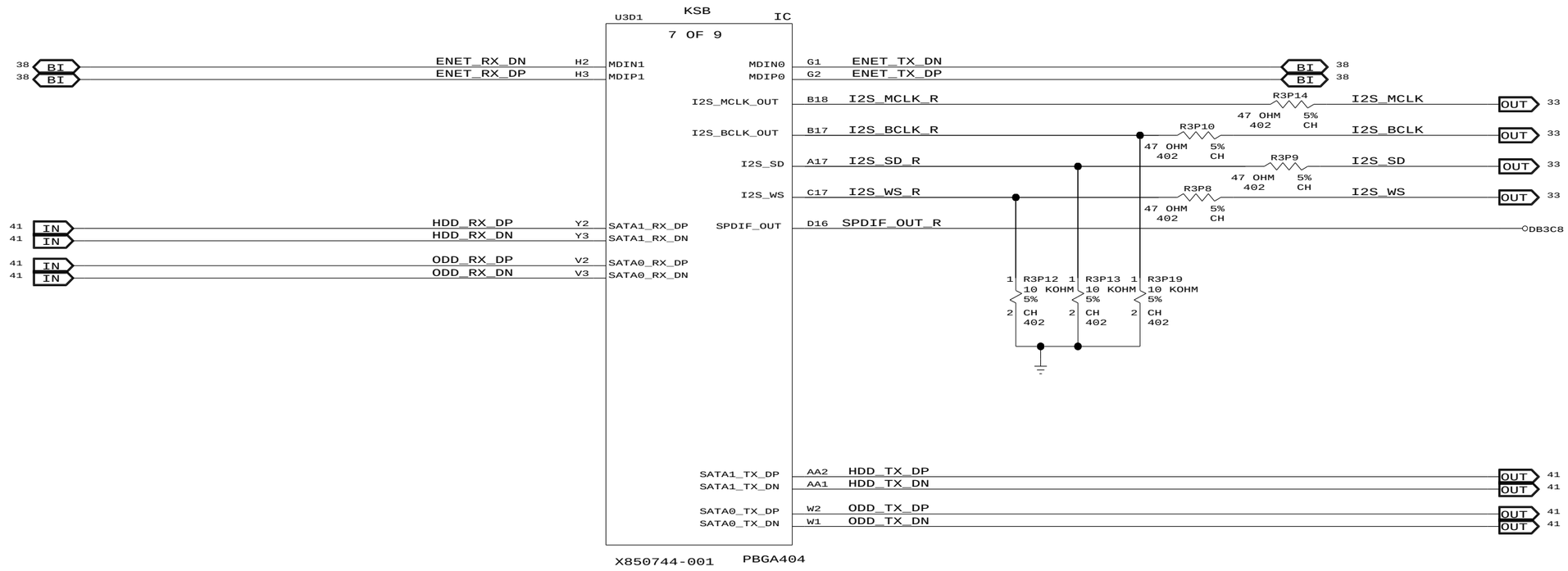
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Fri Jan 04 12:07:17 2008

| MICROSOFT | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| CONFIDENTIAL | STINGRAY | 26/74 | 26/74 | C | 1.0 |

# KSB, ETHERNET + AUDIO + SATA




STITCHING CAPS FOR I2S SIGNALS,PARTICULARLY MCLK.
PLACE AS CLOSE AS POSSIBLE TO I2S_MCLK, I2S_BCLK, I2S_SD, AND I2S_WS.

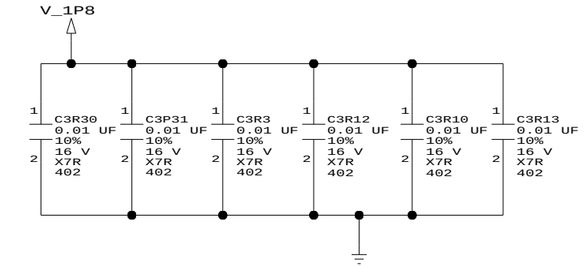


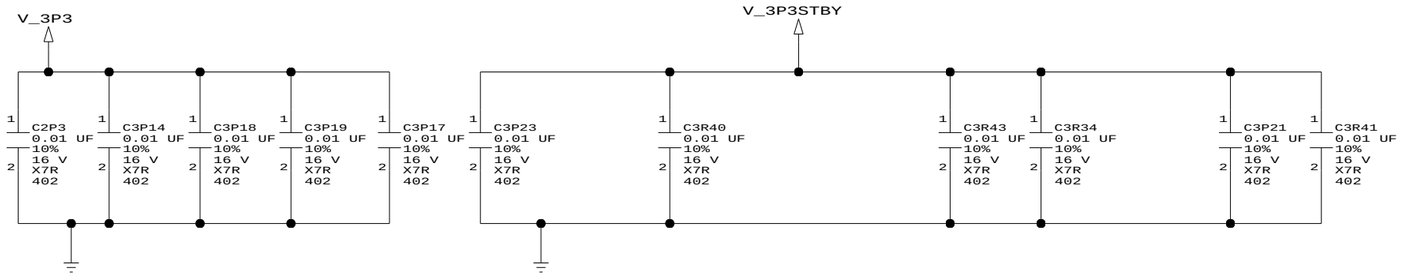

[PAGE_TITLE=KSB, ETHERNET + AUDIO + SATA]

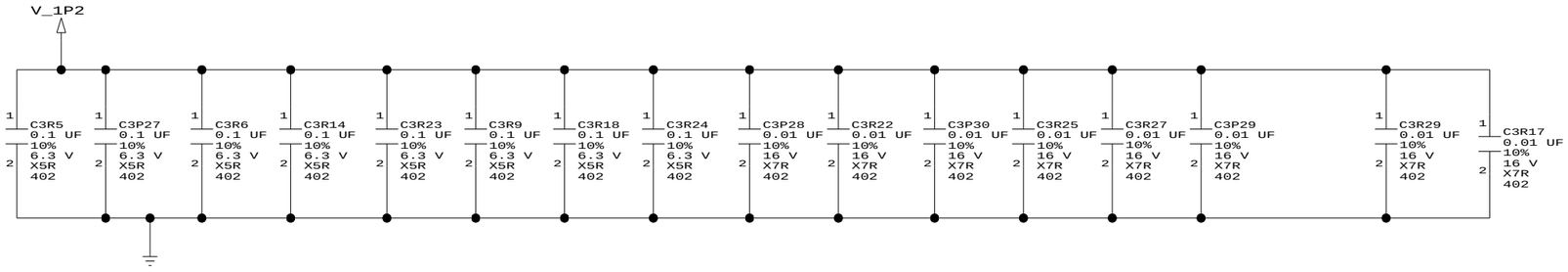
DRAWING
Fri Jan 04 12:07:18 2008

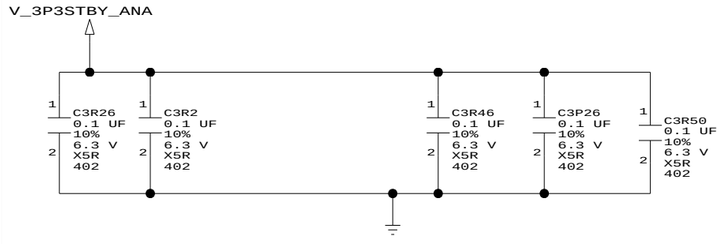
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 27/74 | 27/74 | C | 1.0 |

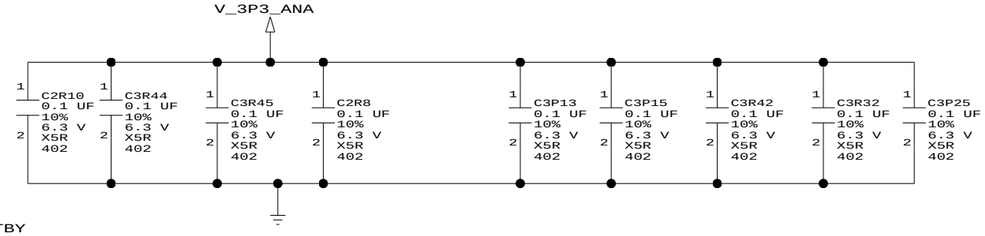
# KSB,DECOUPLING

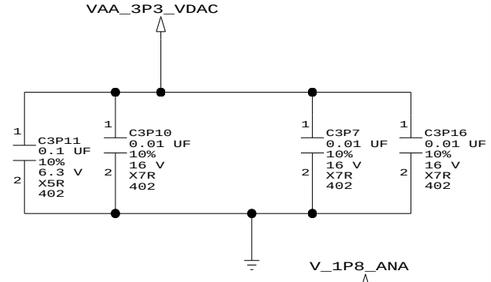


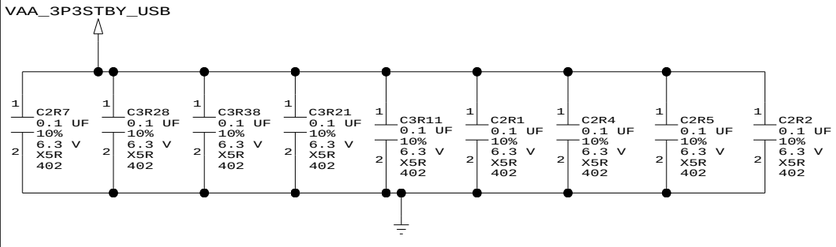



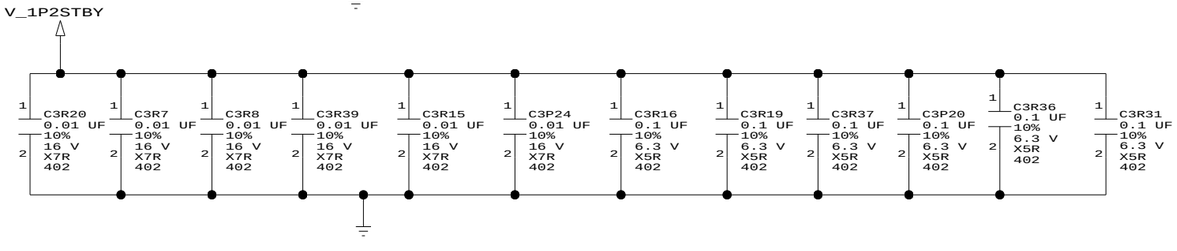



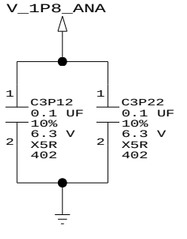



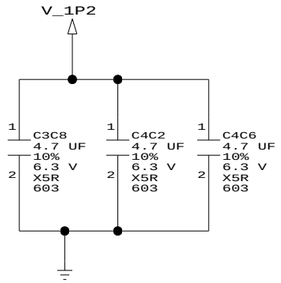



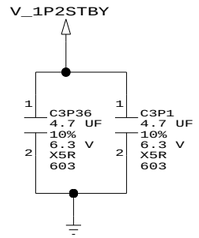

[PAGE_TITLE=KSB, DECOUPLING]

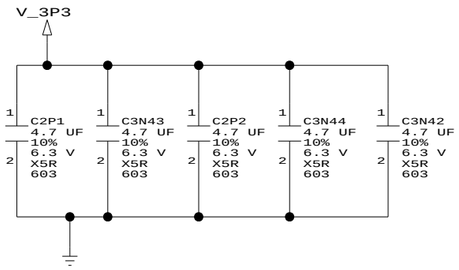
DRAWING
Fri Jan 04 12:07:18 2013

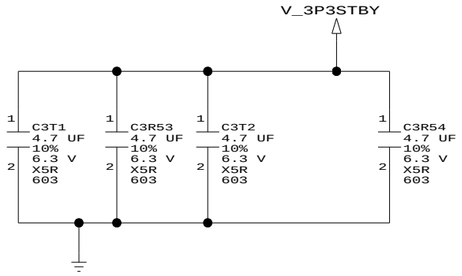
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 28/74 | 28/74 | C | 1.0 |

# KSB, BULK DECOUPLING



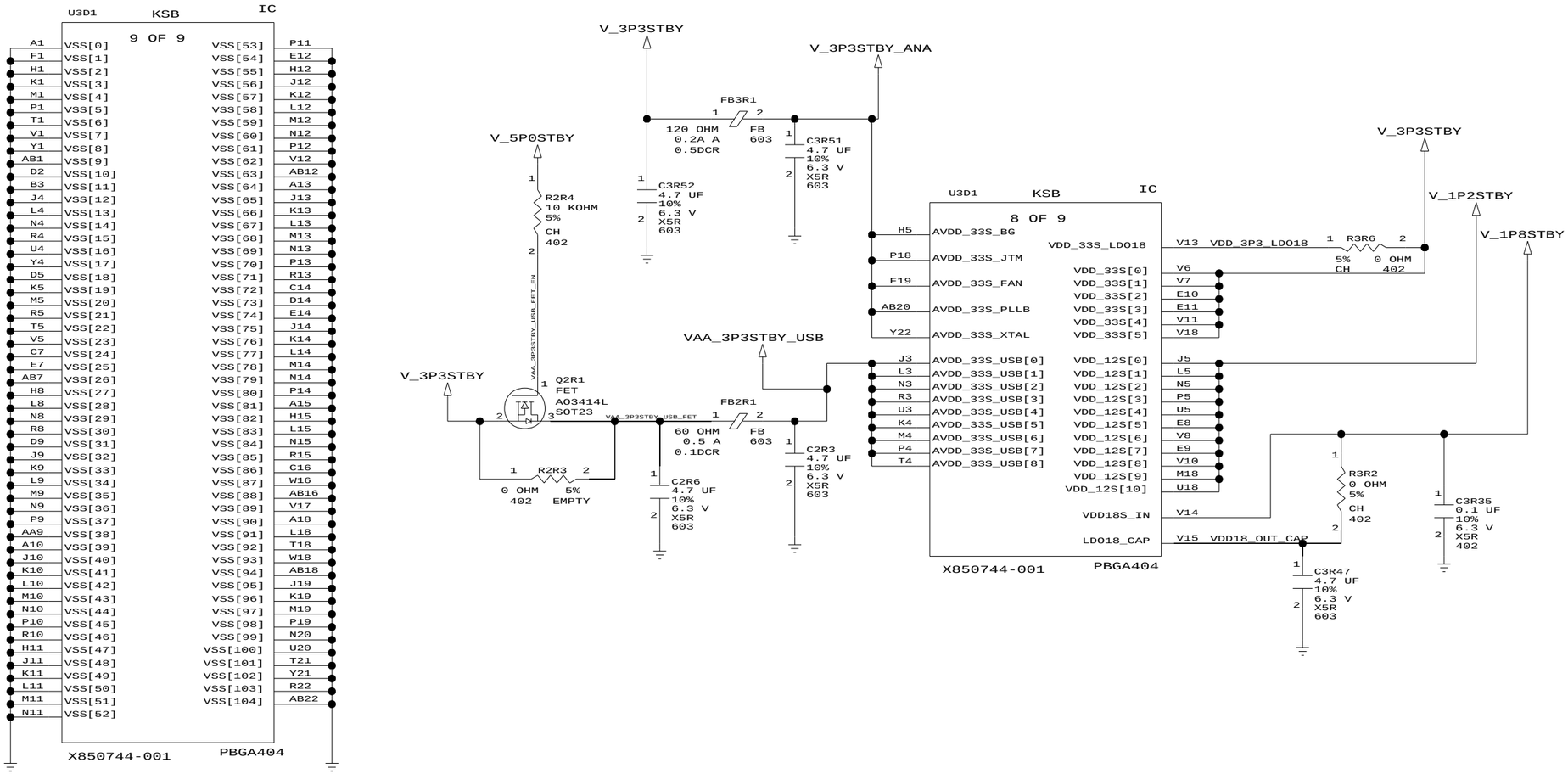

[PAGE_TITLE=KSB, BULK DECOUPLING]

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Fri Jan 04 12:07:18 2008

MICROSOFT CONFIDENTIAL

| PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|
| STINGRAY | 29/74 | 29/74 | C | 1.0 |

# KSB, STANDBY POWER + GROUND

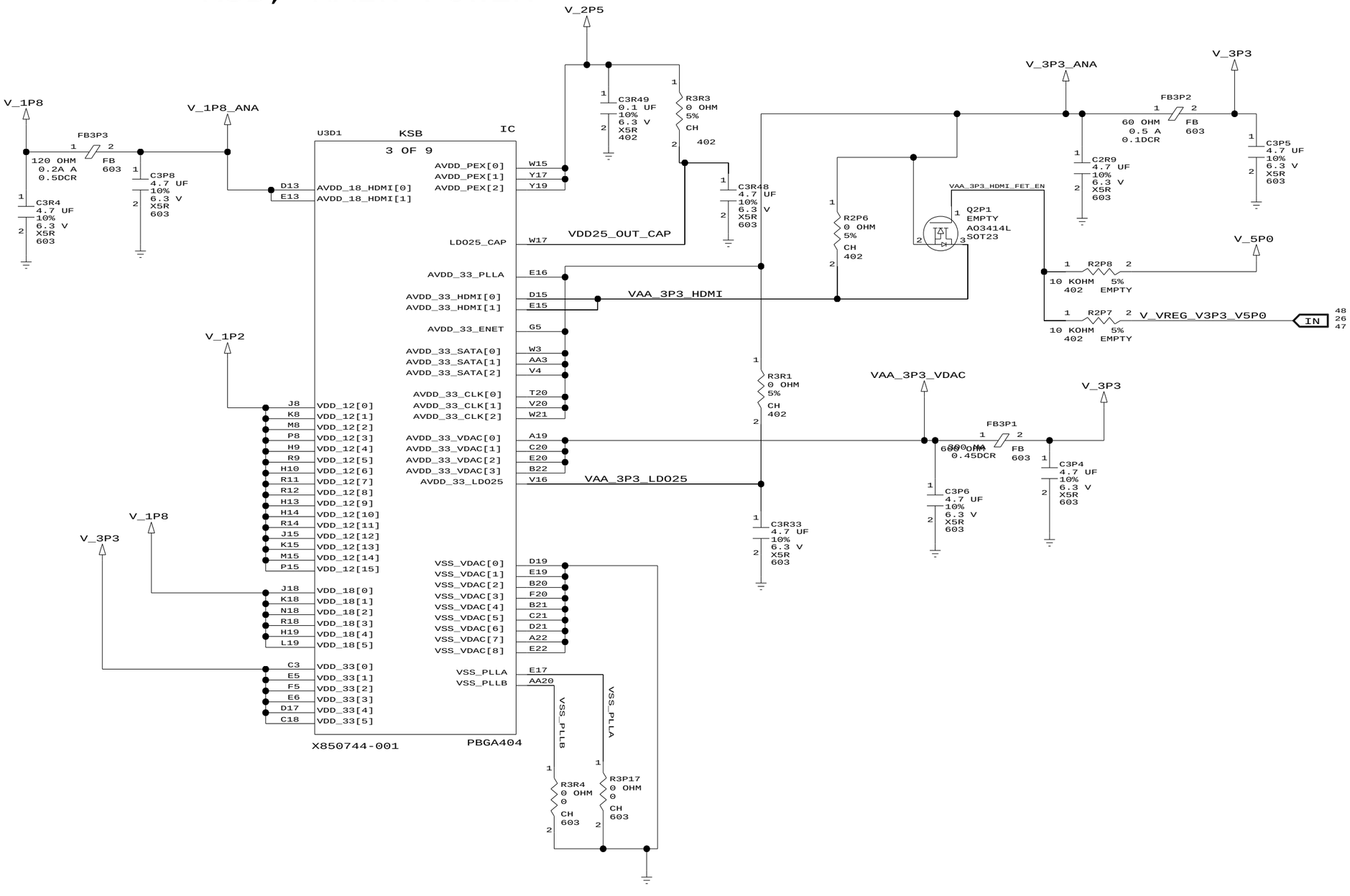



[PAGE_TITLE=KSB, STANDBY POWER]

DRAWING
Fri Jan 04 12:07:18 2008

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 30/74 | 30/74 | C | 1.0 |

# KSB, MAIN POWER




[PAGE_TITLE=KSB, MAIN POWER]

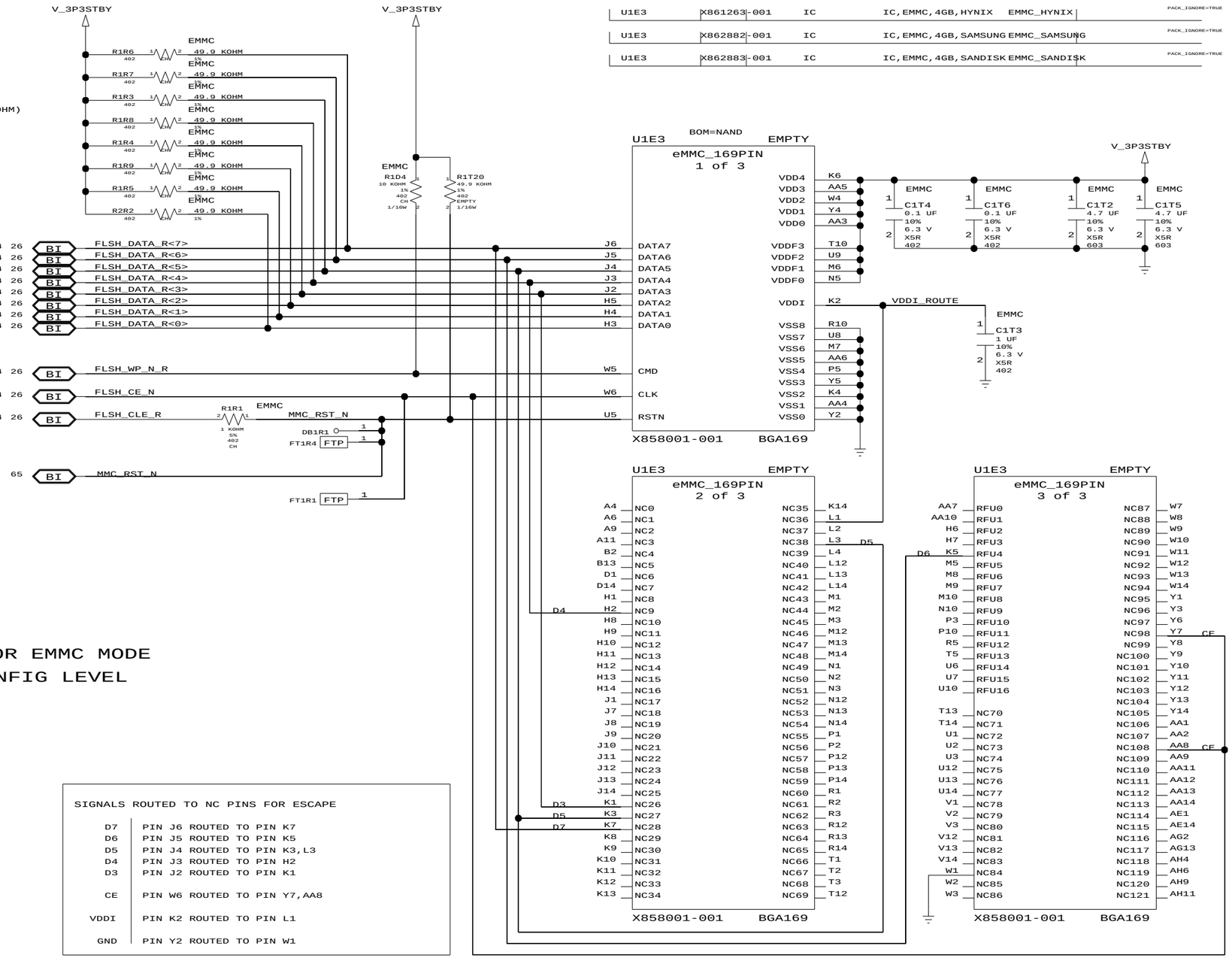
DRAWING
Fri Jan 04 12:07:18 2008

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 31/74 | 31/74 | C | 1.0 |

# EMMC

| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| U1E3 | X861263-001 | IC | IC,EMMC,4GB,HYNIX | EMMC_HYNIX PACK_IGNORE=TRUE |
| U1E3 | X862882-001 | IC | IC,EMMC,4GB,SAMSUNG | EMMC_SAMSUNG PACK_IGNORE=TRUE |
| U1E3 | X862883-001 | IC | IC,EMMC,4GB,SANDISK | EMMC_SANDISK PACK_IGNORE=TRUE |

TOSHIBA 32NM NEEDED STRONGER PULLUPS (20KOHM)




ALL COMPONENTS ON THIS PAGE ARE FOR EMMC MODE
TO BE STUFFED/UNSTUFFED AT CONFIG LEVEL

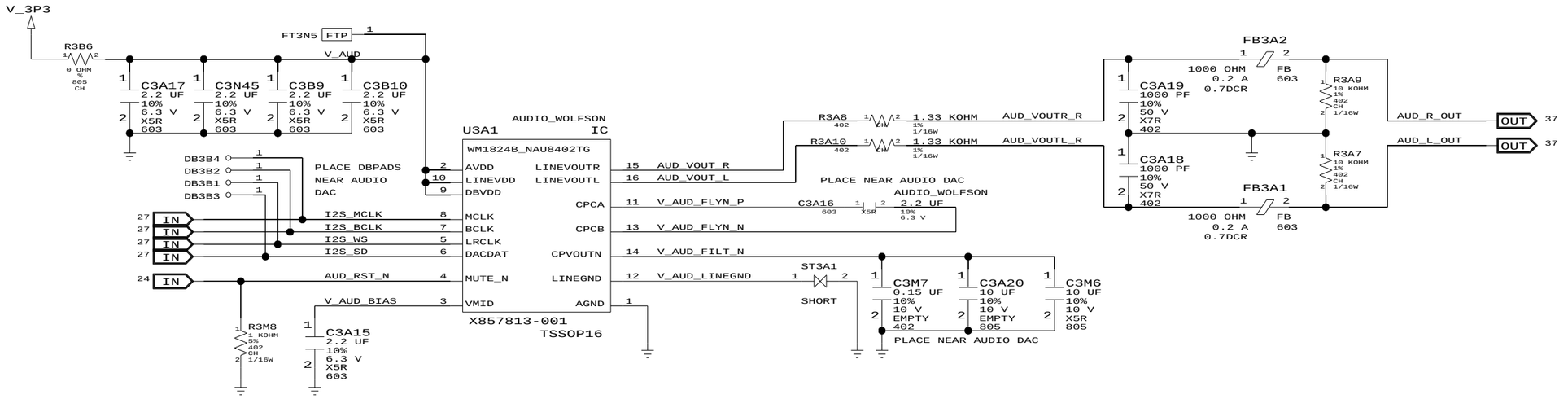
| SIGNALS ROUTED TO NC PINS FOR ESCAPE | |
|---|---|
| D7 | PIN J6 ROUTED TO PIN K7 |
| D6 | PIN J5 ROUTED TO PIN K5 |
| D5 | PIN J4 ROUTED TO PIN K3,L3 |
| D4 | PIN J3 ROUTED TO PIN H2 |
| D3 | PIN J2 ROUTED TO PIN K1 |
| CE | PIN W6 ROUTED TO PIN Y7,AA8 |
| VDDI | PIN K2 ROUTED TO PIN L1 |
| GND | PIN Y2 ROUTED TO PIN W1 |

[PAGE_TITLE=MMC + FLASH]

DRAWING
Fri Jan 04 12:07:19 2013

| MICROSOFT | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| CONFIDENTIAL | STINGRAY | 32/74 | 32/74 | C | 1.0 |

# KSB OUT, AUDIO




| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| U3A1 | X858308-002 | IC | IC,NAU8402WG AUDIO_NUVOTON | PACK_IGNORE=TRUE |
| C3A16 | X800731-001 | X7R | CAP,0.22UF,0603 AUDIO_NUVOTON | PACK_IGNORE=TRUE |

[PAGE_TITLE=PSB OUT, AUDIO]

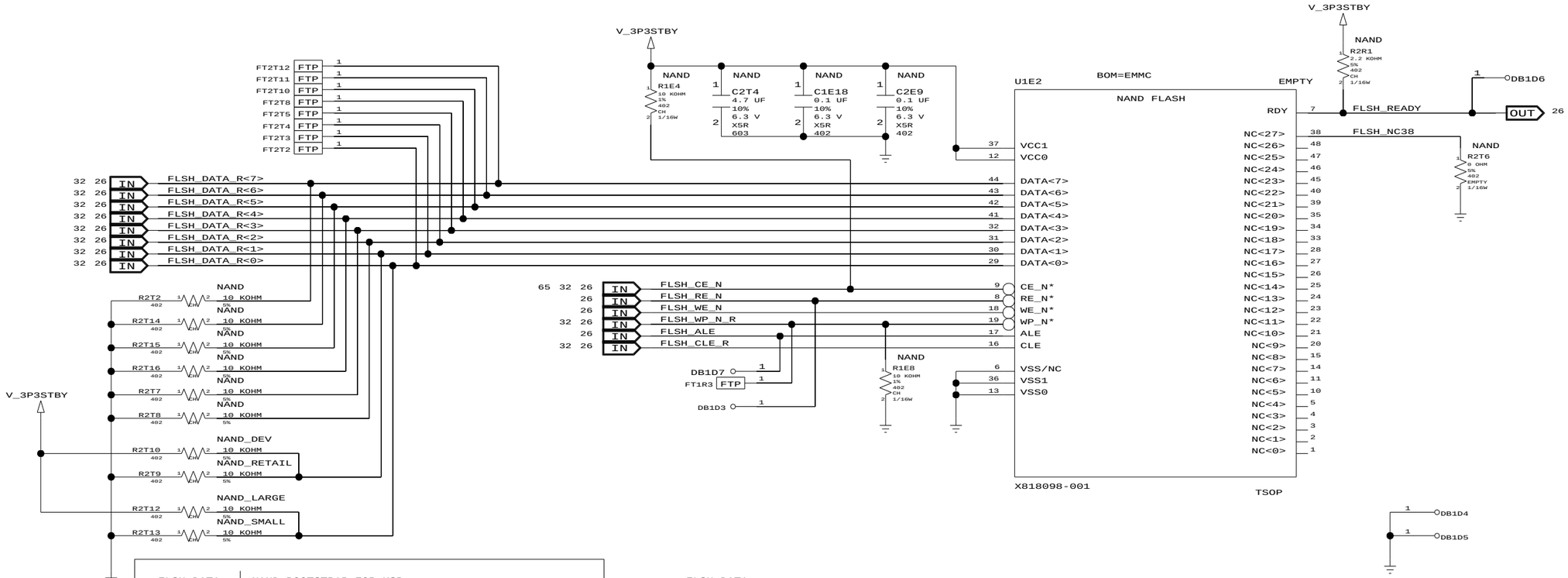
DRAWING
Fri Jan 04 12:07:19 2008

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 33/74 | 33/74 | C | 1.0 |

# KSB OUT, FLASH

| MS_PART# | MATL | REF_DES | | DESCR. | BOM PROPERTY |
|---|---|---|---|---|---|
| U1E2 | X803471-003 | IC | IC,NAND,16MB | NAND_16MB_HYNIX | PACK_IGNORE=TRUE |
| U1E2 | X818098-001 | IC | IC,NAND,512MB | NAND_512MB_HYNIX | PACK_IGNORE=TRUE |

ALL COMPONENTS ON THIS PAGE ARE FOR NAND MODE
TO BE STUFFED/UNSTUFFED AT CONFIG LEVEL




| FLSH_DATA 1 | FLSH_DATA 0 | NAND BOOTSTRAP FOR KSB FLASH CONFIGURATION | CONFIG |
|---|---|---|---|
| LOW | LOW | 0.5KB, 16KB BLOCKS, 128MBIT | RETAIL |
| LOW | HIGH | 0.5KB, 16KB BLOCKS, 512MBIT | RETAIL |
| HIGH | LOW | 2KB PAGES, 128KB BLOCKS, 1/2/4 GBIT | DEV |
| HIGH | HIGH | 4KB PAGES, 256KB BLOCKS, 2/4/8 GBIT | DEV |

FLSH_DATA
2 1 0

0 0 0 Raw NAND mode, 0.5kB, 16kB blocks, 128Mb
0 0 1 Raw NAND mode, 0.5kB, 16kB blocks, 512Mb
0 1 0 Raw NAND mode, 2.0kB, 128kB blocks, vary size
0 1 1 Raw NAND mode,
1 0 0 Raw NAND mode, 4.0kB, 256kB blocks, vary size
1 0 1 Raw NAND mode,
1 1 0 Raw NAND mode,
1 1 1 MMC mode

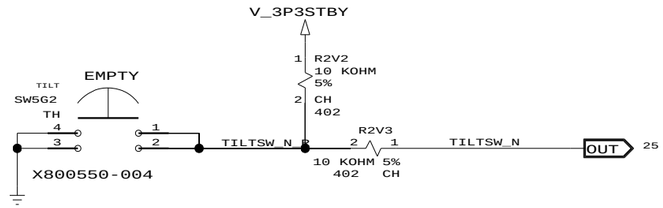
[PAGE_TITLE=PSB OUT, FLASH]

DRAWING
Fri Jan 04 12:07:19 2008

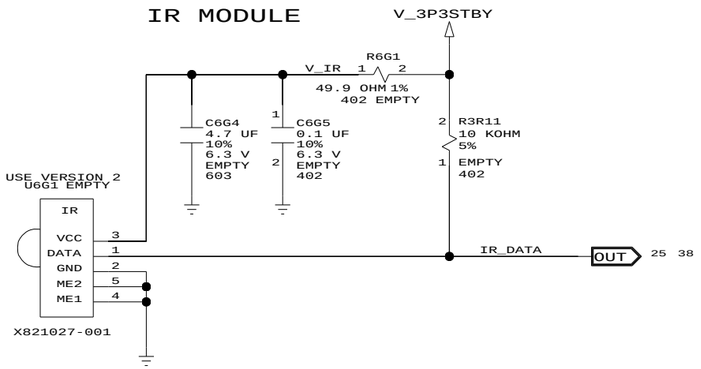
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 34/74 | 34/74 | C | 1.0 |

# CONN, INFARED + SWITCHES + AUDIBLE F/B

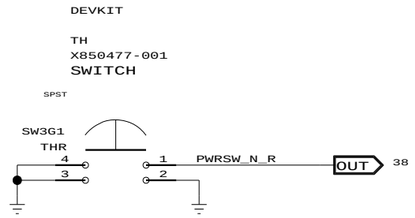
## TILT SWITCH, SOLICO




## IR MODULE

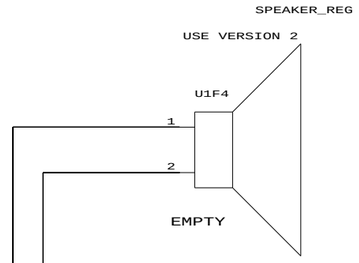



## DEVKIT

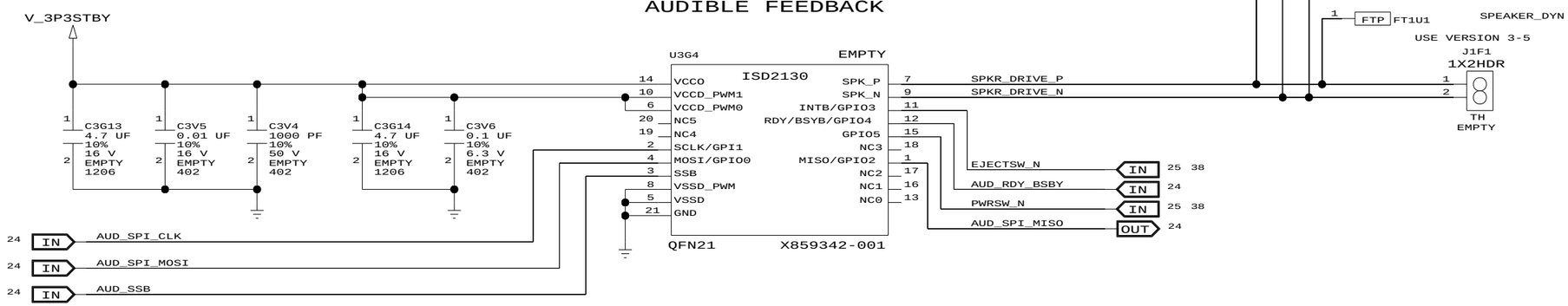



## POWER BUTTON

## SPEAKER_REG




## AUDIBLE FEEDBACK

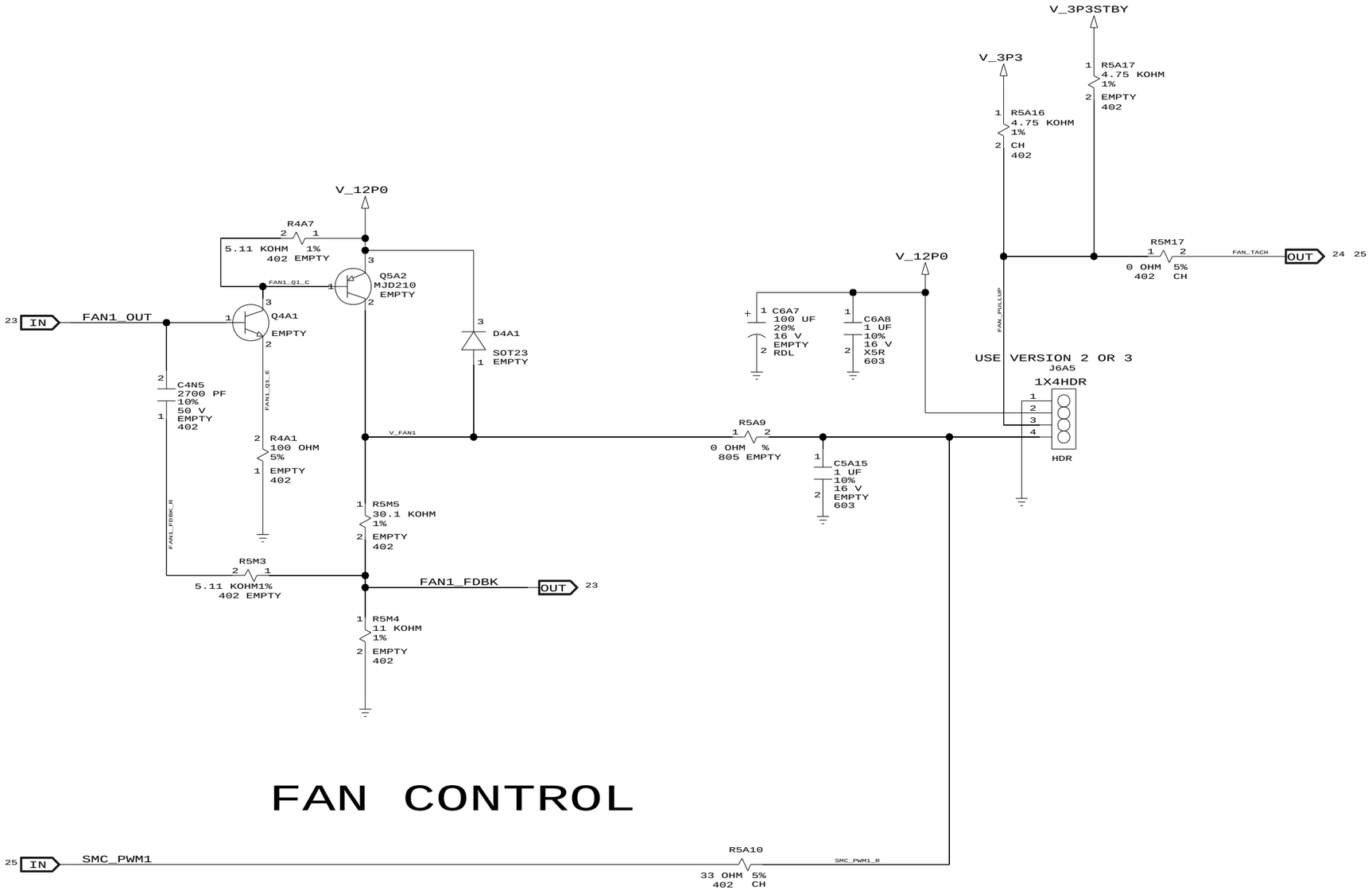



[PAGE_TITLE=CONN,INFRARED+SWITCHES+AUDIBLE F/B]

DRAWING
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| MICROSOFT 2013 CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 35/74 | 35/74 | C | 1.0 |

# CONN, FAN




# FAN CONTROL



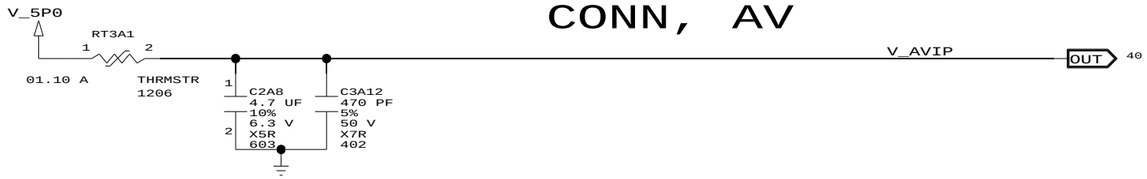

[PAGE_TITLE= CONN, FAN]

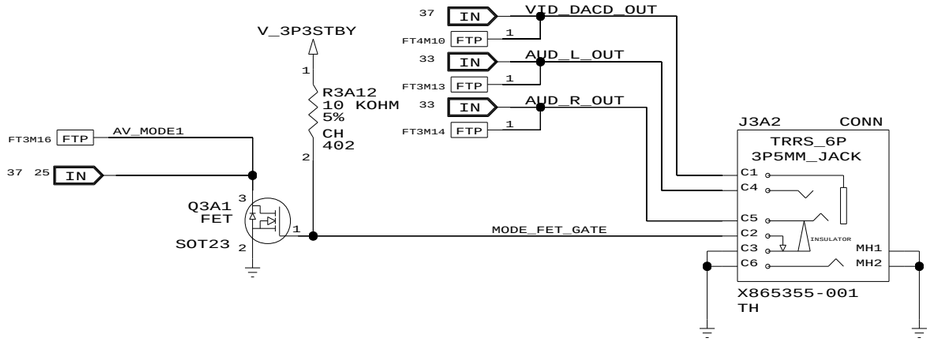
DRAWING
Fri Jan 04 12:07:20 2013

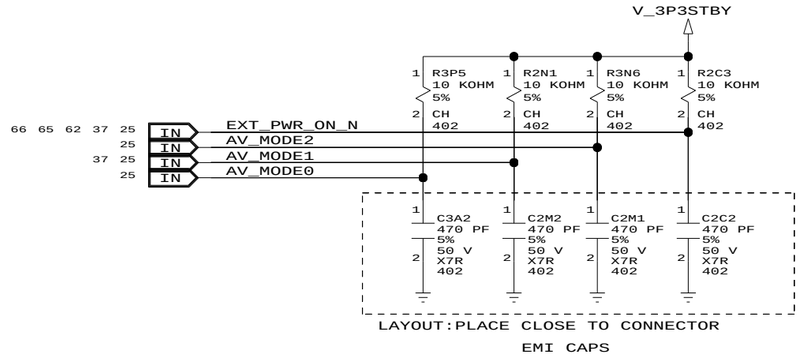
| MICROSOFT CONFIDENTIAL | PROJECT NAME STINGRAY | PAGE 36/74 | CSA PAGE 36/74 | FAB C | REV 1.0 |
|---|---|---|---|---|---|

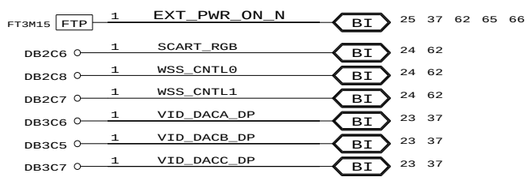
# CONN, AV

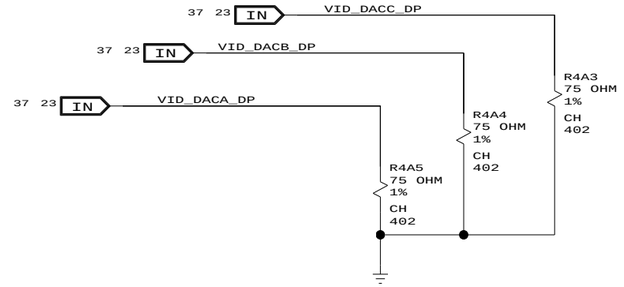
| DAC | STANDARD | ADVANCED | SDTV | HDTV | SCART | VGA |
|---|---|---|---|---|---|---|
| A | N/A | Y(LUMA) | Y | Y | G | G |
| B | N/A | C(CHROMA) | PR | PR | R | R |
| C | N/A | N/A | PB | PB | B | B |
| D | CVBS(COMP) | CVBS(COMP) | CVBS | N/A | CVBS | CVBS |

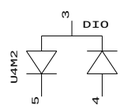


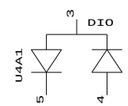



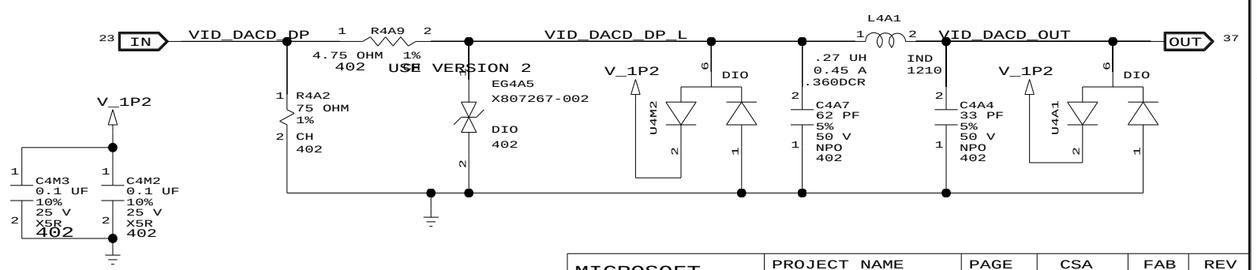



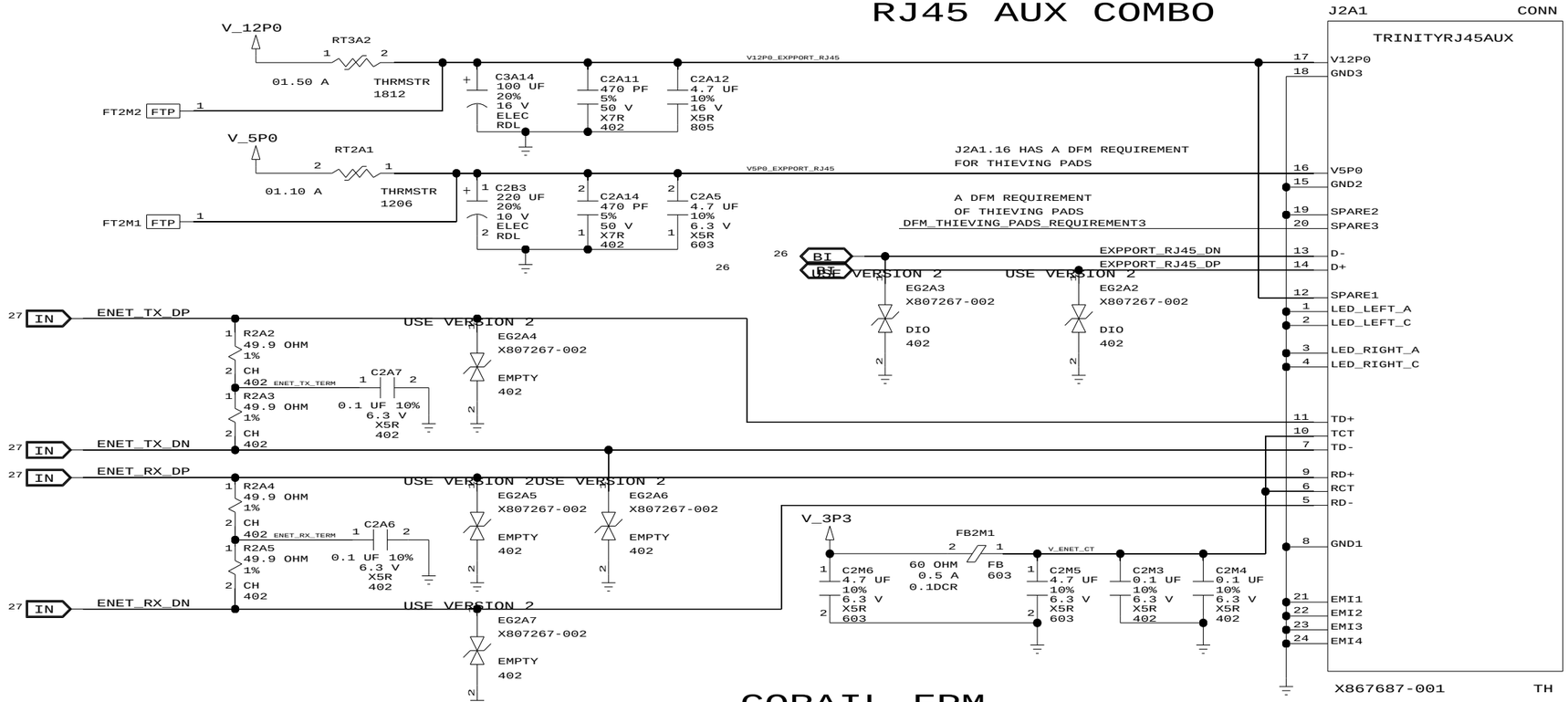

[PAGE_TITLE=[CONN, AVIP]

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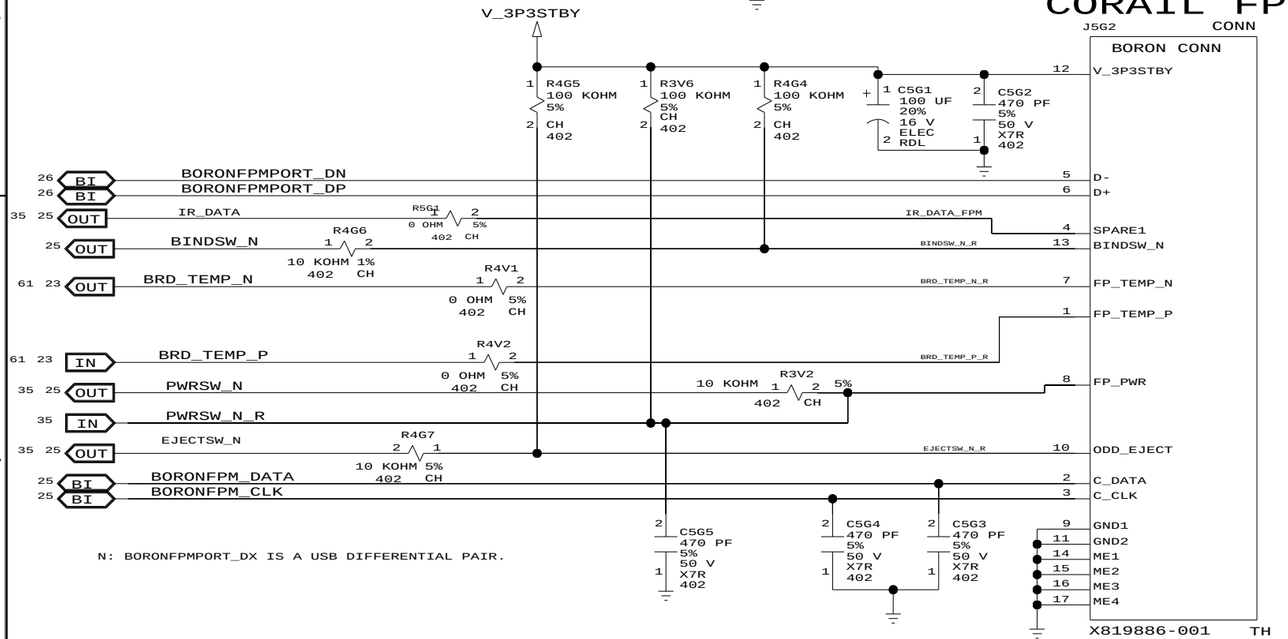
| MICROSOFT | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| CONFIDENTIAL | STINGRAY | 37/74 | 37/74 | C | 1.0 |

# CONN, RJ45 USB AUX COMBO + CORAIL + PWR

## RJ45 AUX COMBO

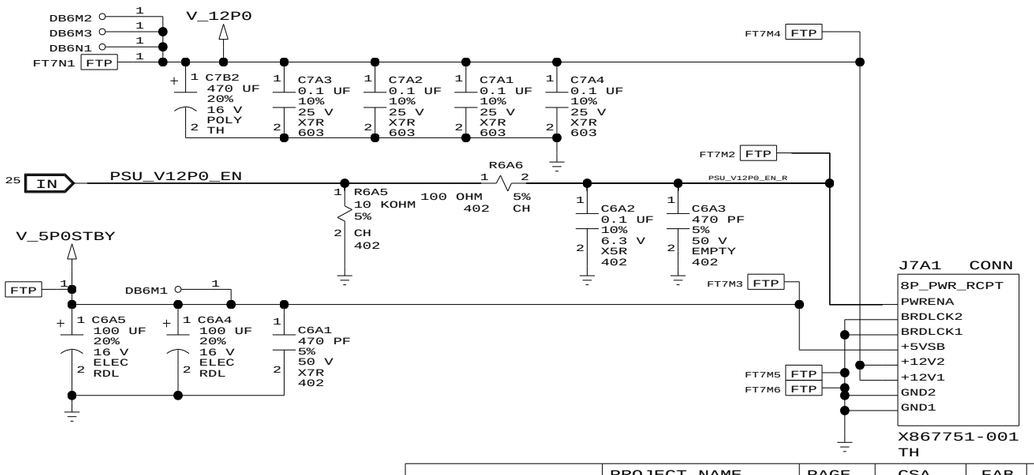



## CORAIL FPM




N: BORONFPMPORT_DX IS A USB DIFFERENTIAL PAIR.

## POWER



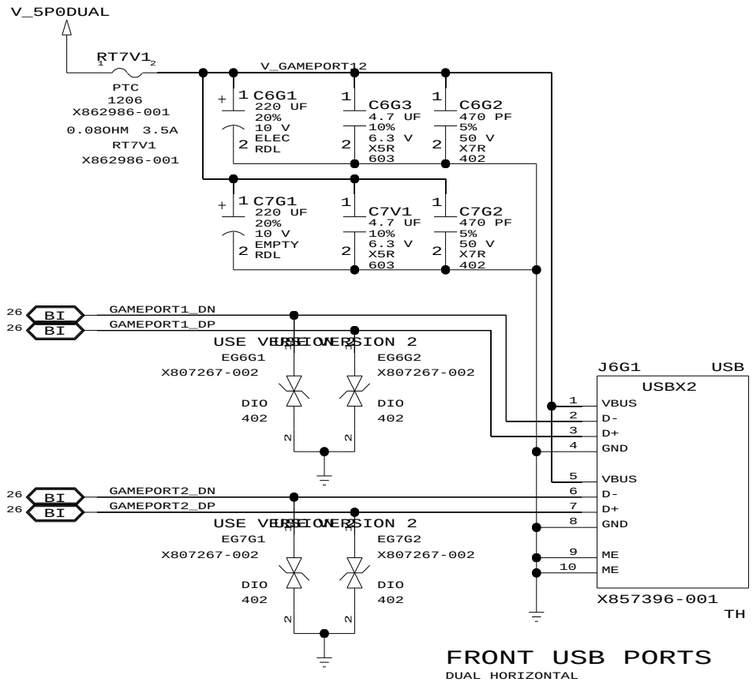

[PAGE_TITLE=CONN, RJ45 AUX COMBO + CORAIL + PWR]

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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 38/74 | 38/74 | C | 1.0 |

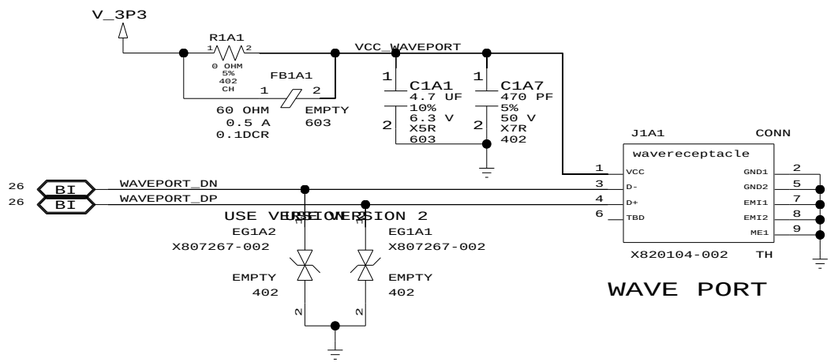
# CONN, USB + MEM PORTS + WAVEPORT

N: ALL DIFFERENTIAL PAIRS ON THIS PAGE ARE USB DIFFERENTIAL PAIRS.

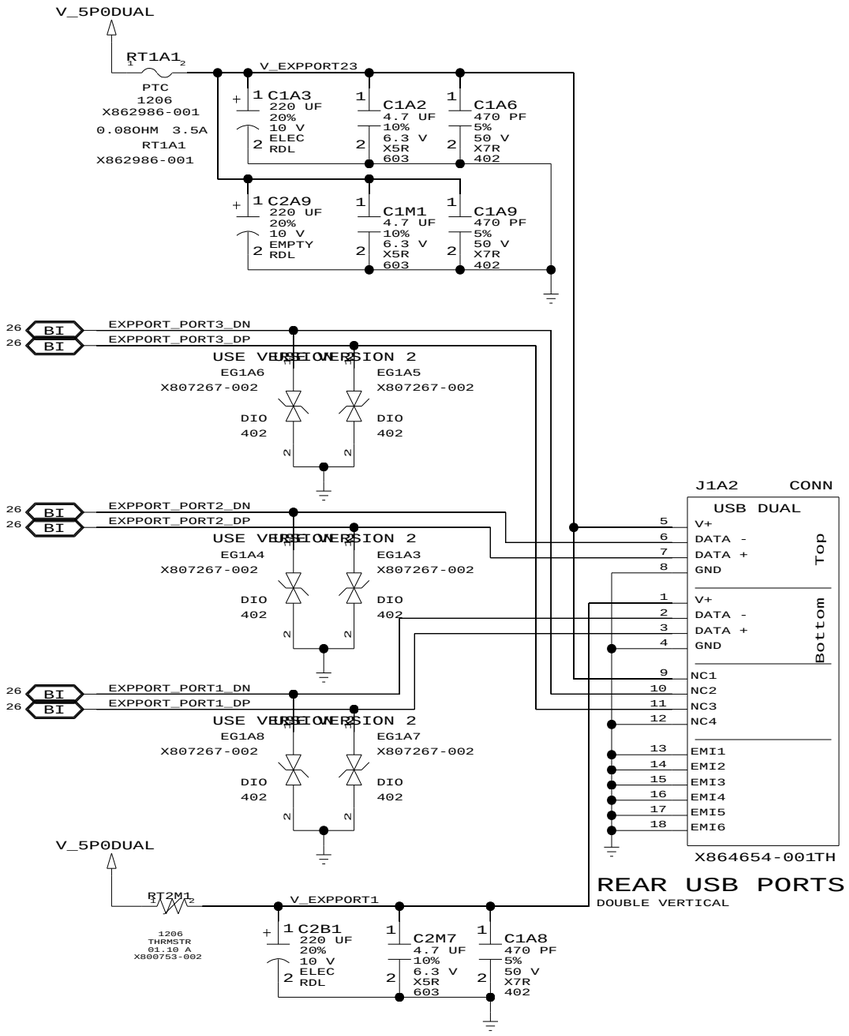



FRONT USB PORTS

DUAL HORIZONTAL




WAVE PORT




REAR USB PORTS

DOUBLE VERTICAL

[PAGE_TITLE=CONN, MEMORY PORTS + GAME PORTS]

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|---|---|---|---|---|---|

# CONN, HDMI

FT3M7 FTP 1
FT3M1 FTP 1
23 IN HDMI_TX2_DP
23 IN HDMI_TX2_DN
USE VERSION 2 — EG3A2 X807267-002 DIO 402
USE VERSION 2 — EG3A1 X807267-002 DIO 402
FT3M2 FTP 1
FT3M8 FTP 1

FT3M9 FTP 1
FT3M3 FTP 1
23 IN HDMI_TX1_DP
23 IN HDMI_TX1_DN
USE VERSION 2 — EG3A4 X807267-002 DIO 402
USE VERSION 2 — EG3A3 X807267-002 DIO 402
FT3M4 FTP 1
FT3M10 FTP 1

FT3M11 FTP 1
FT3M5 FTP 1
23 IN HDMI_TX0_DP
23 IN HDMI_TX0_DN
USE VERSION 2 — EG4A2 X807267-002 DIO 402
USE VERSION 2 — EG3A5 X807267-002 DIO 402
FT4M2 FTP 1
FT4M7 FTP 1

FT4M8 FTP 1
FT4M3 FTP 1
23 IN HDMI_TXC_DP
23 IN HDMI_TXC_DN
USE VERSION 2 — EG4A4 X807267-002 DIO 402
USE VERSION 2 — EG4A3 X807267-002 DIO 402
FT4M4 FTP 1
FT4M9 FTP 1

V_5P0STBY
R4M2 2 KOHM 1% 2 CH 402
R4M1 2 KOHM 1% 2 CH 402
FT4M5 FTP 1
55 23 IN HDMI_DDC_CLK
55 23 BI HDMI_DDC_DATA
USE VERSION 2 — EG4M2 X807267-002 DIO 402
USE VERSION 2 — EG4M1 X807267-002 DIO 402
FT4M1 FTP 1

37 IN V_AVIP
C3A11 0.1 UF 10% 6.3 V X5R 402

J4A5 USE VERSION 3or4 HDR
HDMI
1 TMDS_DATA2_DP
2 TMDS_DATA2_SHD
3 TMDS_DATA2_DN
4 TMDS_DATA1_DP
5 TMDS_DATA1_SHD
6 TMDS_DATA1_DN
7 TMDS_DATA0_DP
8 TMDS_DATA0_SHD
9 TMDS_DATA0_DN
10 TMDS_CLK_DP
11 TMDS_CLK_SHD
12 TMDS_CLK_DN
13 CEC
14 RESERVED
15 SCL
16 SDA
17 DDC_CEC_GND
18 SVCC
19 HOT_PLUG_DET
23 ME4
22 ME3
21 ME2
20 ME1
X806395-002

DB3M1 1 HDMI_CEC

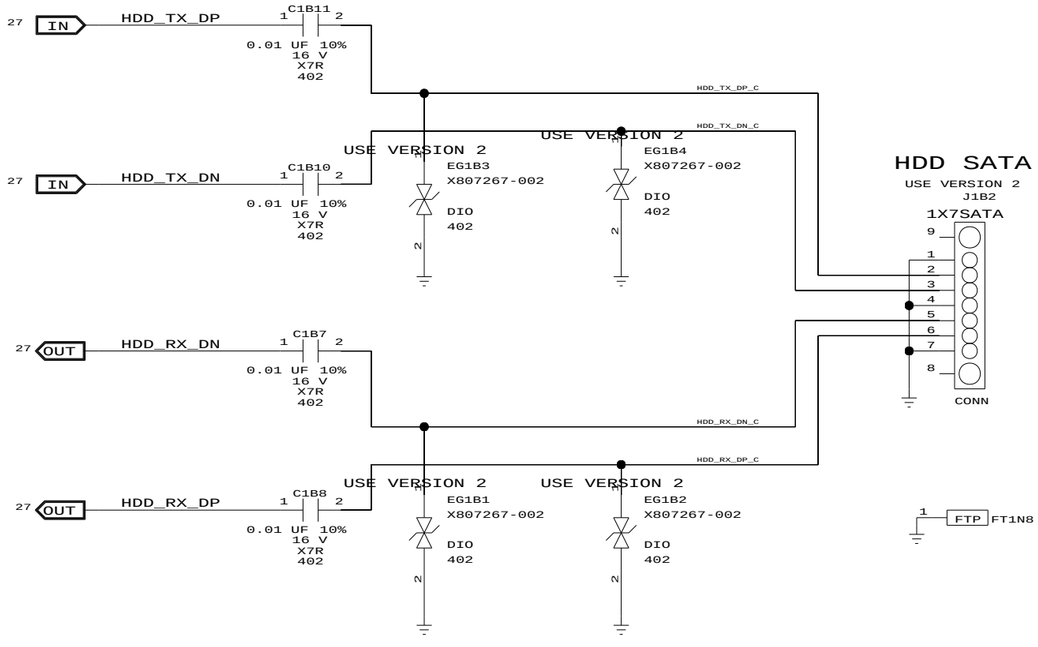
HDMI_HPD_PIN
R4M6 10 KOHM 5% 402 CH
FT4M6 FTP 1
HDMI_HPD OUT 23
USE VERSION 2 — EG4M5 X807267-002 DIO 402
R4M5 47.5 KOHM 1% 2 CH 402

[PAGE_TITLE=CONN, HDMI]

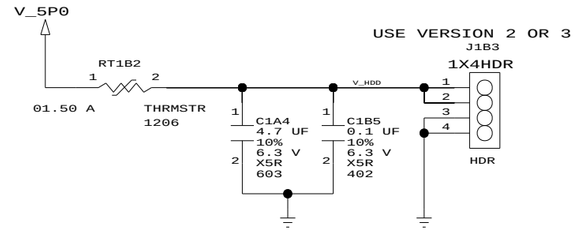
DRAWING
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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 40/74 | 40/74 | C | 1.0 |

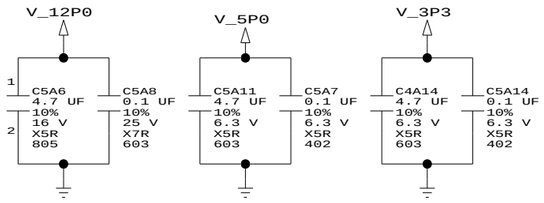
# CONN, ODD + HDD



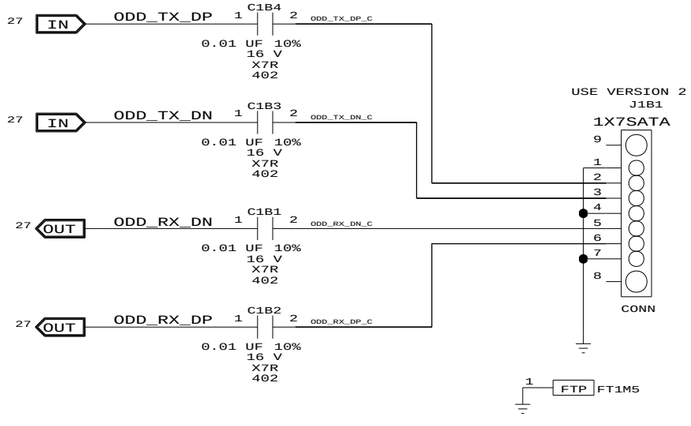

## HDD POWER



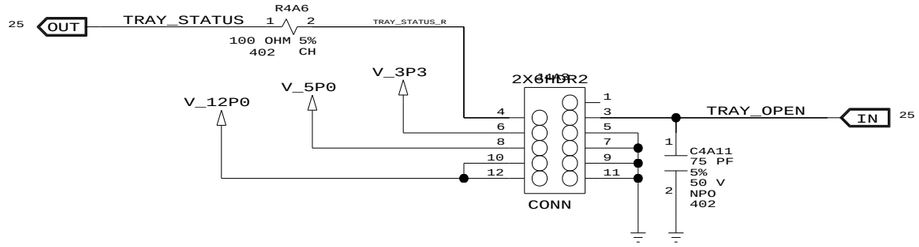

## ODD POWER DECOUPLING




## ODD SATA




## ODD POWER AND CONTROL

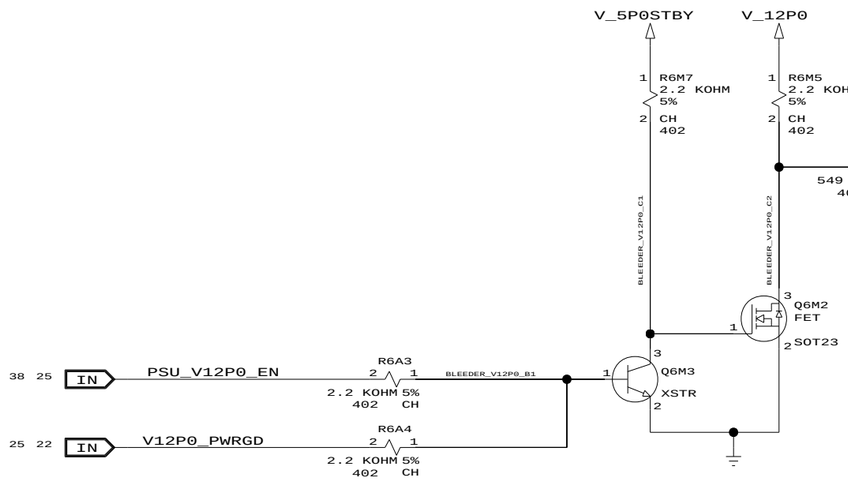



[PAGE_TITLE=CONN, ODD + HDD + MU]

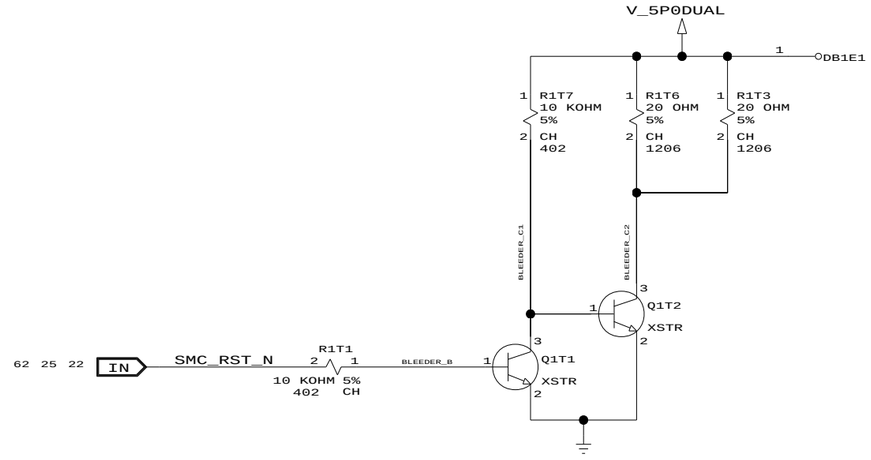
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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 41/74 | 41/74 | C | 1.0 |

# VREG, BLEEDERS




V_12P0 & V_5P0 BLEEDERS



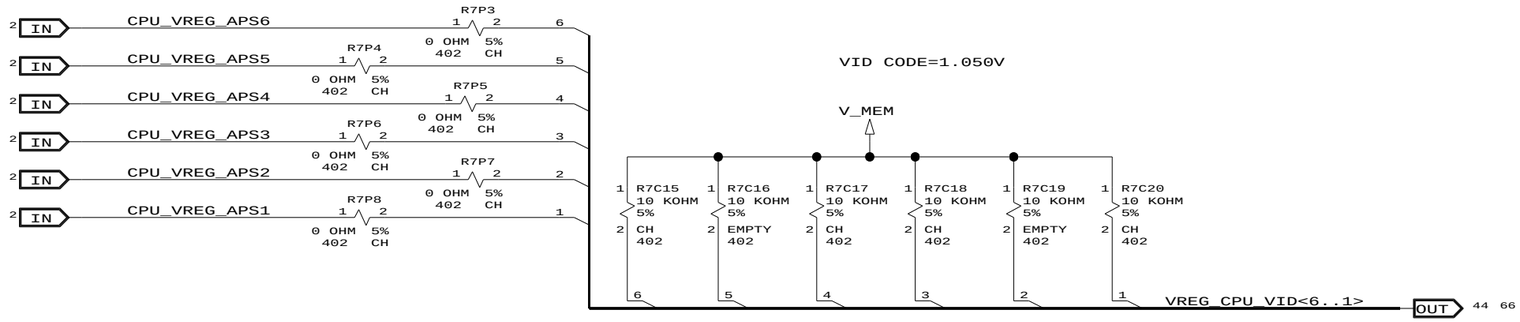

V_5P0DUAL BLEEDER

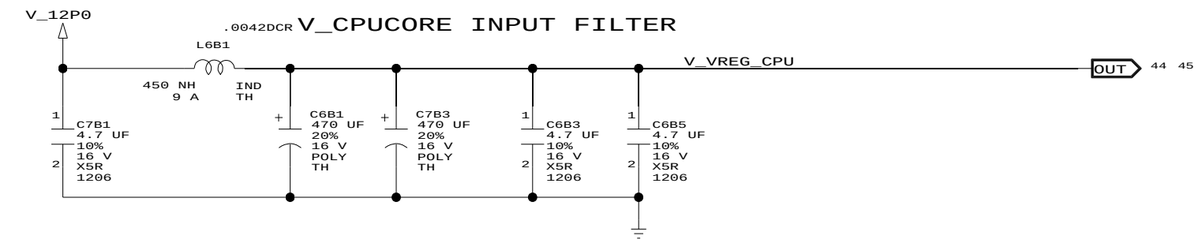
[PAGE_TITLE= VREGS,_BLEEDERS]

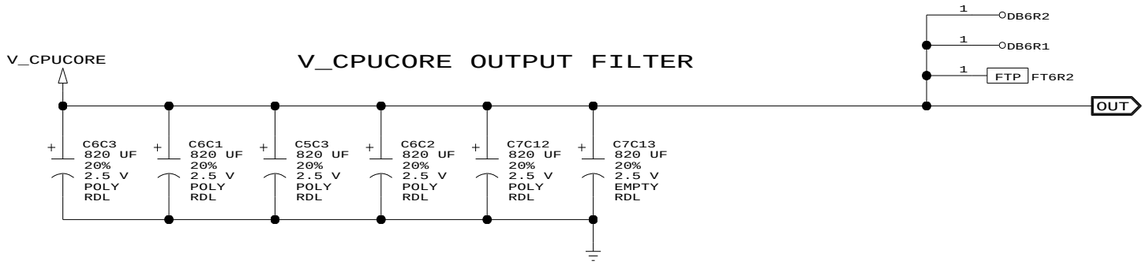
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# VREGS, INPUT + OUTPUT FILTERS




[PAGE_TITLE=VREGS, INPUT + OUTPUT FILTERS]

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|---|---|---|---|---|---|
| | STINGRAY | 43/74 | 43/74 | C | 1.0 |

# VREGS, CPU CONTROLLER

45 43 IN V_VREG_CPU
R7C26 603 681 OHM 1% 1/16W CH
R7C25 603 681 OHM 1% 1/16W CH
VREG_CPU_VCC
R7B5 402 1 KOHM 5% 1/16W CH
VREG_CPU_RAMPADJ_R
C7P2 0.1 UF 10% 25 V X7R 603
RRAMP
R7C22 422 KOHM 1% 402 CH
C7C23 4.7 UF 10% 16 V X5R 1206
C7C22 1 UF 10% 16 V X5R 603
FT3P8 FTP 1
25 IN VREG_CPU_EN
R7C31 10 KOHM 5% 402 CH 1/16W
66 60 59 58 57 56 55 IN PMBUS_CLK
66 60 59 58 57 56 55 BI PMBUS_DATA
45 IN VREG_CPU_PHASE2
R7C23 402 0 OHM 5% 1/16W
45 IN VREG_CPU_PHASE1
R7C24 402 0 OHM 5% 1/16W
66 OUT VREG_CPU_CSCOMP
R7B1 35.7 KOHM 1% 402 CH 1/16W
RT6C1 603 THRMSTR XB01015-001
RCS1
RTH1 TEMP SENSOR
R7C21 6.81 KOHM 1% 402 CH 1/16W
VREG_CPU_CSCOMP_R
R7B3 27.4 KOHM 1% 402 CH 1/16W RPH2
R7B4 27.4 KOHM 1% 402 CH 1/16W RPH1
R7B2 52.3 KOHM 1% 402 CH 1/16W RCS2
C7C15 3300 PF 10% 50 V X7R 402 CCS2
C7C14 3300 PF 10% 50 V X7R 402 CCS1
66 OUT VREG_CPU_CSSUM
66 OUT VREG_CPU_CSREF
V_CPUCORE
C7P1 1000 PF 10% 50 V X7R 402
ST6B3 SHORT OFFGRID
PLACE SHORT NEXT TO OUTPUT OF DUAL INDUCTOR
ST6D1 SHORT OFFGRID
PLACE SHORT NEXT TO POINT OF LOAD
DB7C1
VREG_V_CPUCORE_S
VENABLE
R7C32 0 OHM % 805 EMPTY
C7C16 470 PF 5% 402 X7R 50 V CB
C7C17 10 PF 5% 402 NPO 50 V CFB
R7P9 1.1 KOHM 1% 402 CH RB
C7C18 1500 PF 10% 402 X7R 50 V CA
R7P10 18.2 KOHM 1% 402 CH RA
DB7C2
ST6D2 SHORT OFFGRID
PLACE SHORT NEXT TO POINT OF LOAD

VREG_CPU_VCC3_3P3
C7C20 0.1 UF 10% 6.3 V X5R 402

U7C1 IC
NCP4202

| | | | | | |
|---|---|---|---|---|---|
| 1 | VCC3 | | | | |
| 21 | VCC | | | | |
| VREG_CPU_VCC | | | | | |
| VREG_CPU_RAMPADJ | 8 | RAMPADJ | PWRGD | 28 | VREG_CPUCORE_PWRGD OUT 25 |
| VREG_CPU_EN | 4 | EN | IPD VID6 | 22 | VREG_CPU_VID<6> IN 43 66 |
| | | | IPD VID5 | 23 | VREG_CPU_VID<5> IN 43 66 |
| | | | IPD VID4 | 24 | VREG_CPU_VID<4> IN 43 66 |
| | | | IPD VID3 | 25 | VREG_CPU_VID<3> IN 43 66 |
| | | | IPD VID2 | 26 | VREG_CPU_VID<2> IN 43 66 |
| | | | IPD VID1 | 27 | VREG_CPU_VID<1> IN 43 66 |
| PMBUS_CLK | 3 | SCL | PWM2 | 19 | VREG_CPU_PWM2 OUT 45 |
| PMBUS_DATA | 2 | SDA | PWM1 | 20 | VREG_CPU_PWM1 OUT 45 |
| VREG_CPU_PHASE2_R | 17 | SW2 | | | |
| VREG_CPU_PHASE1_R | 18 | SW1 | | | |
| VREG_CPU_CSCOMP | 14 | CSCOMP | | | |
| VREG_CPU_CSSUM | 13 | CSSUM | OD1_N | 16 | VREG_CPU_DRV_EN OUT 45 |
| VREG_CPU_CSREF | 12 | CSREF | IMON | 5 | VREG_CPU_IMON |
| VREG_CPU_FB | 11 | FB | RT | 7 | VREG_CPU_RT |
| VREG_CPU_COMP | 10 | COMP | | | |
| VREG_CPU_FBRTN | 9 | FBRTN | | | |
| VREG_CPU_IREF | 6 | IREF | | | |
| VREG_CPU_ILIMITFS | 15 | ILIMITFS | MPAD | 29 | |

X857741-001 QFN29

R7C27 10 KOHM 5% 402 CH 1/16W
R7C28 909 KOHM 1% 402 CH 1/16W X812615-001
ROSC=909K
FOSC=250KHZ
R7C29 121 KOHM 1% 402 CH 1/16W
NEED TO SET DURING VALIDATION

PARTS ARE TRIMMED AT FACTORY SUCH THAT VBOOT=1.05V

I2C ADDRESS
1100 000 R/W HEX
WRITE 1100 000 0 0XC0
READ 1100 000 1 0XC1

C7C21 0.1 UF 10% 402 X5R 6.3 V
R7C30 5.76 KOHM 1% 402 CH 1/16W
NOTE: VALID UP TO 58A
C7C19 1000 PF 10% 50 V X7R 402

[PAGE_TITLE=VREGS, CPU CONTROLLER]

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| | STINGRAY | 44/74 | 44/74 | C | 1.0 |

# VREG, CPU OUTPUT PHASE 1 & 2

44 43 IN V_VREG_CPU

R6N1 2.2 OHM 1% 805 CH — VREG_CPU2_VCC — C6B4 1 UF 10% 16 V X5R 603

D6B1 SOT23 DIO

VREG_CPU_BST2 — C6N1 0.01 UF 10% 50 V EMPTY 805

R6N2 2.2 OHM 1% 805 CH — VREG_CPU_BST2_R — C6N2 0.1 UF 10% 25 V X7R 603

U6B1 IC — MOS DRIVER

| | | | |
|---|---|---|---|
| 4 | VCC | BST | 1 |
| 2 | IN | DRVH | 8 |
| 3 | OD_N* | SW | 7 |
| 6 | PGND | DRVL | 5 |

X850497-001

44 IN VREG_CPU_PWM2

44 IN VREG_CPU_DRV_EN

VREG_CPU_DRVH2

VREG_CPU_DRVL2

Q6B2 X857866-001 DPAK_G1D2S3 FET

Q6C2 X857867-001 DPAK_G1D2S3 FET

Q6C1 X857867-001 DPAK_G1D2S3 FET

C7B5 4.7 UF 10% 16 V X5R 1206

C7N1 4.7 UF 10% 16 V EMPTY 1206

R6C1 1 OHM 1% CH 805 — VREG_CPU_SW2_R — C7C8 2200 PF 10% 50 V X7R 805

V_CPUCORE
NOM.VOLTAGE: 0.9-1.2V

VREG_CPU_PHASE2 OUT 44

V_CPUCORE

L6C1 SM EMPTY
DUAL INDUCTOR
L1 L4 L3 L2
1 4 3 2
DB7C4

X820048-001
420NH

DCR +/-8%

V_CPUCORE

L6C2 IND
DUAL INDUCTOR
L2 L3 L4 L1
2 3 4 1

X850592-001
420NH
40 A

+/-4%

NOTE: THRUHOLE PART IS 2%
IMPROVEMENT ON DCR

R5N7 2.2 OHM 1% 805 CH — VREG_CPU1_VCC — C5B10 1 UF 10% 16 V X5R 603

D5B1 SOT23 DIO

VREG_CPU_BST1 — C5N5 0.01 UF 10% 50 V EMPTY 805

R5N8 2.2 OHM 1% 805 CH — VREG_CPU_BST1_R — C5N6 0.1 UF 10% 25 V X7R 603

U5B5 IC — MOS DRIVER

| | | | |
|---|---|---|---|
| 4 | VCC | BST | 1 |
| 2 | IN | DRVH | 8 |
| 3 | OD_N* | SW | 7 |
| 6 | PGND | DRVL | 5 |

X850497-001

44 IN VREG_CPU_PWM1

VREG_CPU_DRVH1

VREG_CPU_DRVL1

Q6B1 X857866-001 DPAK_G1D2S3 FET

Q5C3 X857867-001 DPAK_G1D2S3 FET

Q5C1 X857867-001 DPAK_G1D2S3 FET

C6B2 4.7 UF 10% 16 V X5R 1206

C6N3 4.7 UF 10% 16 V EMPTY 1206

R5C1 1 OHM 1% CH 805 — VREG_CPU_SW1_R — C5C1 2200 PF 10% 50 V X7R 805

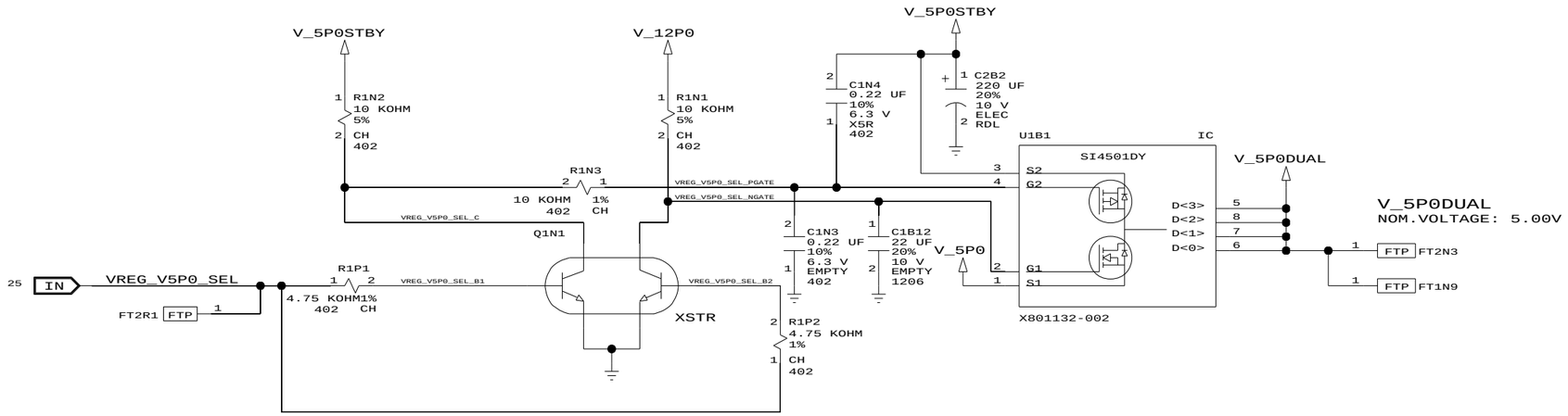
VREG_CPU_PHASE1 OUT 44

[PAGE_TITLE=VREGS, CPU OUTPUT PHASE 1,2]

DRAWING
Fri Jan 04 12:07:21 2008

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 45/74 | 45/74 | C | 1.0 |

# VREGS, V5P0 DUAL



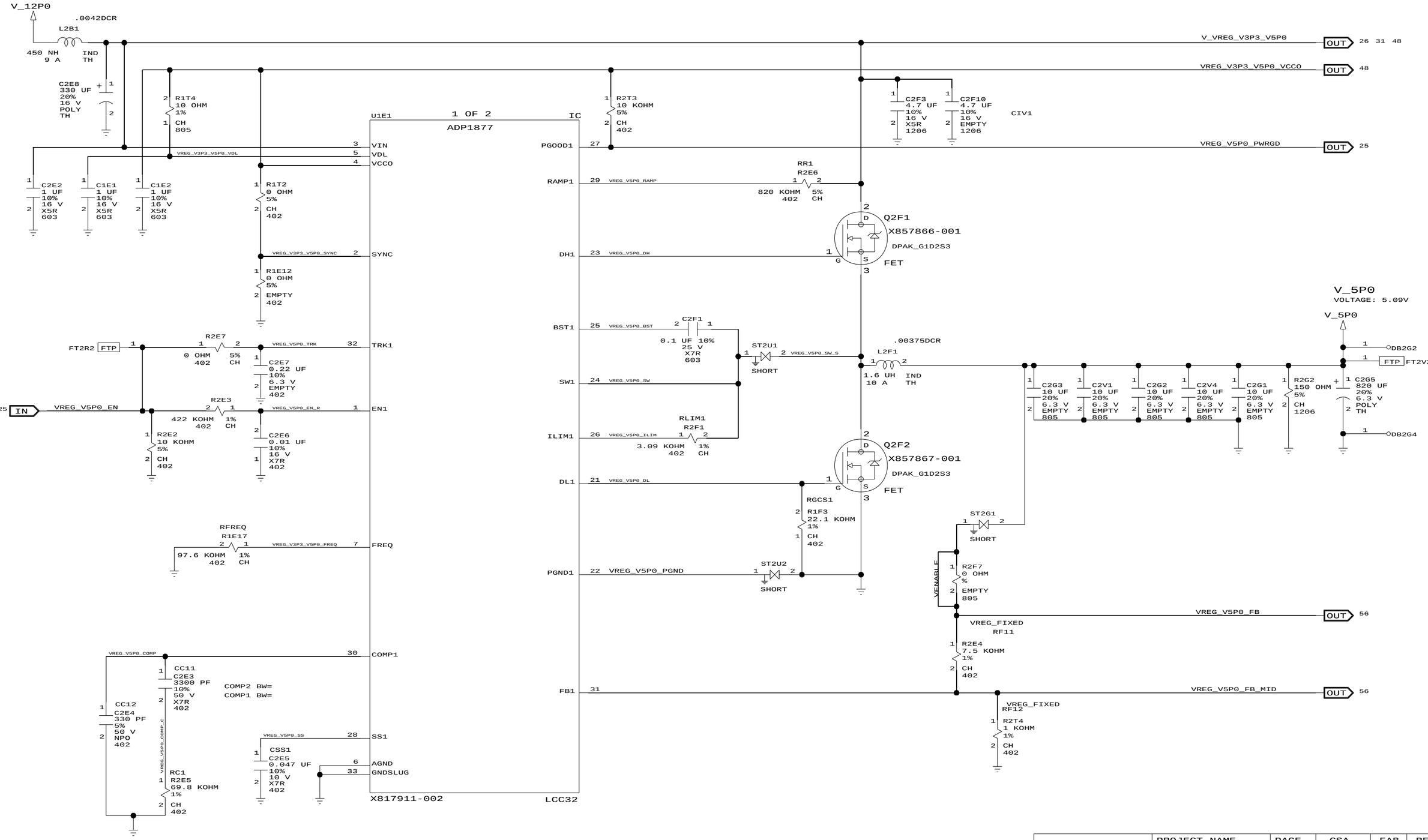

| VREG_5P0_SEL | VREG_5P0_SEL NGATE/PGATE | V_5P0DUAL |
|---|---|---|
| HIGH | LOW | V_5P0STBY |
| LOW | HIGH | V_5P0 |

[PAGE_TITLE=VREGS, V5P0 DUAL]

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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 46/74 | 46/74 | C | 1.0 |

# VREGS, V5P0

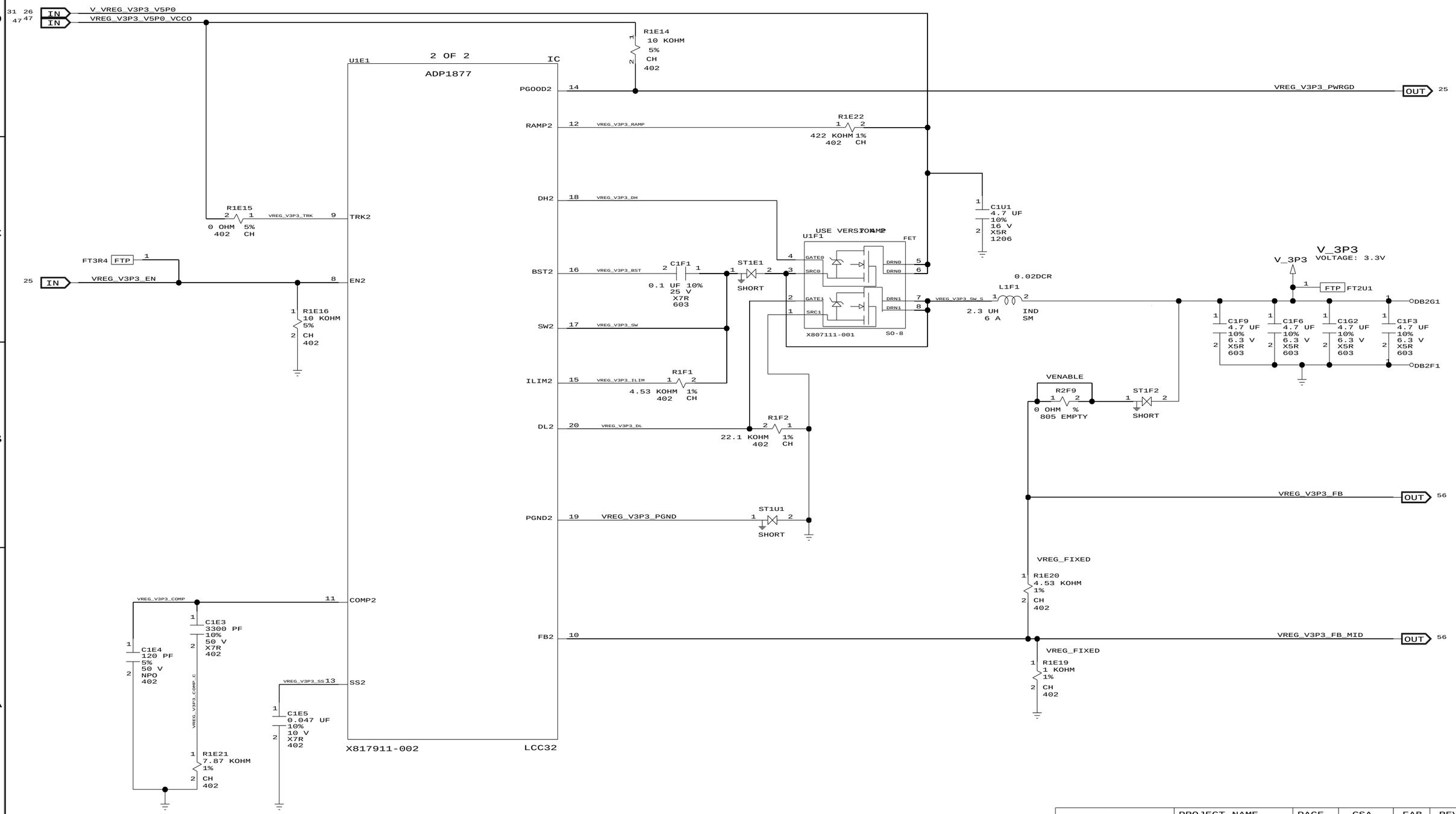



[PAGE_TITLE=VREGS, V5P0]

DRAWING
Fri Jan 04 12:07:22 2013

| MICROSOFT CONFIDENTIAL | PROJECT NAME STINGRAY | PAGE 47/74 | CSA PAGE 47/74 | FAB C | REV 1.0 |
|---|---|---|---|---|---|

# VREGS, V3P3




[PAGE_TITLE=VREGS, V3P3]

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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 48/74 | 48/74 | C | 1.0 |

# VREGS, VEDRAM

V_12P0
.0042DCR
L7B1
450 NH
9 A
IND
TH
C3F8
330 UF
20%
16 V
POLY
TH
C7B8
470 UF
20%
16 V
POLY
TH
R3G2
10 OHM
1%
CH
805
U3G1
1 OF 2
ADP1877
IC
R3G16
10 KOHM
5%
CH
402
C3F1
4.7 UF
10%
16 V
X5R
1206
C3F7
4.7 UF
10%
16 V
EMPTY
1206
CIV1
V_VREG_VMEM_VEDRAM
OUT 50
VREG_VMEM_VEDRAM_VCCO
OUT 50
VREG_VEDRAM_PWRGD
OUT 25
VIN
VDL
VCCO
VREG_VMEM_VEDRAM_VDL
C3G1
1 UF
10%
16 V
X5R
603
C3G3
1 UF
10%
16 V
X5R
603
C3V1
1 UF
10%
16 V
X5R
603
SYNC HIGH=FORCED PWM MODE
SYNC LOW =PULSE SKIP MODE
R3F12
0 OHM
5%
CH
402
R3F14
0 OHM
5%
CH
402
VREG_VMEM_VEDRAM_SYNC
SYNC
R3F13
0 OHM
5%
EMPTY
402
VREG_VEDRAM_TRK
TRK1
PGOOD1
RAMP1
VREG_VEDRAM_RAMP
RR1
R3F10
475 KOHM
402
1%
CH
DH1
VREG_VEDRAM_DH
Q3F1
X857866-001
DPAK_G1D2S3
FET
V_CPUEDRAM
V_CPUEDRAM
VOLTAGE: 1.075V
FTP FT4T2
DB4E3
.0034DCR
L4F1
1.7 UH
13.8 A
IND
TH
C4E3
820 UF
20%
2.5 V
POLY
RDL
C4E6
820 UF
20%
2.5 V
EMPTY
RDL
C4F7
820 UF
20%
2.5 V
EMPTY
RDL
BST1
VREG_VEDRAM_BST
C3F5
0.1 UF
10%
25 V
X7R
603
ST3F1
SHORT
VREG_VEDRAM_SW_S
SW1
VREG_VEDRAM_SW
FT2T9 FTP
25 IN
VREG_VEDRAM_EN
EN1
R3F15
10 KOHM
5%
CH
402
RLIM1
R3F7
3.09 KOHM
402
1%
CH
ILIM1
VREG_VEDRAM_ILIM
Q4F1
X857867-001
DPAK_G1D2S3
FET
C5T52
4.7 UF
10%
16 V
EMPTY
1206
C5E11
4.7 UF
10%
16 V
EMPTY
1206
C4T7
4.7 UF
10%
16 V
EMPTY
1206
C5E10
4.7 UF
10%
16 V
EMPTY
1206
DL1
VREG_VEDRAM_DL
RGCS1
R3V1
22.1 KOHM
1%
EMPTY
402
ST5E1
SHORT
DB4E4
DB4E5
RFREQ
R3G4
97.6 KOHM
402
1%
CH
VREG_VEDRAM_FREQ
FREQ
PGND1
VREG_VEDRAM_PGND
ST4U1
SHORT
VENABLE
R3F11
0 OHM
%
EMPTY
805
VREG_VEDRAM_FB
OUT 55
VREG_FIXED
RF11
R3F8
7.87 KOHM
1%
CH
402
VREG_VEDRAM_COMP
COMP1
CC11
C3F2
1000 PF
10%
50 V
X7R
402
CC12
C3F3
10 PF
5%
50 V
NPO
402
VREG_VEDRAM_COMP_C
RC1
R3F6
169 KOHM
1%
CH
402
RC1=120K
VREG_VEDRAM_SS
SS1
CSS1
C3F4
0.047 UF
10%
10 V
X7R
402
AGND
GNDSLUG
X817911-002
LCC32
FB1
VREG_VEDRAM_FB_MID
OUT 55
VREG_FIXED
RF12
R3F9
10 KOHM
1%
CH
402

[PAGE_TITLE=[VREGS, VEDRAM]

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| MICROSOFT | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| CONFIDENTIAL | STINGRAY | 49/74 | 49/74 | C | 1.0 |

# VREGS, VMEM

49 IN V_VREG_VMEM_VEDRAM

49 IN VREG_VMEM_VEDRAM_VCCO

R3G15 10 KOHM 5% CH 402

U3G1 2 OF 2 IC ADP1877

PGOOD2 14 VREG_VMEM_PWRGD OUT 25

CIV2 C7F3 4.7 UF 10% 16 V X5R 1206

C7U1 4.7 UF 10% 16 V EMPTY 1206

RR2 R3G13 422 KOHM 1% 402 CH

RAMP2 12 VREG_VMEM_RAMP

Q7G1 X857866-001 DPAK_G1D2S3 FET

DH2 18 VREG_VMEM_DH

R3G5 0 OHM 5% 402 CH

VREG_VMEM_TRK 9 TRK2

V_MEM VOLTAGE: 1.8V

V_MEM

DB7F1

FT7U1 FTP

.0034DCR

L7F1 IND TH 1.7 UH 13.8 A

FT2T1 FTP 1

25 IN VREG_VMEM_EN 8 EN2

R3G14 10 KOHM 5% CH 402

BST2 16 VREG_VMEM_BST

C3G9 0.1 UF 10% 25 V X7R 603

ST6V2 SHORT

VREG_VMEM_SW_S

C7F2 820 UF 20% 2.5 V POLY RDL

C7F1 4.7 UF 10% 6.3 V EMPTY 603

DB7F2

SW2 17 VREG_VMEM_SW

V_MEM

C7R1 330 UF 20% 2.5 V EMPTY SM

RLIM2 R3G10 3.09 KOHM 1% 402 CH

ILIM2 15 VREG_VMEM_ILIM

Q7G2 X857867-001 DPAK_G1D2S3 FET

DL2 20 VREG_VMEM_DL

V_MEM

C4U4 330 UF 20% 2.5 V EMPTY SM

RGCS2 R4G3 22.1 KOHM 1% EMPTY 402

ST7F1 SHORT

ST6V1 SHORT

PGND2 19 VREG_VMEM_PGND

VENABLE R3G17 0 OHM % EMPTY 805

VREG_VMEM_FB OUT 55

VREG_FIXED RF21

R3G11 20 KOHM 1% CH 402

FB2 10

VREG_VMEM_FB_MID OUT 55

VREG_FIXED RF22

R3G12 10 KOHM 1% CH 402

VREG_VMEM_COMP 11 COMP2

CC21 C3G6 1000 PF 10% 50 V X7R 402

CC22 C3G7 47 PF 5% 50 V NPO 402

VREG_VMEM_COMP_C

RC2 R3V4 169 KOHM 1% CH 402

VREG_VMEM_SS 13 SS2

CSS1 C3G8 0.047 UF 10% 10 V X7R 402

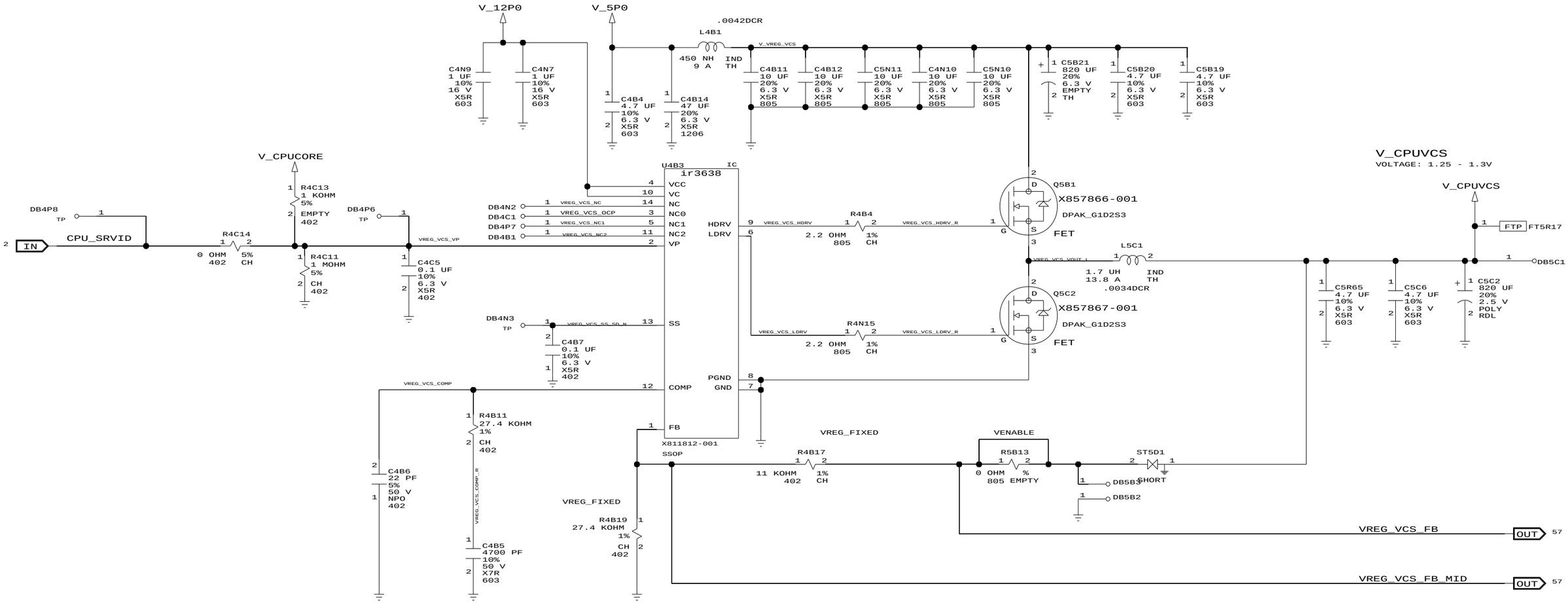
X817911-002 LCC32

[PAGE_TITLE=[VREGS, VMEM]

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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 50/74 | 50/74 | C | 1.0 |

# VREGS, VCS



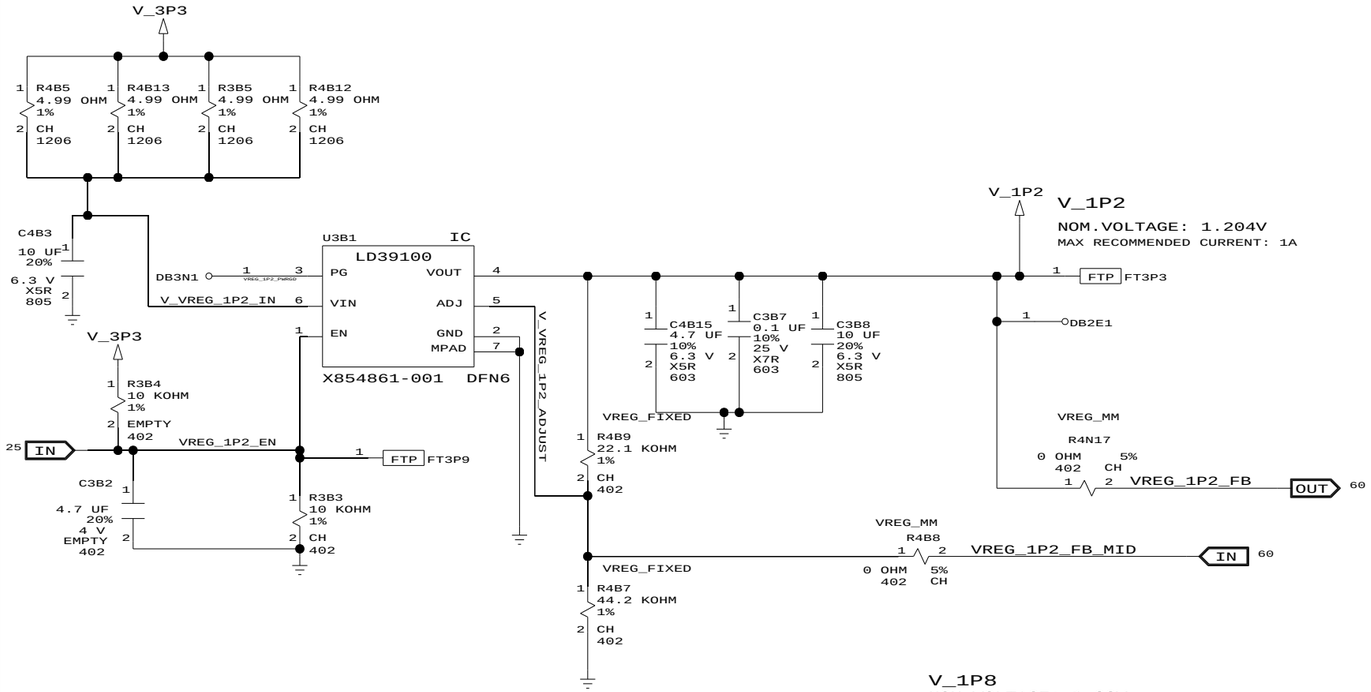

GAIN=0.4 WITH R4B17 = 11K, R4B19 = 27.4K
OUTPUT = CPU_SRVID(1+GAIN)

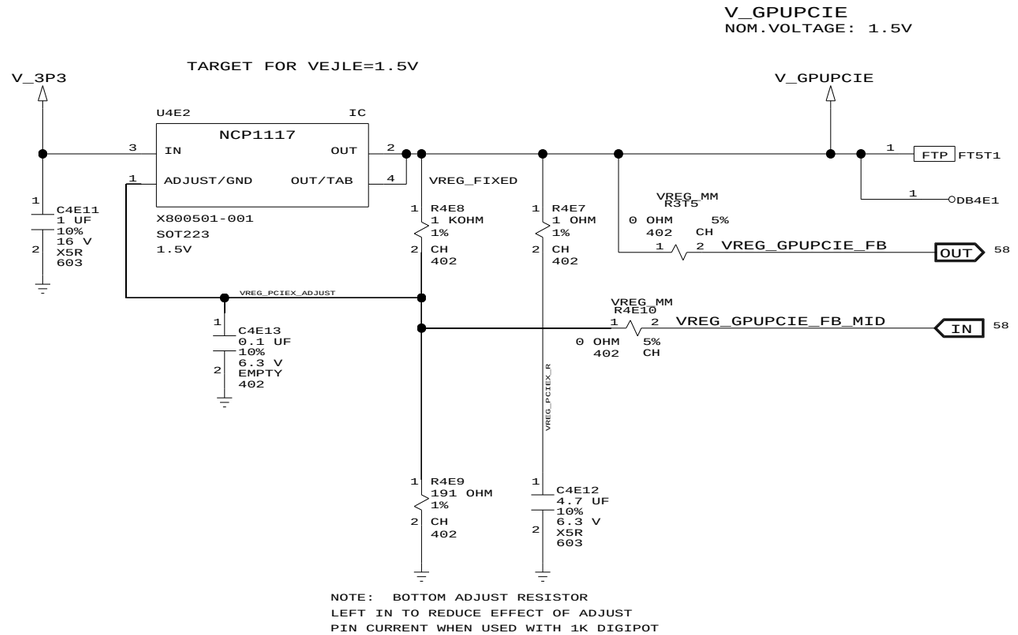
[PAGE_TITLE=VREGS, VCS]

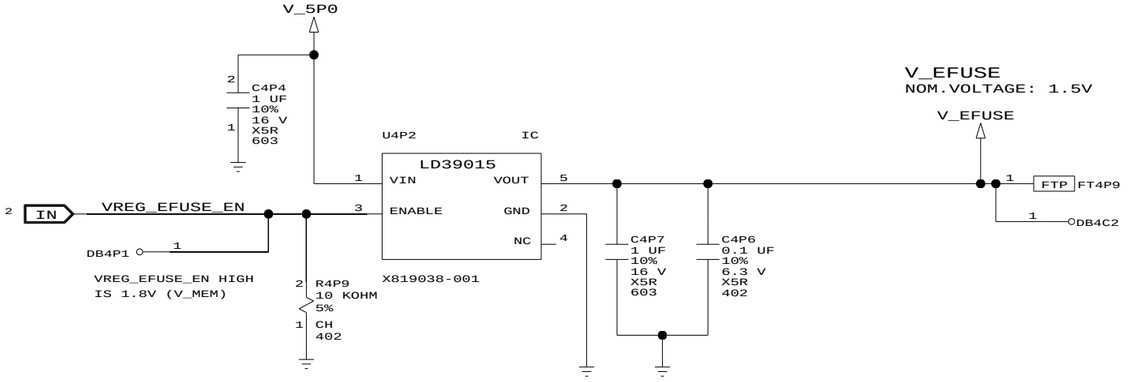
DRAWING
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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 51/74 | 51/74 | C | 1.0 |

# VREGS, 1P2+1P8+GPUPCIE+CPUPLL+EFUSE



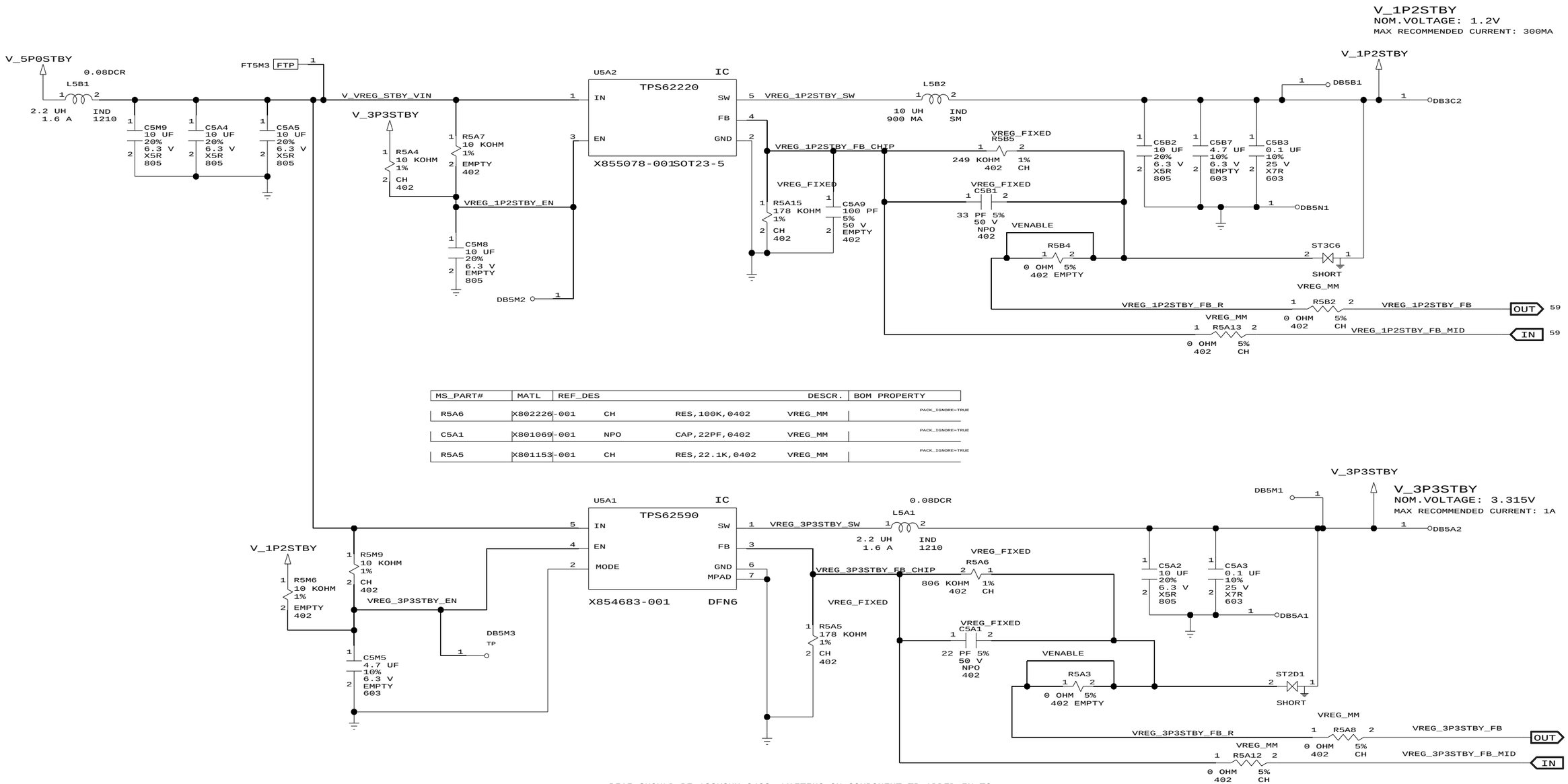

[PAGE_TITLE=VREGS, 1P8+GPUPCIE+SBPCIE+CPUPLL+EFUSE]

DRAWING
Fri Jan 04 12:07:23 2013

| MICROSOFT CONFIDENTIAL | PROJECT NAME STINGRAY | PAGE 52/74 | CSA PAGE 52/74 | FAB C | REV 1.0 |
|---|---|---|---|---|---|

# VREGS, STANDBY SWITCHERS

| MS_PART# | MATL | REF_DES | | DESCR. | BOM PROPERTY |
|---|---|---|---|---|---|
| R5B5 | X855279-001 | CH | RES,42.2K,0402 | VREG_MM | PACK_IGNORE=TRUE |
| C5B1 | X800440-001 | NPO | CAP,75PF,0402 | VREG_MM | PACK_IGNORE=TRUE |
| R5A15 | X800467-001 | CH | RES,30.1K,0402 | VREG_MM | PACK_IGNORE=TRUE |




| MS_PART# | MATL | REF_DES | | DESCR. | BOM PROPERTY |
|---|---|---|---|---|---|
| R5A6 | X802226-001 | CH | RES,100K,0402 | VREG_MM | PACK_IGNORE=TRUE |
| C5A1 | X801069-001 | NPO | CAP,22PF,0402 | VREG_MM | PACK_IGNORE=TRUE |
| R5A5 | X801153-001 | CH | RES,22.1K,0402 | VREG_MM | PACK_IGNORE=TRUE |




R5A5 SHOULD BE 182KOHM 0402, WAITING ON COMPONENT TB ADDED IN TC

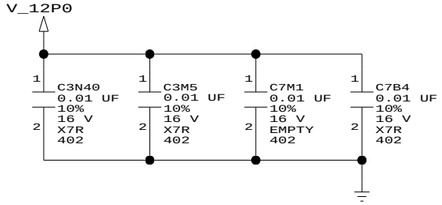
[PAGE_TITLE=VREGS, STANDBY SWITCHERS]

DRAWING
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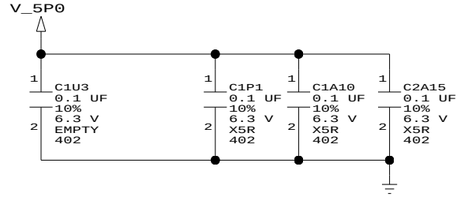
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 53/74 | 53/74 | C | 1.0 |

# BOARD LEVEL DECOUPLING

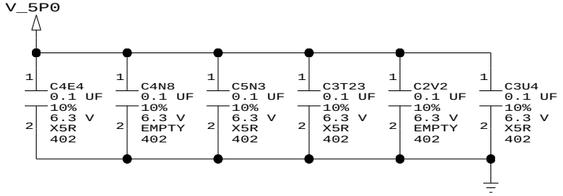
V_12P0



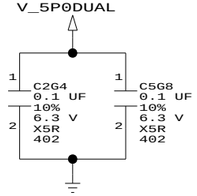

V_5P0



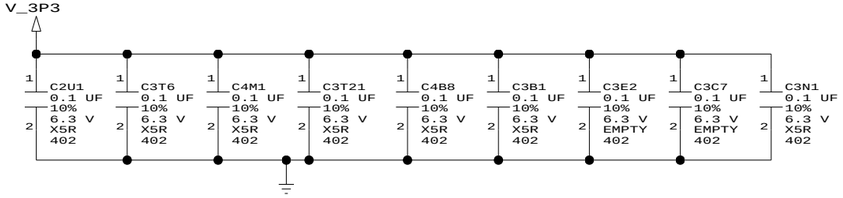

V_5P0



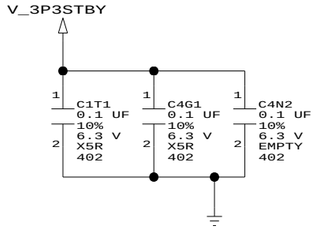

V_5P0DUAL



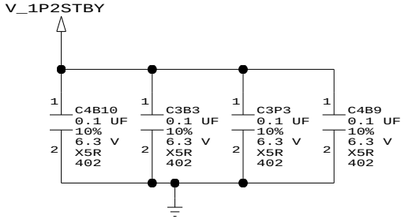

V_3P3



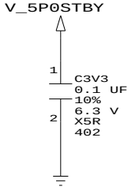

V_3P3STBY



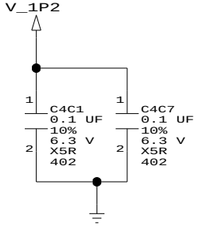

V_1P2STBY




V_5P0STBY




V_1P2

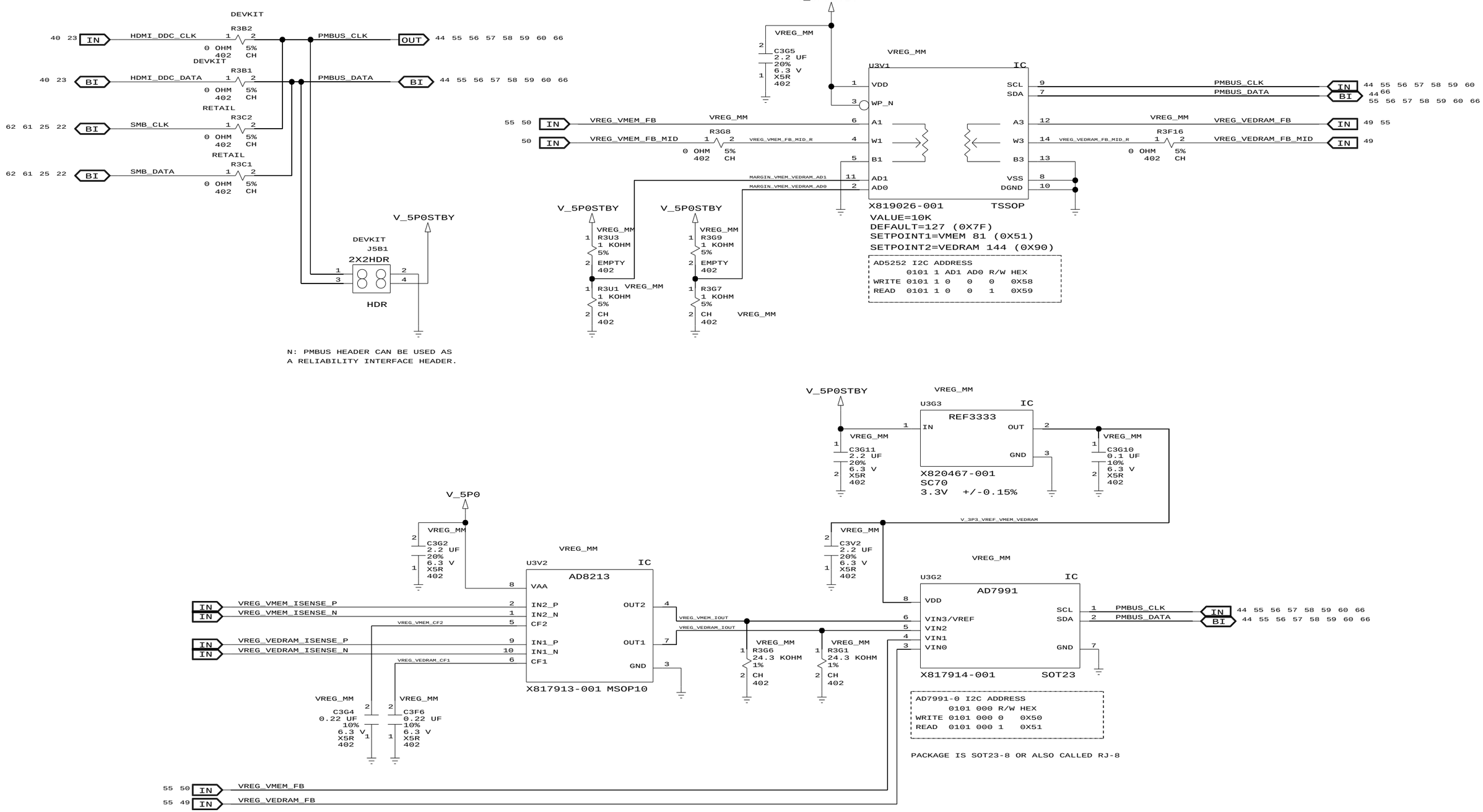



[PAGE_TITLE=BOARD, DECOUPLE]

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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 54/74 | 54/74 | C | 1.0 |

# MARGIN, VMEM + VEDRAM

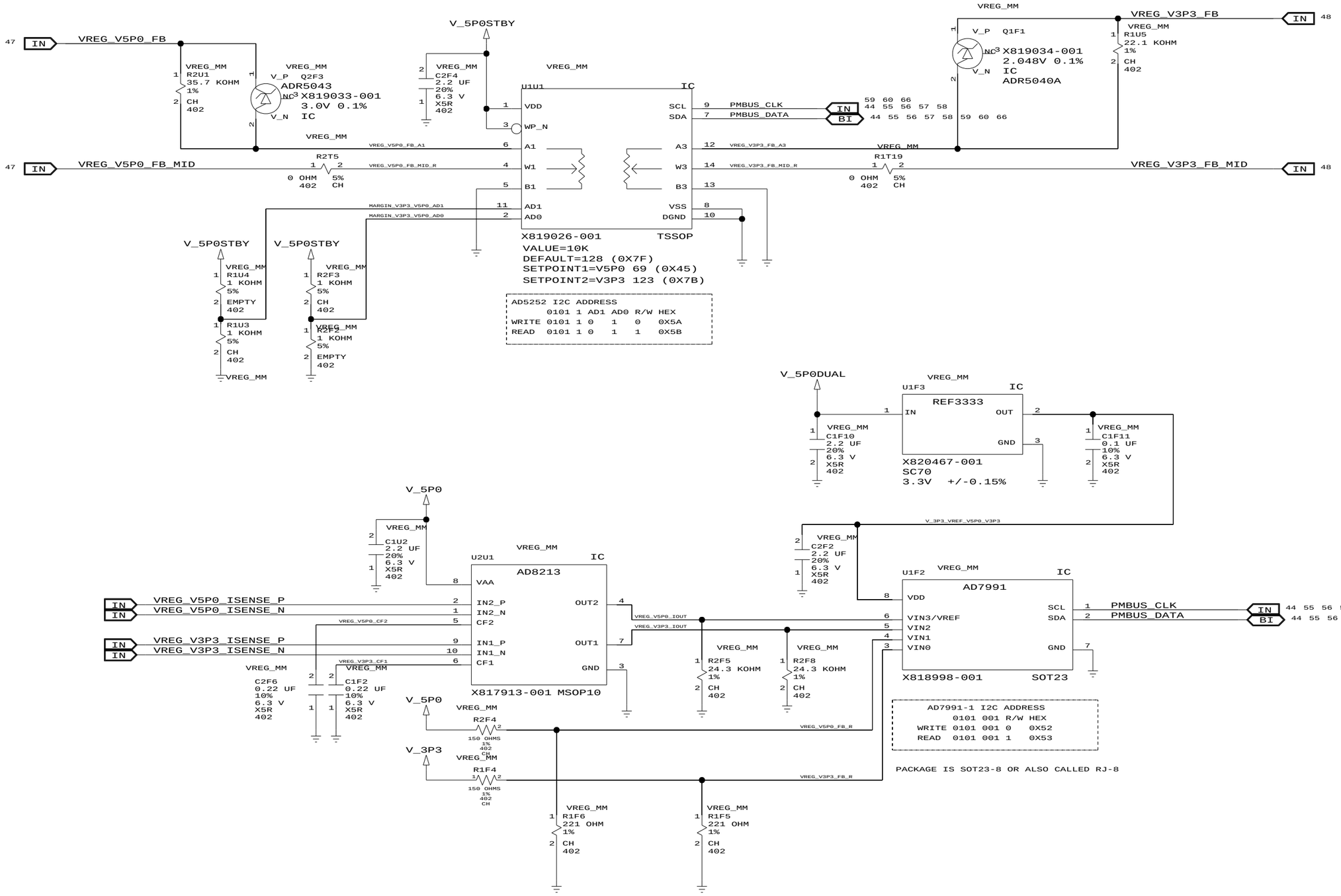



[PAGE_TITLE=MARGIN, VMEM + VEDRAM]

DRAWING
Fri Jan 04 12:07:24 2008

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 55/74 | 55/74 | C | 1.0 |

# MARGIN, V3P3 + V5P0

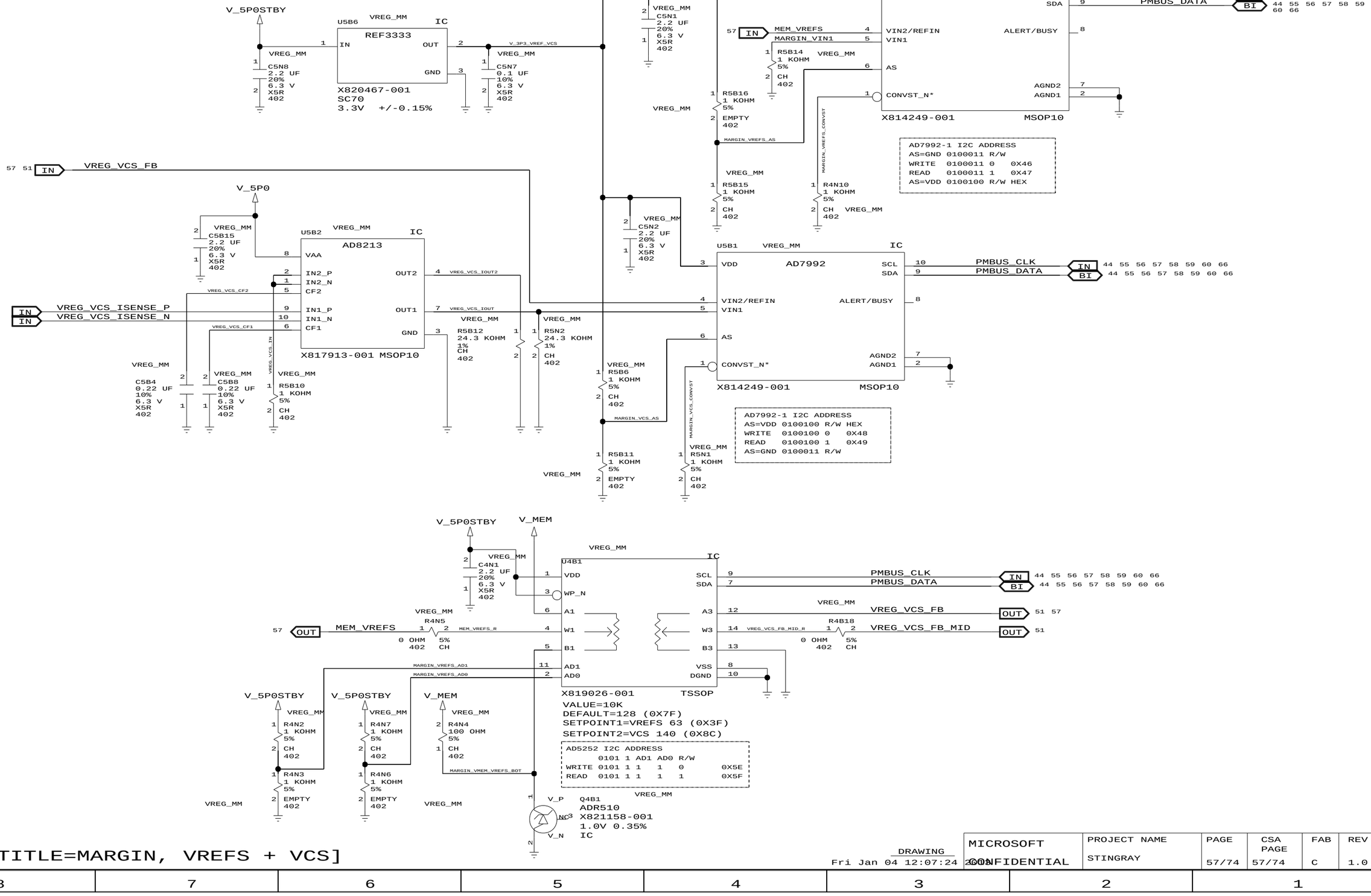



[PAGE_TITLE=MARGIN, V3P3 + VP50]

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Fri Jan 04 12:07:24 2008

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 56/74 | 56/74 | C | 1.0 |

# MARGIN, VREFS + VCS

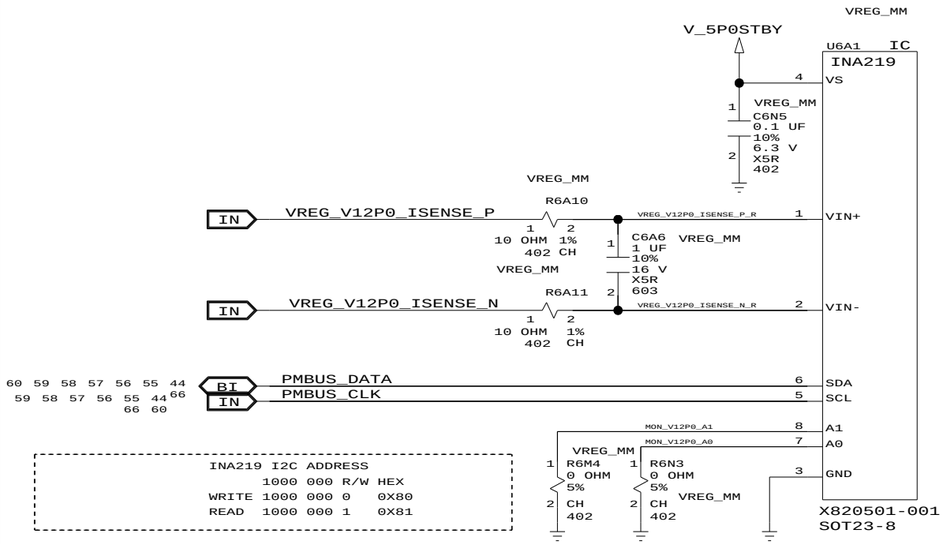


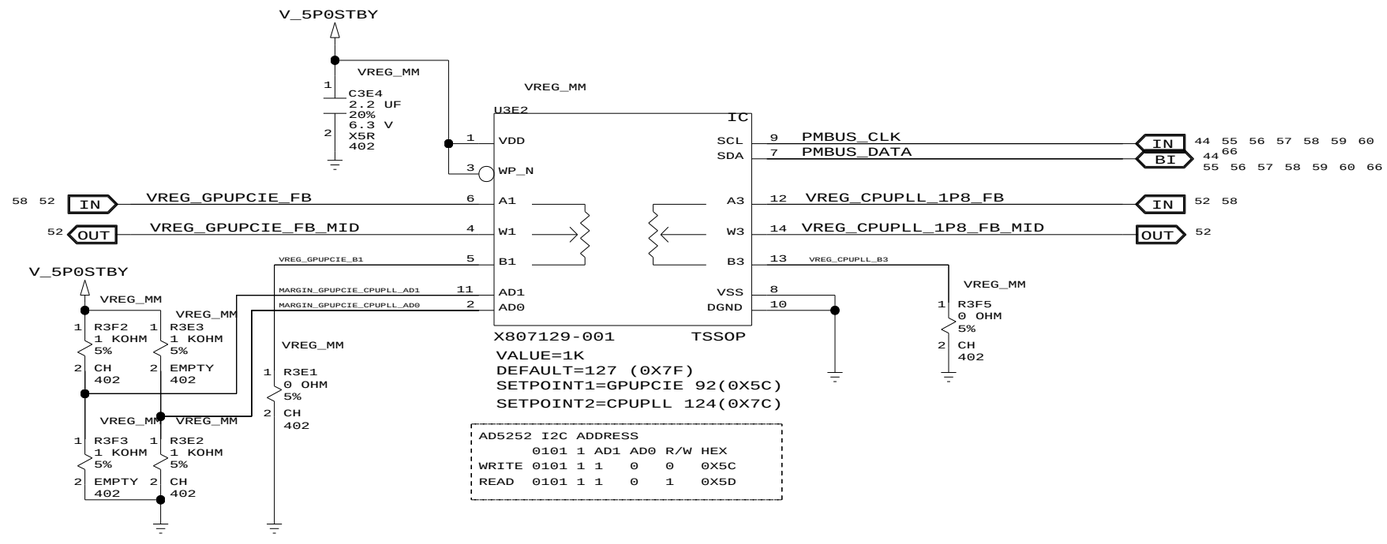

[PAGE_TITLE=MARGIN, VREFS + VCS]

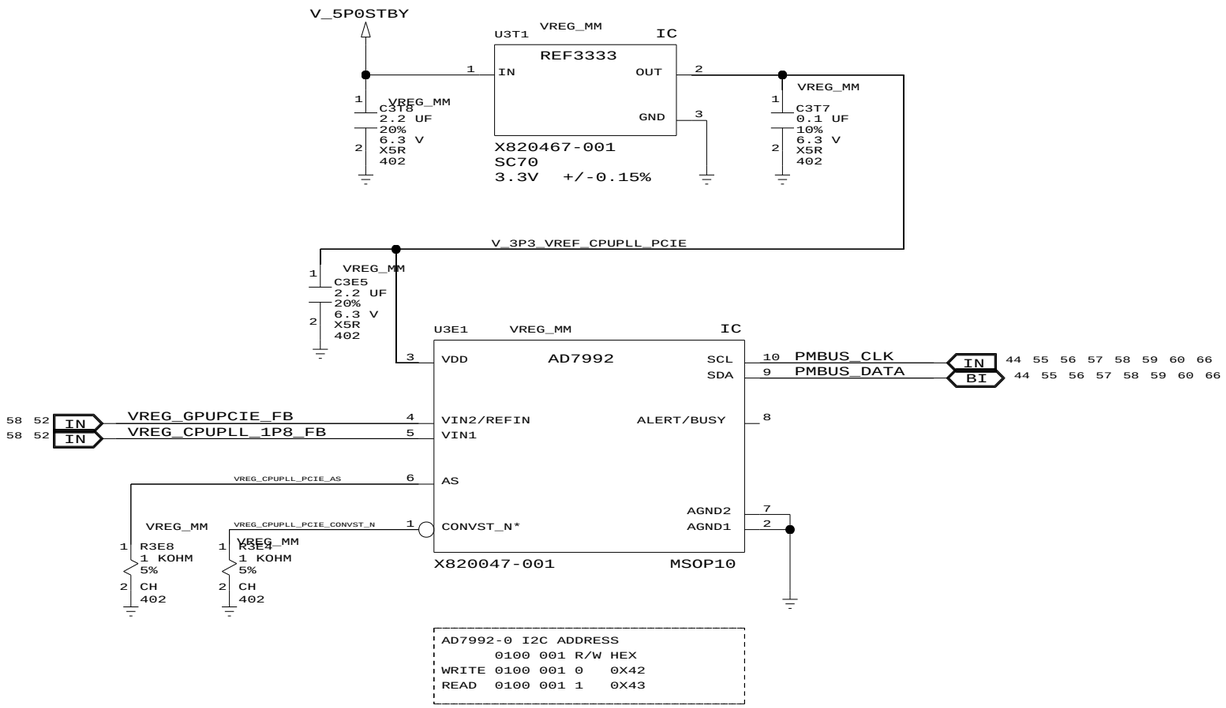
DRAWING
Fri Jan 04 12:07:24 2008

| MICROSOFT CONFIDENTIAL | PROJECT NAME STINGRAY | PAGE 57/74 | CSA PAGE 57/74 | FAB C | REV 1.0 |
|---|---|---|---|---|---|

# MARGIN, VGPUPCIE,1P8, VCPUPLL, V12P0, TEMP



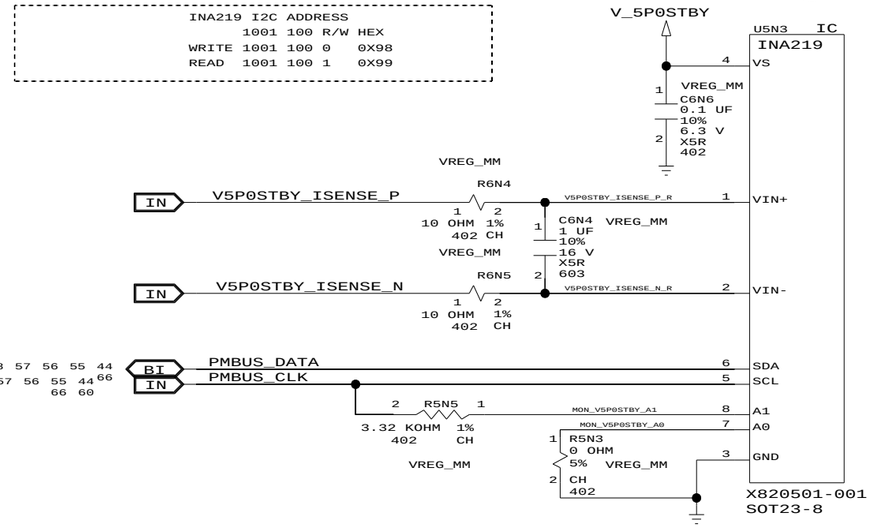



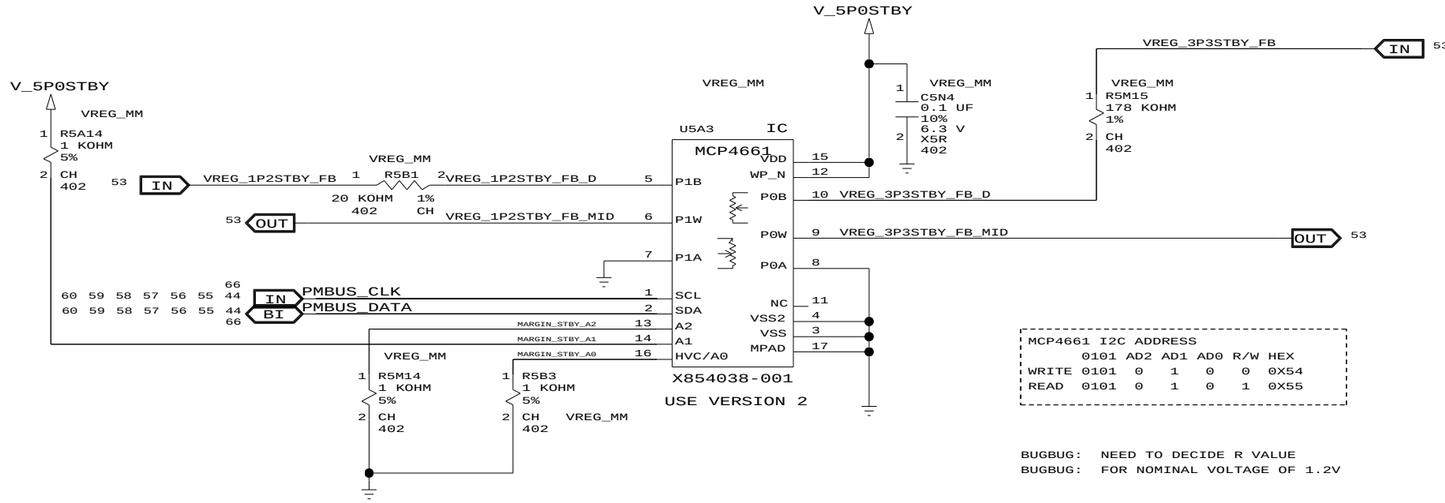

[PAGE_TITLE=MARGIN, VGPUPCIE, VSBPCIE, VCPUPLL, V12P0, TEMP]

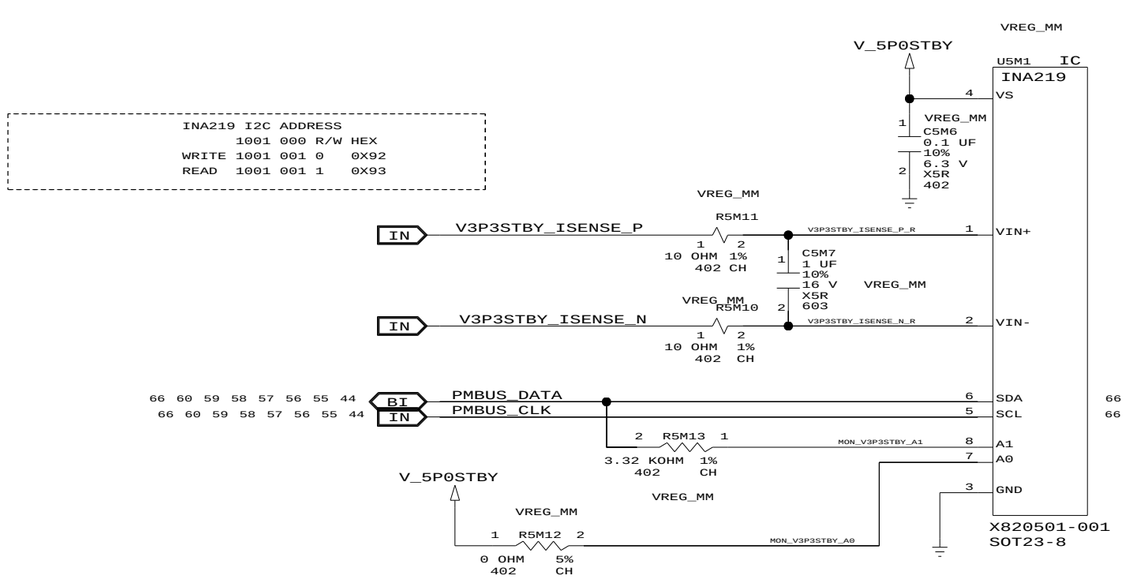
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Fri Jan 04 12:07:24 2008

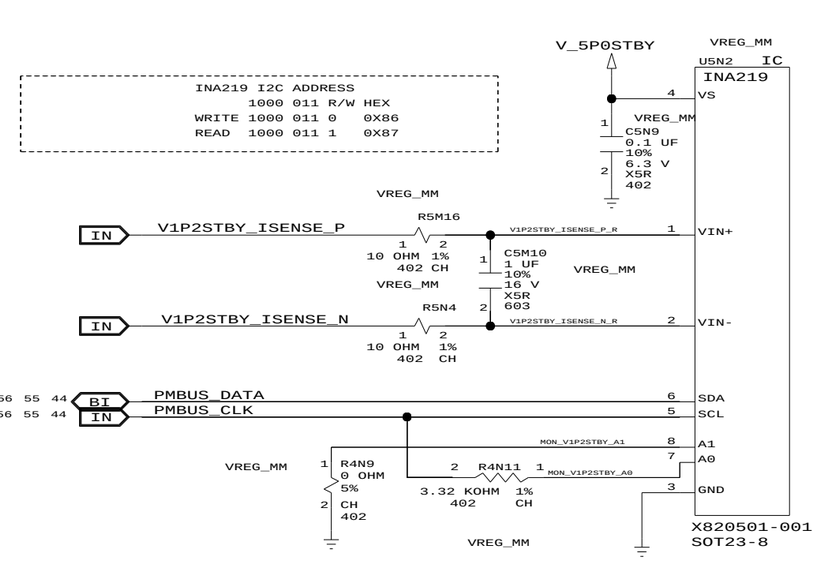
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 58/74 | 58/74 | C | 1.0 |

# MARGIN, STANDBY



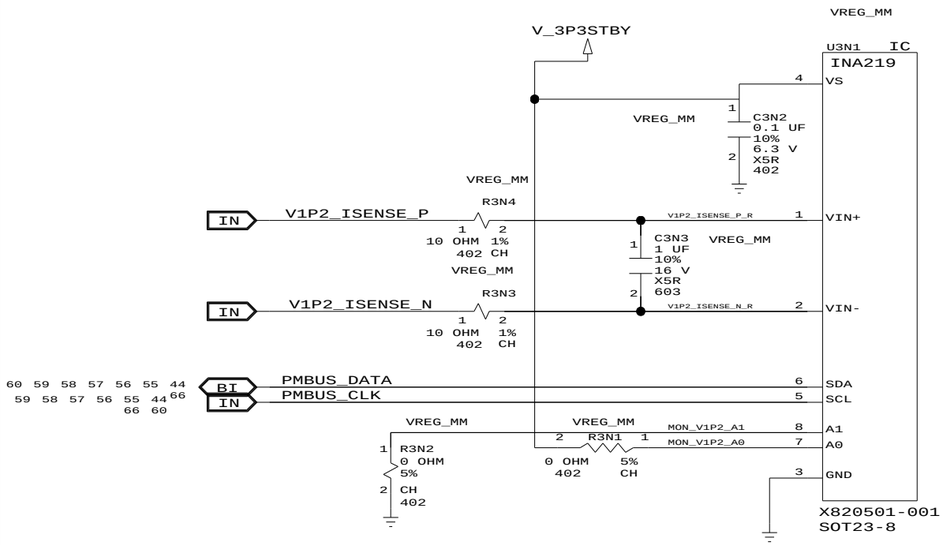




[PAGE_TITLE=MARGIN, VGPUPCIE, VSBPCIE, VCPUPLL, V12P0, TEMP]

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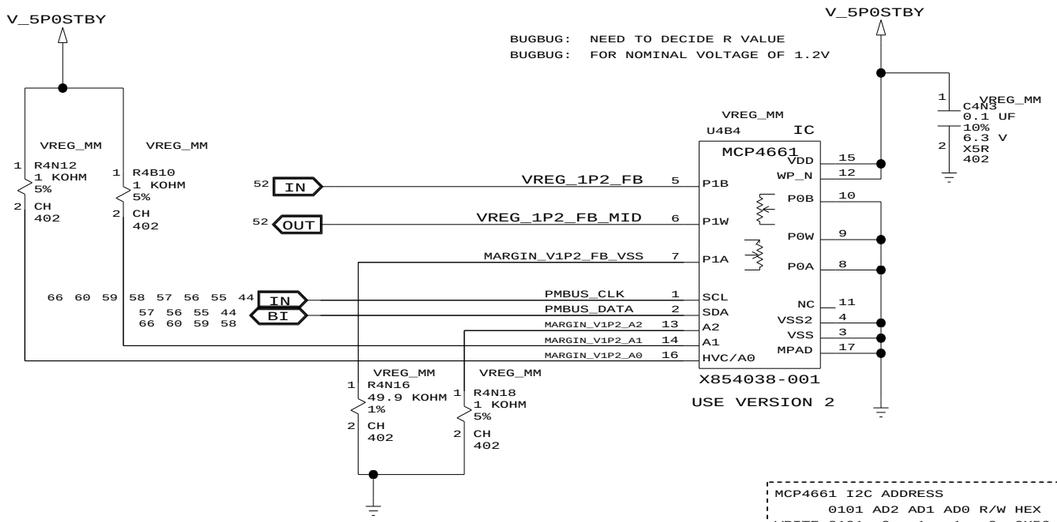
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 59/74 | 59/74 | C | 1.0 |

# MARGIN, V1P2




INA219 I2C ADDRESS

| | 1000 | 100 | R/W | HEX |
|---|---|---|---|---|
| WRITE | 1000 | 001 | 0 | 0X82 |
| READ | 1000 | 001 | 1 | 0X83 |




MCP4661 I2C ADDRESS

| | 0101 | AD2 | AD1 | AD0 | R/W | HEX |
|---|---|---|---|---|---|---|
| WRITE | 0101 | 0 | 1 | 1 | 0 | 0X56 |
| READ | 0101 | 0 | 1 | 1 | 1 | 0X57 |

[PAGE_TITLE=MARGIN, VGPUPCIE, VSBPCIE, VCPUPLL, V12P0, TEMP]

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| PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|
| STINGRAY | 60/74 | 60/74 | C | 1.0 |

# EXTERNAL TEMP SENSORS

CPU_TEMP_P — R4D7 (CPU_TEMP_P_KSB) — R4R28 (CPU_TEMP_P_EXT) — C4D3 100 PF 5% 50 V EMPTY 402
CPU_TEMP_N — R4D8 (CPU_TEMP_N_KSB) — R4R29 (CPU_TEMP_N_EXT) — JTM_INT / JTM_EXT
GPU_TEMP_P — R4D6 (GPU_TEMP_P_KSB) — R4R27 (GPU_TEMP_P_EXT) — C4D2 100 PF 5% 50 V EMPTY 402
GPU_TEMP_N — R4D5 (GPU_TEMP_N_KSB) — R4R26 (GPU_TEMP_N_EXT)
BRD_TEMP_P — R3R18 (BRD_TEMP_P_KSB) — R3D30 (BRD_TEMP_P_EXT) — 402 EMPTY 50 V 5% 100 PF C3R56
BRD_TEMP_N — R3D31 (BRD_TEMP_N_EXT)
EDRAM_TEMP_P — R4D3 (EDRAM_TEMP_P_KSB) — R4R24 (EDRAM_TEMP_P_EXT) — C4D1 100 PF 5% 50 V EMPTY 402
EDRAM_TEMP_N — R4D4 (EDRAM_TEMP_N_KSB) — R4R25 (EDRAM_TEMP_N_EXT)

All: 0 OHM 5% 402 CH; JTM_INT / JTM_EXT

ST4R1 SHORT
ST4D1 SHORT
ST4R2 SHORT

CPU_TEMP_P_EXT — C4R4 100 PF 5% 50 V EMPTY 402 — PLACE CLOSE TO SENSOR
GPU_TEMP_P_EXT — C4R5 100 PF 5% 50 V EMPTY 402 — PLACE CLOSE TO SENSOR
BRD_TEMP_P_EXT — C4R1 100 PF 5% 50 V EMPTY 402 — PLACE CLOSE TO SENSOR

EXT_TEMP_N — JTM_EXT R4R35 0 OHM 5% CH 402

ISOLATE CLOSE TO KSB

SMB_CLK — R3P7 — SMB_CLK_EXTTEMP_R — R3R13 — SMB_CLK_EXTTEMP
SMB_DATA — R3P6 — SMB_DATA_EXTTEMP_R — R3R14 — SMB_DATA_EXTTEMP
All: 0 OHM 5% 402 CH JTM_EXT

ISOLATE CLOSE TO EXT TEMP

V_EXTTEMP

U4R1 IC TMP423 — JTM_EXT C4R2 0.1 UF 10% 6.3 V X5R 402
DXP1, DXP2, DXP3, DXN; V+, SCL, SDA, GND
SMB_CLK_EXTTEMP, SMB_DATA_EXTTEMP
X858059-001 SOT23-8

TMP423B I2C ADDRESS
1001 100 R/W HEX
WRITE 1001 101 0 0X9A
READ 1001 101 1 0X9B

V_EXTTEMP

Q3R1 EMPTY 1.1 OHM SOT23 — R3R20 10 KOHM 1% EMPTY 402 — R4R36 0 OHM 5% CH 402 — FTP FT3R18
EXT_TEMP_PWR_CTRL

V_3P3STBY

V_3P3STBY

JTM_EXT C3D10 0.1 UF 10% 6.3 V X5R 402 — JTM_EXT C3D11 0.1 UF 10% 6.3 V X5R 402

STITCHING CAPS FOR BRD_TEMP PLANE SPLIT

V_EXTTEMP

U3R1 IC TMP441 — JTM_EXT C3D9 0.1 UF 10% 6.3 V X5R 402
DXP, DXN, A1, A0; V+, SCL, SDA, GND
SMB_CLK_EXTTEMP, SMB_DATA_EXTTEMP
X858003-001 SOT23-8

EDRAM_TEMP_P_EXT — C4R3 100 PF 5% 50 V EMPTY 402 — PLACE CLOSE TO SENSOR
EXT_TMP_A1
EXT_TMP_A0

R4R34 0 OHM 5% EMPTY 402
JTM_EXT R4R31 0 OHM 5% CH 402
R4R30 0 OHM 5% EMPTY 402
R4R33 0 OHM 5% EMPTY 402
R4R32 0 OHM 5% CH 402

TMP441 I2C ADDRESS
10011 AD1 AD0 R/W HEX
WRITE 10011 1 0 0 0X9C
READ 10011 1 0 1 0X9D

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 61/74 | 61/74 | C | 1.0 |

# XDK, DEBUG CONNECTORS

KSB DEBUG BUS

J2C2 EMPTY

2X13SOFTTOUCH

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 24 IN | FAN_TACH_SMM_R | A1 | A_D1 | B_D0 | B1 | WSS_CNTL1 | IN 24 37 | |
| 37 24 IN | WSS_CNTL0 | A2 | A_D3 | B_D2 | B2 | SCART_RGB | IN 24 37 | |
| | | A3 | A_GND1 | B_GND0 | B3 | | | |
| 24 IN | SB_GPIO<4> | A4 | A_D5 | B_D4 | B4 | SB_GPIO<1> | IN 24 | |
| | | A5 | A_D7 | B_D6 | B5 | SB_GPIO<0> | IN 24 | |
| | | A6 | A_GND3 | B_GND2 | B6 | | | |
| 24 IN | SB_GPIO<2> | A7 | A_D9 | B_D8 | B7 | SB_GPIO<3> | IN 24 | |
| 24 IN | AUD_SPI_CLK_R | A8 | A_D11 | B_D10 | B8 | AUD_SPI_MOSI_R | IN 24 | |
| | | A9 | A_GND5 | B_GND4 | B9 | | | |
| 24 IN | AUD_RDY_BSBY_R | A10 | A_D13 | B_D12 | B10 | SB_GPIO<5> | IN 24 | |
| 24 IN | AUD_SSB_R | A11 | A_D15 | B_D14 | B11 | AUD_SPI_MISO_R | IN 24 | |
| | | A12 | A_GND7 | B_GND6 | B12 | | | |
| | | A13 | A_NC | B_CLK | B13 | | | |
| | | 27 | MH1 | MH3 | 29 | | | |
| | | 28 | MH2 | MH4 | 30 | | | |

X854421-001

## PHY MDIO INTERFACE

| NET NAME | MDIO SIGNAL |
|---|---|
| SB_GPIO<1> | PCIE MDC |
| SB_GPIO<0> | PCIE MDIO |
| SB_GPIO<2> | FPHY MDIO |
| SB_GPIO<3> | FPHY MDC |
| SB_GPIO<5> | SATA0 MDC |
| SB_GPIO<4> | SATA0 MDIO |
| AUD_SPI_MOSI_R | SATA1 MDC |
| AUD_SPI_CLK_R | SATA1 MDIO |
| WSS_CNTL0 | USGBRPA MDC |
| WSS_CNTL1 | USGBRPA MDIO |
| AUD_RDY_BSBY_R | USGBRPB MDC |
| FAN_TACH | USGBRPB MDIO |

26 OUT SPI_SS_N — R2C28 0 OHM 5% 402 CH — SPI_SS_N_CONN — DEVKIT J2C1 2X5HDR10 HDR

SPI_MOSI_CONN — R2C29 0 OHM 5% 402 CH — SPI_MOSI OUT 26

26 IN SPI_MISO — 4 — SPI_CLK_CONN — R2C30 0 OHM 5% 402 CH — SPI_CLK OUT 26

V_5P0STBY

DEVKIT C2C3 0.1 UF 10% 6.3 V X5R 402

V_3P3

DEVKIT C4A12 0.1 UF 10% 6.3 V X5R 402

DEVKIT J2C3 2X7HDR14 HDR

24 IN KER_DBG_TXD — 2
25 IN SMC_DBG_TXD — 4
25 OUT SMC_DBG_EN — 6
8
SMB_CLK_R — 10
12

KER_DBG_RXD OUT 24

SMC_RST_XDK_N — R2C10 100 OHM 5% 402 CH — SMC_RST_N IN 22 25 42

V_3P3STBY

DEVKIT C2C4 0.1 UF 10% 6.3 V X5R 402

DEVKIT C2B4 1 UF 10% 16 V X5R 603

61 55 25 22 BI SMB_CLK — DEVKIT R2C6 100 OHM 5% 402 CH — 2

SMB_DATA_R — DEVKIT R2C7 100 OHM 5% 402 CH — SMB_DATA BI 22 25 55 61

V_5P0STBY

DEVKIT C2C5 0.1 UF 10% 6.3 V X5R 402

DEVKIT C2C6 1 UF 10% 16 V X5R 603

EXT_PWR_ON_DBG — DEVKIT R2C4 1 KOHM 5% 402 CH — EXT_PWR_ON_N OUT 25 37 65 66

[PAGE_TITLE=XDK. DEBUG CONN]

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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 62/74 | 62/74 | C | 1.0 |

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| MICROSOFT CONFIDENTIAL | PROJECT NAME STINGRAY | PAGE 63/74 | CSA PAGE 63/74 | FAB C | REV 1.0 |
|---|---|---|---|---|---|

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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 64/74 | 64/74 | C | 1.0 |

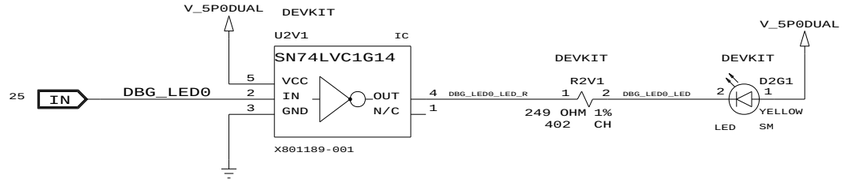
# XDK DEBUG

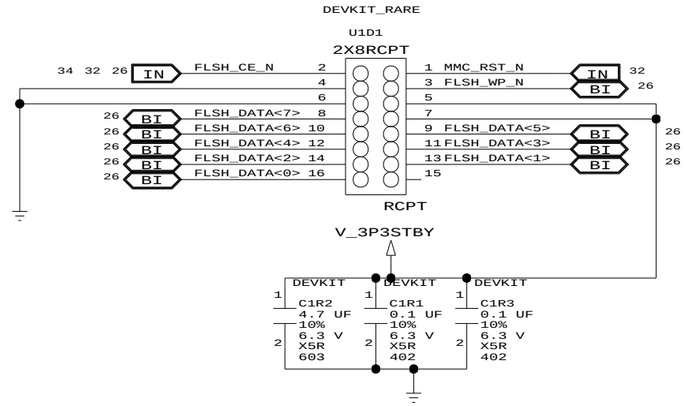


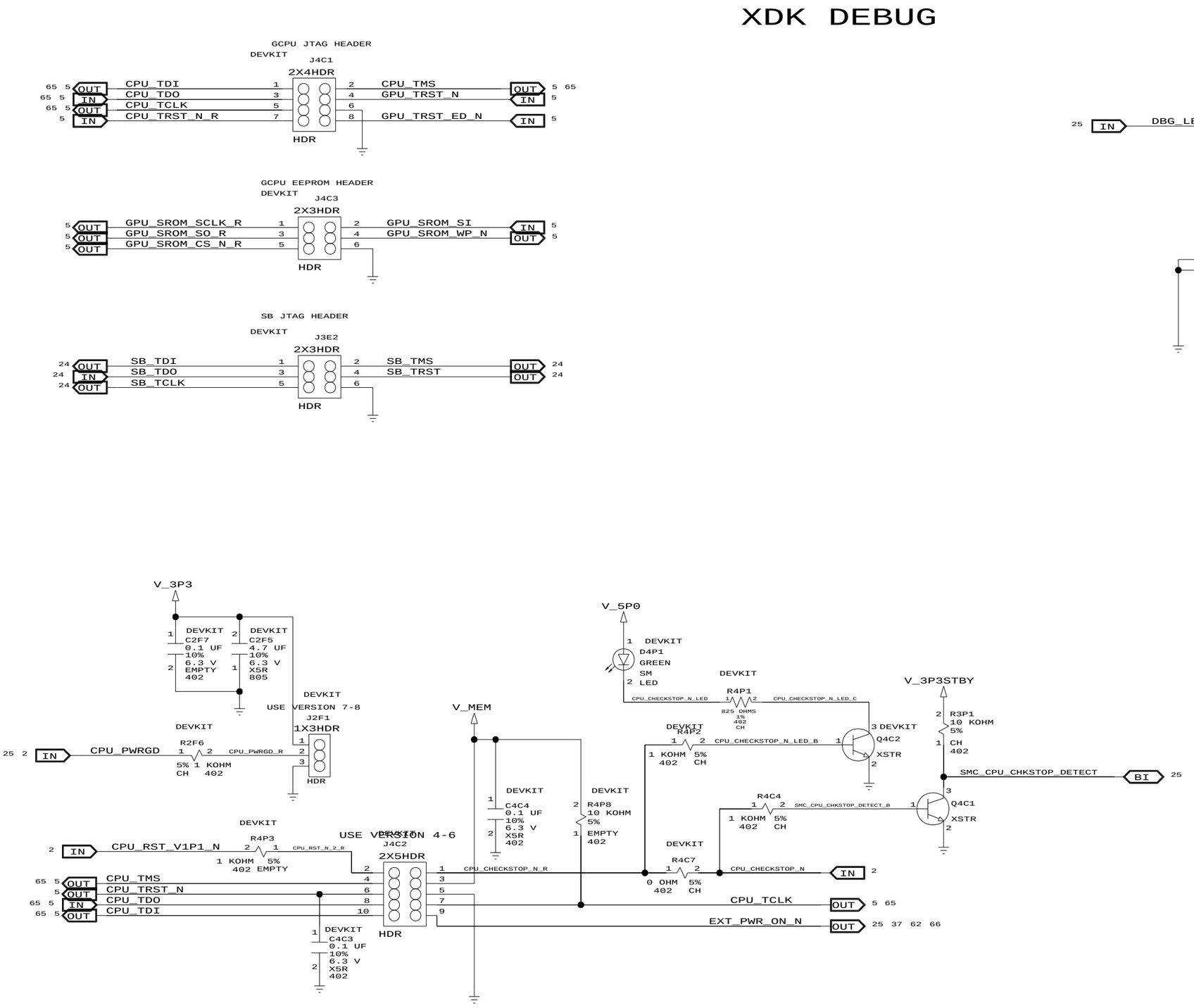


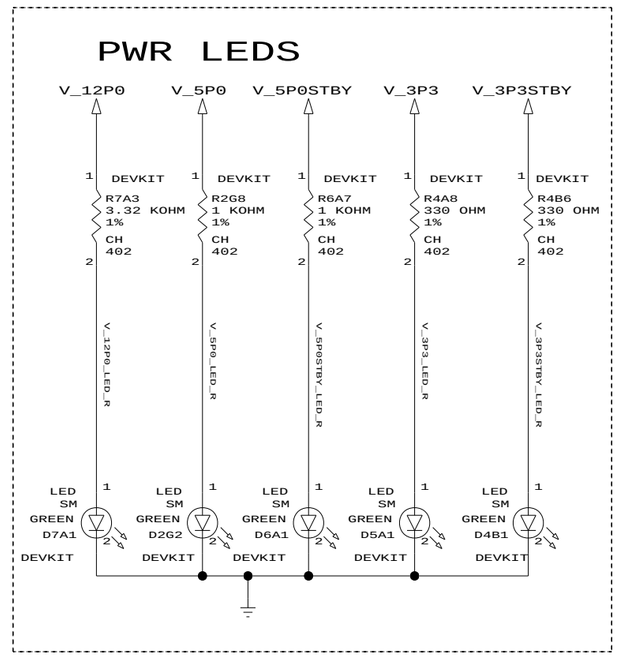




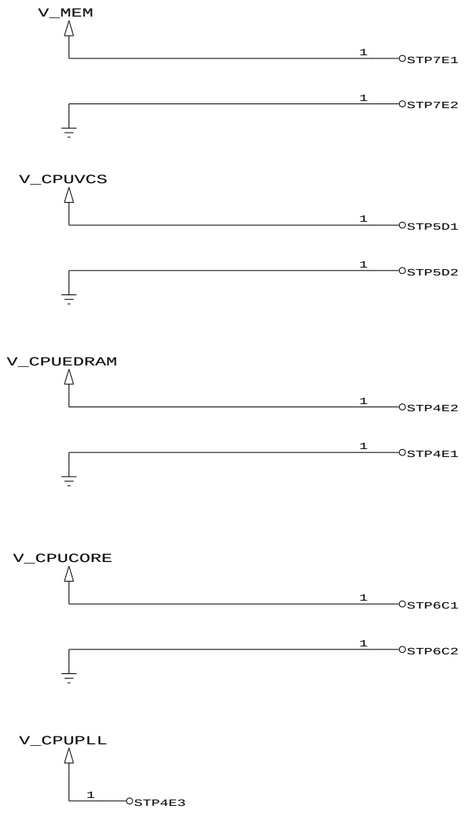

[PAGE_TITLE=XDK, DEBUG TITAN]

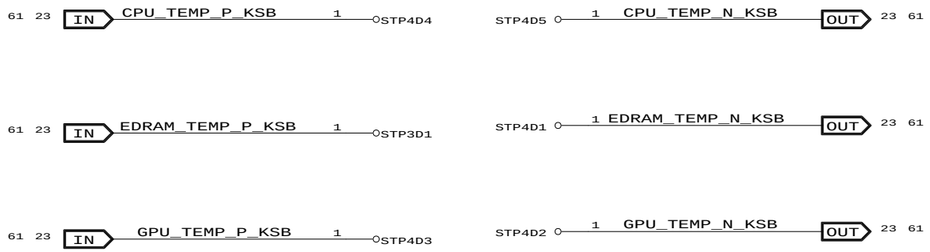
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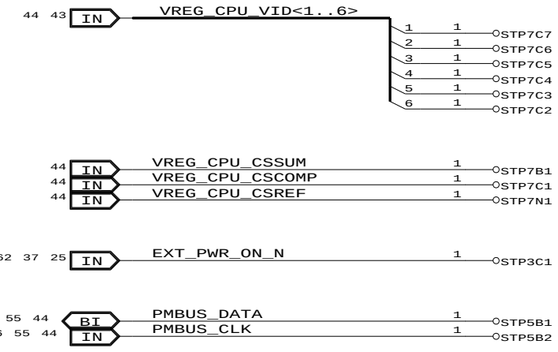
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 65/74 | 65/74 | C | 1.0 |

# DEBUG BOARD, SPYDER CONN

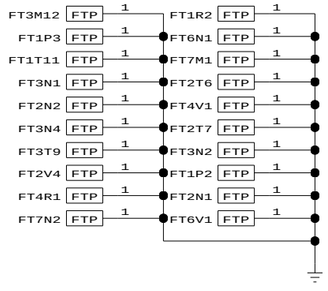
ALL STP POINTS SHALL BE ADDED TO TOP SIDE IN LAYOUT




GND FTS FOR TEST FIXTURES



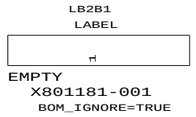

[PAGE_TITLE=DEBUG BOARD, SPYDER CONN]

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Fri Jan 04 12:07:26 2008

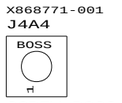
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 66/74 | 66/74 | C | 1.0 |

# LABELS AND MOUNTING

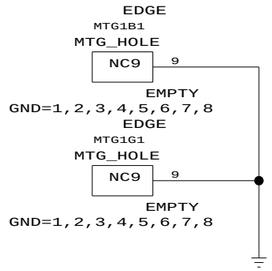
INTELLIGENT SERIAL NUMBER TARGET.

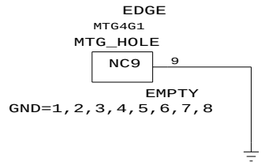



| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| LB2B1 | X863442-001 137SX250_TARGET | LABEL,WAITSBURG | TEST | PACK_IGNORE=TRUE |

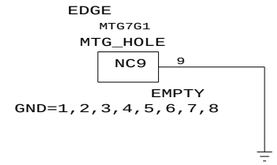



WEST PCB MOUNTING HOLES

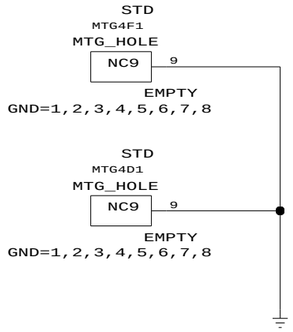


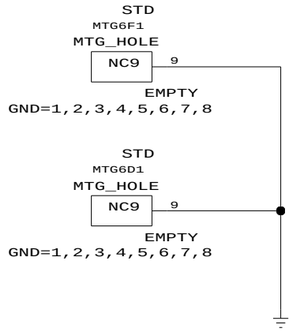




EAST PCB MOUNTING HOLES




HEAT SINK MOUNTING HOLES




[PAGE_TITLE=LABELS AND MOUNTING]

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|---|---|---|---|---|
| STINGRAY | 67/74 | 67/74 | C | 1.0 |

# POWER ARCHITECTURE

+12V
I=10.83A
P=130W

L1
0.4W
450nH/8A
Iin_max 5.8A
0.4W
2x470uF/
16V
Poly
10mOhm
4x4.7uF
1x1uF
MLCC

2.9A
1.62W
DPAK
1.38W
DPAK
1.15W
420nH/dual
Isat 40A
250KHz/ph
L2
1.38W
DPAK
0.12W
6x820uF/
2.5V
Poly
7mOhm

V_CPUCORE
0.9V-1.2V
43W TD/48A TD
51.6W Max/57A Max
54A TD
Min load 20%
η=82%

2.9A
1.62W
DPAK
1.15W
L2
1.38W
DPAK
1.38W
DPAK

V_5P0
5.09V,
43.3W TD/ 8.1A TD
47.6W Max/ 8.9 A
Max
Min load 50%
η=93%

L1
0.26W
450nH/8A
Iin_max 5.51A
0.18W
1x470uF/
16V
Poly
10mohm
1x4.7uF
1x0.1uF
MLCC
4.6A
1.4W
DPAK
1.4W
1.6uH 600KHz
Ipk 13A
L3
1.3W*
(1.05W)
DPAK
0.25W
820uF/
6.3V
E-Al
11mOhm

+5POSTBY
I=1.0A
P=5W

0.09W
2.2uH
Iin_max 0.91A
V_5P0DUAL
1.5A TD
8.4W Max
DUALSO8
7W
3x10uF/
6.3V
MLCC
8mOhm
1x0.1uF/
16V
MLCC
xmOhm

V_EFUSE
0.02A TD
0.03W TD
V_EFUSE(1.5V)
Linear SOT23
0.08W Max

0.06A
SOT23-5
0.24W
0.1W
2.2uH/
1.7MHz
Isat 1.3A
1x10uF/
6.3V
MLCC
8mOhm

V_1P2STBY
0.20W TD/0.15A TD
0.24W Max/0.20mA Max
η=85%

Measured data shows
v_3p3 may be 330mA
less than budget

0.66A
SOT23-5
0.4W
0.08W
2.2uH/
1.7MHz
Isat 0.9A
1x10uF/
6.3V
MLCC
8mOhm

V_3P3STBY
2.1W TD/0.64A TD
2.64W Max/0.80A Max
η=85%

L1
0.12W
450nH/8A
Iin_max 3.4A
3x4.7uF/
6.3V
MLCC
11mOhm
2x1uF/
6.3V
MLCC

0.5W
DPAK
1.21W*
(0.81W)
DPAK
L3
1.4W
1.7uH/
400KHz
Ipk 12.4A
820uF/
2.5V
Poly
7mOhm

V_VCS
1.25-1.3V
12W TD/9.6A TD
13.2W Max/10.6A Max
η=82%

1.0 Ohm/
0.0625W
each, 0402

V_1P2
Linear DPAK
1.08W MAX
V_1P2
0.76A TD
0.9W TD

FB

0.91A
0.13W
2.3uH/600KHz
Isat 5.6A
L4
DUALSO8
1W
4x4.7uF/
6.5V
MLCC
8mohm

V_3P3
3.3V
2.88A TD
9.5W TD
10.56W Max
Min load 50%
η=88%

V_CPUPLL
0.45A TD
0.83W TD
V_1P83PLL
Linear DPAK
1.01W Max(loss)
V_1P83 Main
0.031A TD
0.057W TD

V_GPUPCIE
0.11A TD
0.165W TD
V_GPUPCIE(VTT)
(1.5V)
Linear SOT223
0.22W Max(loss)

**MICROSOFT CONFIDENTIAL**
**CORONA XDK MB POWER ARCHITECTURE**
**MAX PWR4 DISSIPATION(+10% LOAD)**
**JAN 12, 2010**

**Legend**
- Power rail from Switching Regulator
- Power rail from External PSU
- Power rail from Linear Regulator

0.91W*
*=using same FET like HS

Expected
Airflow=0.3m/s

L1
0.16W
450nH/8A
Iin_max 3.1A
0.19W
1x470uF/
16V
Poly
10mohm
1x330uF/
16V
Poly
10mohm
4x4.7uF
1x1uF
MLCC

1.25A
0.98W
DPAK
1.5W*
(1W)
DPAK
1.2W
1.7uH/600KHz
Isat 11A
L3
2x820uF/
2.5V
Poly
7mohm
Option
1x820uF/
2.5V
Poly
7mohm

V_EDRAM
1.075V
9.7W TD/9A TD
10.6W Max/9.9A Max
Min load 50%
η=88%

1.9A
1W
DPAK
1.4W*
(1W)
DPAK
1.2W
1.7uH/600KHz
Isat 11.5A
L3
1x820uF/
2.5V
Poly
7mohm
Option
2x330uF
OsCon
7mOhm
Option
8x10uF
MLCC
11mOhm

V_MEM
1.8V
16W TD/9A
17.8W Max/10A max
Min load 50%
η=84%

<PAGE_TITLE=POWER ARCHITECTURE>

DRAWING
TRINITY_FAB_A
Fri Jan 04 12:07:26 2013
CONFIDENTIAL

| MICROSOFT | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 68/74 | 68/74 | C | 1.0 |

= No Change from Trinity, Regression Testing Only

= Change/New, Needs Special Attention on Corona

= Alternate Part Stuffing

# Corona System

**Rev 0.55, 4/16/10**

HDMI

TOSLINK

AVIP

TMDS

S/PDIF

DDC

Analog Video

I2S

Audio DACs

SMBus

PHY

HDMI

DACs

PLLs

DVE

JTM

Clocks to CGPU

DVO

Board R-Cal

Thermal diode info 3

PCI-Express Bitrate: 2.5 Gbps

2

2

PCI-Express PHY

RJ-45

PHY

Ethernet

AUX Port

Expansion Port Interface (x3)

USB 2.0

USB 2.0

PHY

EHCI

OHCI

USB Group A

Controller Port Interface (x2)

Spare

USB 2.0

USB 2.0

Wave 802.11n

USB 2.0

USB 2.0

PHY

EHCI

OHCI

USB Group B

IR

Capacitive Touch

FPBus

2

ROL

Radio

Baseband

Boron

5

Audio Chip

Internal Speaker

System Control, DVD Tray

Serial

SMBus

PWM Fan

SMC

GPIOs

SFC (NAND /MMC)

SMM

SATA PHY

SATA

Internal HDD

SATA PHY

SATA

ODD Drive

24

Parallel Debug

Serial

8

8

16 MB NAND System Flash

SMC: 16 kB
Kernel: 1 MB
Config: 256B
Dash: < 7 MB

Phison MMC Controller

4 GB NAND Flash

Host Core

32 kB L1 I$
32 kB L1 D$

Dataflow Cores

32 kB L1 I$
32 kB L1 D$

1 MB Shared L2$

64 kB Boot ROM

3D Core (GPU)

10 MB emDRAM

Production Fuses

CGPU

18

Parallel Debug

JTAG

Serial Debug

POST

5

Debug Input

SPI

SROM

1 Gb, x32

64

1 Gb, x32

64

110 / 220 AC

External PSU

12V @ 130W
5V @ 5W
62% eff @ 1/2W

## Power Delivery

V_CPUCORE 55.7A @ APSV
V_CPUEDRAM 7.3A @ 1.075V
V_CPUVCS 10.3A @ 1.25V
V_MEM 6A @ 1.8V
V_5P0 7A @ 5.09V
V_3P3 3.2A @ 3.3V
V_CPUPLL 0.6A @ 1.8V
V_3P3STBY 0.8A @ 3.3V
V_1P2STBY 0.2A @ 1.2V
V_EFUSE 0.020A @ 1.5V
V_1P2 0.9A @ 1.2V
V_GPUPCIE 0.2A @ 1.5V

## Software

SMC

HWInit

Kernel

## Mechanicals

Vejle HS+Fan

Minor tweaks

Plastics/ Ind Des

KSB Heatsink

**Value SKU**

16MB Flash will be replaced with 4GB/8GB flash using new MMC interface on KSB.

**KSB**

Process: 0.15u
Die Size: ~43mm^2
Frequency: 150-250 MHz
Core VDD: 1.2V
PD Power: 4W
Signal I/O: 212
Package: 23mm x 23mm 404 pin PBGA

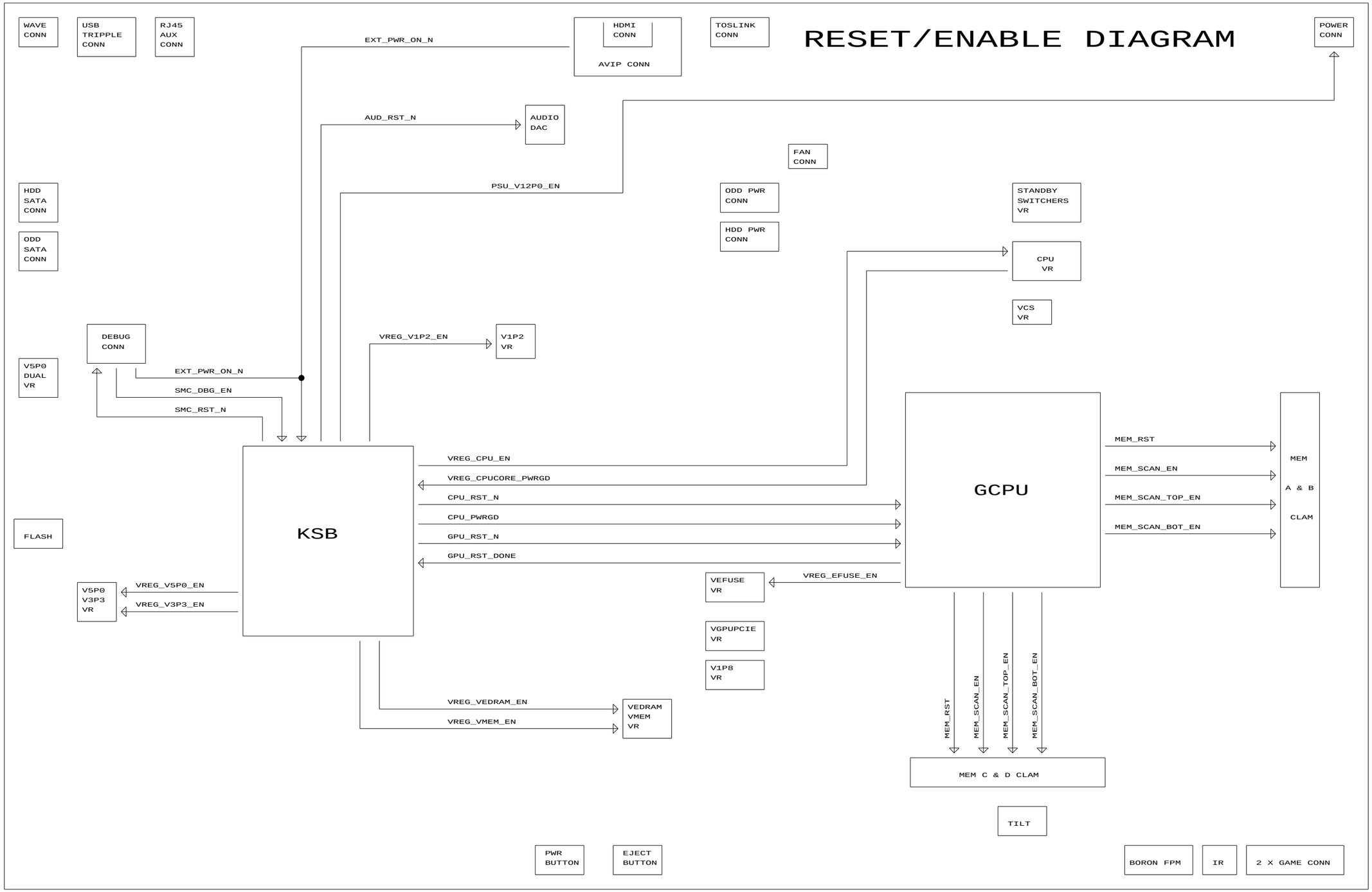
**MS Confidential**

<PAGE_TITLE=SYSTEM BLOCK DIAGRAM>

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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 69/74 | 69/74 | C | 1.0 |

# RESET/ENABLE DIAGRAM




<PAGE_TITLE=CLOCK DIAGRAM>

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| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | STINGRAY | 70/74 | 70/74 | C | 1.0 |

# COMPONENT STUFFING TABLES

| MARGIN REF DES TO STUFF REFDES | DBG/XDK | RETAIL |
|---|---|---|
| C1F2 | STUFF | NO-STUFF |
| C1F10 | STUFF | NO-STUFF |
| C1F11 | STUFF | NO-STUFF |
| C1G7 | STUFF | NO-STUFF |
| C1G8 | STUFF | NO-STUFF |
| C1G9 | STUFF | NO-STUFF |
| C1G10 | STUFF | NO-STUFF |
| C1G11 | STUFF | NO-STUFF |
| C1G12 | STUFF | NO-STUFF |
| C1G13 | STUFF | NO-STUFF |
| C1U2 | STUFF | NO-STUFF |
| C2F2 | STUFF | NO-STUFF |
| C2F4 | STUFF | NO-STUFF |
| C2F6 | STUFF | NO-STUFF |
| C2E4 | STUFF | NO-STUFF |
| C2E5 | STUFF | NO-STUFF |
| C2F6 | STUFF | NO-STUFF |
| C3G11 | STUFF | NO-STUFF |
| C3G2 | STUFF | NO-STUFF |
| C3G4 | STUFF | NO-STUFF |
| C3U2 | STUFF | NO-STUFF |
| C3U3 | STUFF | NO-STUFF |
| C3V2 | STUFF | NO-STUFF |
| C4B1 | STUFF | NO-STUFF |
| C4N1 | STUFF | NO-STUFF |
| C4N4 | STUFF | NO-STUFF |
| C5B15 | STUFF | NO-STUFF |
| C5B5 | STUFF | NO-STUFF |
| C5B8 | STUFF | NO-STUFF |
| C5N7 | STUFF | NO-STUFF |
| C5N8 | STUFF | NO-STUFF |
| C6A6 | STUFF | NO-STUFF |
| C6M5 | STUFF | NO-STUFF |
| J1G3 | STUFF | NO-STUFF |
| Q1F1 | STUFF | NO-STUFF |
| Q2F3 | STUFF | NO-STUFF |
| Q4B1 | STUFF | NO-STUFF |
| R1F4 | STUFF | NO-STUFF |
| R1F5 | STUFF | NO-STUFF |
| R1F6 | STUFF | NO-STUFF |
| R1G1 | STUFF | NO-STUFF |
| R1T19 | STUFF | NO-STUFF |
| R1U3 | STUFF | NO-STUFF |
| R1U4 | STUFF | NO-STUFF |

| MARGIN CONTINUED... REFDES | DBG/XDK | RETAIL |
|---|---|---|
| R1U5 | STUFF | NO-STUFF |
| R2B1 | STUFF | NO-STUFF |
| R2B3 | STUFF | NO-STUFF |
| R2F2 | STUFF | NO-STUFF |
| R2F3 | STUFF | NO-STUFF |
| R2F4 | STUFF | NO-STUFF |
| R2F5 | STUFF | NO-STUFF |
| R2F8 | STUFF | NO-STUFF |
| R2T5 | STUFF | NO-STUFF |
| R2U1 | STUFF | NO-STUFF |
| R3E1 | STUFF | NO-STUFF |
| R3E2 | STUFF | NO-STUFF |
| R3E3 | STUFF | NO-STUFF |
| R3E4 | STUFF | NO-STUFF |
| R3F16 | STUFF | NO-STUFF |
| R3F2 | STUFF | NO-STUFF |
| R3F3 | STUFF | NO-STUFF |
| R3F5 | STUFF | NO-STUFF |
| R3G1 | STUFF | NO-STUFF |
| R3G6 | STUFF | NO-STUFF |
| R3G7 | STUFF | NO-STUFF |
| R3G8 | STUFF | NO-STUFF |
| R3G9 | STUFF | NO-STUFF |
| R3N6 | STUFF | NO-STUFF |
| R3U1 | STUFF | NO-STUFF |
| R3U2 | STUFF | NO-STUFF |
| R3U3 | STUFF | NO-STUFF |
| R4B1 | STUFF | NO-STUFF |
| R4B18 | STUFF | NO-STUFF |
| R4B2 | STUFF | NO-STUFF |
| R4B3 | STUFF | NO-STUFF |
| R4N1 | STUFF | NO-STUFF |
| R4N10 | STUFF | NO-STUFF |
| R4N13 | STUFF | NO-STUFF |
| R4N14 | STUFF | NO-STUFF |
| R4N2 | STUFF | NO-STUFF |
| R4N3 | STUFF | NO-STUFF |
| R4N4 | STUFF | NO-STUFF |
| R4N5 | STUFF | NO-STUFF |
| R4N7 | STUFF | NO-STUFF |
| R4N8 | STUFF | NO-STUFF |
| R5B6 | STUFF | NO-STUFF |
| R5N2 | STUFF | NO-STUFF |
| R6A10 | STUFF | NO-STUFF |

| MARGIN CONTINUED... REFDES | DBG/XDK | RETAIL |
|---|---|---|
| R6A11 | STUFF | NO-STUFF |
| R6M3 | STUFF | NO-STUFF |
| R6M4 | STUFF | NO-STUFF |
| U1F2 | STUFF | NO-STUFF |
| U1F3 | STUFF | NO-STUFF |
| U1G2 | STUFF | NO-STUFF |
| U1U1 | STUFF | NO-STUFF |
| U2U1 | STUFF | NO-STUFF |
| U3E1 | STUFF | NO-STUFF |
| U3E2 | STUFF | NO-STUFF |
| U3G2 | STUFF | NO-STUFF |
| U3G3 | STUFF | NO-STUFF |
| U3T1 | STUFF | NO-STUFF |
| U3V1 | STUFF | NO-STUFF |
| U4B1 | STUFF | NO-STUFF |
| U4B2 | STUFF | NO-STUFF |
| U5B2 | STUFF | NO-STUFF |
| U5B6 | STUFF | NO-STUFF |
| U5N1 | STUFF | NO-STUFF |
| U6A1 | STUFF | NO-STUFF |

PLEASE SEE BOM VARIANTS FOR STUFFING INSTUCTIONS ON V1P2 AND STBY MARGINING

| GDDR MEM. REFDES TO STUFF REFDES | TOP ONLY (4GB) | TOP & BOT (8GB) |
|---|---|---|
| C5F1 | STUFF | STUFF |
| C5U9 | STUFF | STUFF |
| C6F3 | STUFF | STUFF |
| C6U7 | STUFF | STUFF |
| C7D12 | STUFF | STUFF |
| C7E9 | STUFF | STUFF |
| C7R6 | STUFF | STUFF |
| C7T7 | STUFF | STUFF |
| R5F5 | STUFF | STUFF |
| R5T6 | STUFF | STUFF |
| R5T7 | STUFF | STUFF |
| R5U1 | NO-STUFF | STUFF |
| R5U6 | STUFF | STUFF |
| R6F5 | STUFF | STUFF |
| R6R1 | STUFF | STUFF |
| R6R2 | STUFF | STUFF |
| R6T7 | STUFF | STUFF |
| R6T8 | STUFF | STUFF |
| R6T9 | STUFF | STUFF |
| R6U1 | NO-STUFF | STUFF |
| R7E1 | STUFF | STUFF |
| R7E8 | STUFF | STUFF |
| R7T1 | NO-STUFF | STUFF |
| R7T6 | NO-STUFF | STUFF |
| U5F1 | STUFF | STUFF |
| U5U1 | NO-STUFF | STUFF |
| U5U2 | NO-STUFF | STUFF |
| U6F1 | STUFF | STUFF |
| U6U1 | NO-STUFF | STUFF |
| U7D1 | STUFF | STUFF |
| U7E1 | STUFF | STUFF |
| U7T1 | NO-STUFF | STUFF |
| U7T1 | NO-STUFF | STUFF |

| SROM MEMORY REFDES TO STUFF REFDES | DEBUG | CPU DBG | XDK | RETAIL |
|---|---|---|---|---|
| C4R4 | STUFF | STUFF | STUFF | NO-STUFF |
| R4R7 | STUFF | STUFF | STUFF | NO-STUFF |
| R4R9 | STUFF | STUFF | STUFF | NO-STUFF |
| R4R10 | NO-STUFF | NO-STUFF | NO-STUFF | NO-STUFF |
| R4R11 | STUFF | STUFF | STUFF | NO-STUFF |
| R4R12 | STUFF | STUFF | STUFF | NO-STUFF |

[PAGE_TITLE=COMPONENT STUFFING TABLES]

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| MICROSOFT CONFIDENTIAL | PROJECT NAME STINGRAY | PAGE 71/74 | CSA PAGE 71/74 | FAB C | REV 1.0 |
|---|---|---|---|---|---|

# I2C REFERENCE TABLES

## DIGITAL POTENTIOMETERS

| VOLTAGE RAIL | STEPS | STEP SIZE | I2C R/W ADDRESS |
|---|---|---|---|
| VMEM | 256 | 0.007031V | W: 01011000 0X58, R: 01011001 0X59 |
| VEDRAM | 256 | 0.004199V | W: 01011000 0X58, R: 01011001 0X59 |
| V5P0 | 256 | 0.011719V | W: 01011010 0X5A, R: 01011011 0X5B |
| V3P3 | 256 | 0.008V | W: 01011010 0X5A, R: 01011011 0X5B |
| VREF | 256 | 0.007031V | W: 01011110 0X5E, R: 01011111 0X5F |
| VCS | 256 | ?V | W: 01011110 0X5E, R: 01011111 0X5F |
| GPUPCIE | 256 | 0.005859V | W: 01011100 0X5C, R: 01011101 0X5D |
| CPUPLL_1P8 | 256 | 0.007148V | W: 01011100 0X5C, R: 01011101 0X5D |
| V1P2STBY | ? | ?V | W: 01010100 0X54, R: 01010101 0X55 |
| V3P3STBY | ? | ?V | W: 01010100 0X54, R: 01010101 0X55 |
| V1P2 | ? | ?V | W: 01010110 0X56, R: 01010111 0X57 |

STEPS/STEPSIZE TB UPDATED

## ANALOG TO DIGITAL CONVERTERS

| VOLTAGE RAIL | STEPS | STEP SIZE | I2C R/W ADDRESS |
|---|---|---|---|
| VMEM | 4096 | 0.001221V | W: 01010000 0X50, R: 01010001 0X51 |
| VEDRAM | 4096 | 0.001221V | W: 01010000 0X50, R: 01010001 0X51 |
| V5P0 | 4096 | 0.000806V | W: 01010010 0X52, R: 01010011 0X53 |
| V3P3 | 4096 | 0.000806V | W: 01010010 0X52, R: 01010011 0X53 |
| VCS | 4096 | 0.000806V | W: 01001000 0X48, R: 01001001 0X49 |
| MEM_VREF | 4096 | 0.000806V | W: 01000110 0X46, R: 01000111 0X47 |
| GPUPCIE | 4096 | 0.000806V | W: 01000010 0X42, R: 01000011 0X43 |
| CPUPLL_1P8 | 4096 | 0.000806V | W: 01000010 0X42, R: 01000011 0X43 |
| V12P0 | ? | ?V | W: 10000000 0X80, R: 10000001 0X81 |
| TEMP SENSOR | ? | ?V | W: 10011100 0X9C, R: 10011101 0X9D |
| V5P0STBY | ? | ?V | W: 10011000 0X98, R: 10011001 0X99 |
| V1P2STBY | ? | ?V | W: 10000110 0X86, R: 10000111 0X87 |
| V3P3STBY | ? | ?V | W: 10010010 0X92, R: 10010011 0X93 |
| V1P2 | ? | ?V | W: 10000010 0X82, R: 10000011 0X83 |

STEPS/STEPSIZE TB UPDATED

# WAITSBURG DOC TRACKER

| Document# | Document Title |
|---|---|
| H104359 | Waitsburg Document Tree |
| H101502 | Corona Game Console Product Specification |
| H08676 | Console Usage Model And Reliability Budget |
| H101667 | Corona Console Specifications Template |
| H101671 | Corona System Block Diagram |
| H09424 | Vjele Datasheet |
| H08769 | Trinity Console PLL Specification |
| H101637 | Southbridge Security Review |
| H101786 | KSB Programmer's Model |
| H101787 | KSB Register Specification |
| H101665 | Corona Power On Reset Timing Diagram |
| H08754 | Trinity Console Audio / Video Specification |
| H101664 | Corona SMC Firmware Requirements |
| H102138 | Winchester Console Flash Specification |
| H104793 | DataIO Flash Programming Job Creation for Waitsburg eMMC |
| H104815 | Waitsburg Console USB Specification |
| H08770 | Trinity Console Tilt Switch Specification |
| H09297 | Spec,System Fan,High Speed PWM |
| H09169 | Specification,RJ45+Auxiliary Power Connector |
| H08776 | Trinity Console FPM Requirements Document |
| H08775 | Trinity Console FPM LED Display Spec |
| H09443 | Boron Front Panel Module Product Specification |
| H100727 | Trinity Front Panel Module SMC Serial Protocol Specification |
| H101005 | Wave Module Specification |
| H09553 | 802.11n Wi-Fi Module Supplier Qualification Requirements |
| H08757 | Trinity Console IR Specification |
| H02235 | Spec,Xenon,Optical Disc Drive Anti Piracy Specification |
| H08938 | Xbox360 Optical Disc Drive Component Specification,Mustang |
| H08939 | Xbox360 Optical Disc Drive Interface Specification,Mustang |
| H08753 | Hard Drive Component Specification |
| H101663 | Corona System Power Budget |
| H101672 | Corona 1/2w Standby Power Budget |
| H101662 | Corona Console VR Architecture |
| H101572 | Spec,ACPT,Sys PS,120w,Small Form Factor;2-Outp W/Fan,1/2w |
| H08945 | Spec,DC Output Cord Assy Acceptance Specification,Dual Barrel |
| ?H08758 | Trinity Console V_CPUCORE Regulator Specification (Obsolete?) |
| H08762 | Trinity Console V_5P0 Regulator Specification |
| H08763 | Trinity Console V_3P3 Regulator Specification |
| H08765 | Trinity Console V_EDRAM Regulator Specification |
| H08764 | Trinity Console V_MEM Regulator Specification |
| H08759 | Trinity Console V_CPUVCS Regulator Specification |
| H101661 | Corona Console Linear Regulators Specification |
| H101659 | Corona Console V_3P3STBY Regulator Specification |
| H101660 | Corona Console V_1P2STBY Regulator Specification |
| H05204 | Spec,Xenon PCB Specification,TG150 |
| H07678 | Specification Odin Mechanical Design |

[PAGE_TITLE=DOC TRACKER]

Change list

- started from Corona Fab E

Sep 26, 2011
- retail:p3, removed CPU_DBGx_POSTx connections to FTPs
- all:p22, modified R3R5 because of updated .ptf entry (was X814857-001, is X814857-001) (was 84.52 KOHM, is 84.5 KOHM)
- all:p30, modified FB2R1 because of updated .ptf entry (was X801122-001, is X801122-001) (was 0.5A, is 500)
- all:p31, modified FB3P2 because of updated .ptf entry (was X801122-001, is X801122-001) (was 0.5A, is 500)
- all:p31, modified FB3P1 because of updated .ptf entry (was X857387-001, is X857387-001) (was 0.45 OHM, is 0.45DCR)
- all:p35, modified SW5G2 because of updated .ptf entry (was X800550-004, is X800550-004) (was SM, is TH)
- all:p35, modified SW3G1 because of updated .ptf entry (was X850477-001, is X850477-001) (was _, is SPST)
- all:p38, modified FB2M1 because of updated .ptf entry (was X801122-001, is X801122-001) (was 0.5A, is 500)
- all:p39, modified FB1A1 because of updated .ptf entry (was X801122-001, is X801122-001) (was 0.5A, is 500)

Oct 11,2011
- all:p26, added series term to FLSH_DATA<0:7>,FLSH_WP_N,FLSH_CLE,FLSH_CE_N
- all:p26, replaced stitching caps to 25v
- all:p32, removed Phison,NAND, added eMMC
- all:p34, connected NAND after the series term
- all:p33, replaced WM1824 with WM1824B
- all:p44, replaced NCP4201 with NCP4202
- all:p39, combined V_GAMEPORT2 with V_GAMEPORT1,replaced thermistor with X800753-001,removed V_GAMEPORT2 thermistor
- all:p39, combined V_EXPPORT2 with V_EXPPORT1, replaced thermistor with X800753-001, removed V_EXPPORT2 thermistor
- all:p39, updated usbtriple connector symbol

Oct 21, 2011
- xdk:p1, title page
- retail:p1, title page
- xdk:p2, MATERIAL=EMPTY:R4P5,R4P4, BOM=DEVKIT:R4R20, BOM=GCPU_VEJLE_BASE:U5E1
- all:p4, BOM=GDDR_BOTTOM:U5U2,R6U6
- all:p5, BOM=DEVKIT: all parts this page
- all:p9: removed BOM=CORE: all parts this page
- all:p10: removed BOM=CORE: all parts this page
- all:p11: removed BOM=CORE: all parts this page
- all:p14: MATERIAL=EMPTY: U7E1, added PART_LINE_ITEMs for U7E1, BOM=GDDR_TOP_SAMSUNG&GDDR_TOP_WINBOND:R7T4, added PART_LINE_ITEM for R7T4
- all:p15: MATERIAL=EMPTY: U7T1, added PART_LINE_ITEMs for U7T1, BOM=GDDR_TOP_SAMSUNG&GDDR_TOP_WINBOND:R7E7, added PART_LINE_ITEM for R7E7, BOM=GDDR_BOTTOM:R7T6
- all:p16: MATERIAL=EMPTY: U7D1, added PART_LINE_ITEMs for U7D1, BOM=GDDR_TOP_SAMSUNG&GDDR_TOP_WINBOND:R7R4, added PART_LINE_ITEM for R7R4
- all:p17: MATERIAL=EMPTY: U7R1, added PART_LINE_ITEMs for U7R1, BOM=GDDR_TOP_SAMSUNG&GDDR_TOP_WINBOND:R7D5, added PART_LINE_ITEM for R7D5, BOM=GDDR_BOTTOM:R7T1
- all:p18: MATERIAL=EMPTY: U5F1, added PART_LINE_ITEMs for U5F1, BOM=GDDR_TOP_SAMSUNG&GDDR_TOP_WINBOND:R5U4, added PART_LINE_ITEM for R5U4
- all:p19: MATERIAL=EMPTY: U5U1, added PART_LINE_ITEMs for U5U1, BOM=GDDR_TOP_SAMSUNG&GDDR_TOP_WINBOND:R5F2, added PART_LINE_ITEM for R5F2, BOM=GDDR_BOTTOM:R5U1
- all:p20: MATERIAL=EMPTY: U6F1, added PART_LINE_ITEMs for U6F1, BOM=GDDR_TOP_SAMSUNG&GDDR_TOP_WINBOND:R6U4, added PART_LINE_ITEM for R6U4
- all:p21: MATERIAL=EMPTY: U6U1, added PART_LINE_ITEMs for U6U1, BOM=GDDR_TOP_SAMSUNG&GDDR_TOP_WINBOND:R6F2, added PART_LINE_ITEM for R6F2, BOM=GDDR_BOTTOM:R6U1
- all:p24: BOM=RETAIL:R2C12, BOM=DEVKIT:R2C11
- all:p25: BOM=RETAIL:R2G4, BOM=DEVKIT:R2G3, BOM=JTM_INT:R3R12, BOM=JTM_EXT:R3R10
- all:p35: BOM=DEVKIT:SW3G1
- xdk:p38: BOM=DEVKIT:R7A1,R7A2,R6B1
- all:p45: X857866-001:Q6B2,Q6B1, X857867-001:Q6C2,Q6C1,Q5C3,Q5C1
- all:p47: X857866-001:Q2F1, X857867-001:Q2F2, BOM=DEVKIT:R2G1, BOM=VREG_FIXED:R2E4,R2T4, X821117-001:R2E5, X801000-001:R2F1
- all:p48: BOM=DEVKIT:R1F19, BOM=VREG_FIXED:R1E20, BOM=VREG_FIXED:R1E19
- all:p49: X857866-001:Q3F1, X857867-001:Q4F1, BOM=DEVKIT:R4F1, BOM=VREG_FIXED:R3F8,R3F9, Part# change:C3F3,C3F2,R3F6,R3F10,R3F7,R3V1
- all:p50: X857866-001:Q7G1, X857867-001:Q7G2, BOM=DEVKIT:R7F1, BOM=VREG_FIXED:R3G11,R3G12, Part# change:C3G7,C3G6,R3V4,R3G13,R3G10,R4G3
- all:p51: X857866-001:Q5B1, X857867-001:Q5C2, BOM=DEVKIT:R5C2, BOM=VREG_FIXED:R4B17,R4B19
- all:p52: BOM=RETAIL:R4B14, added PART_LINE_ITEM for R4B14, BOM=VREG_FIXED:R4B9,R4B7,R4E8,R4E6
- all:p53: BOM=DEVKIT:R4B7,R5A11, BOM=VREG_FIXED:R5B5,C5B1,R5A15,R5A6,C5A1,R5A5, added PART_LINE_ITEMs, BOM=VREG_MM:R5B2,R5A13,R5A8,R5A12
- all:p54: MATERIAL=EMPTY:C7M1,C4N8,C2V2,C3E2,C3C7,C1U3,C4N2
- all:p55: BOM=DEVKIT:R3B1,R3B2,J5B1, BOM=RETAIL:R3C2,R3C1, BOM=VREG_MM: various parts
- all:p56: BOM=VREG_MM: various parts
- all:p57: BOM=VREG_MM: various parts
- all:p58: BOM=VREG_MM: various parts
- all:p59: BOM=VREG_MM: various parts
- all:p60: BOM=VREG_MM: various parts
- all:p61: BOM=JTM_INT: various parts, BOM=JTM_EXT: various parts
- all:p62: BOM=DEVKIT: all parts this page
- all:p65: BOM=DEVKIT: various parts
- all:p67: MATERIAL=EMPTY:LB2B1, added PART_LINE_ITEM for LB2B1
- all:p73: updated to Waitsburg Document tree
- all:p74: this page

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- all:p32: changed R1D4 from X801814-001 to X806693-001
- all:p53: changed R5A4 from NO-STUFF to STUFF, changed R5A7,C5M8,C5M5 to NO-STUFF

Schematic Transitions for Waitsburg

This document lists the schematic changes needed to transform Waitsburg Fab A from Debug to XDK to Retail.
The base schematic is Waitsburg Fab A Debug

Waitsburg Debug to XDK changes
Start from Waitsburg Fab A Debug and replace the following files/pages:
page1.csa: Rename title page to XDK
page12.csa: Delete nets MA_A<12>, MB_A<12>
page13.csa: Delete nets MC_A<12>, MD_A<12>
page14.csa: Delete nets MA_A<12>, stuff R7T5,R7T4 (MEM_A_VREF1 resistor divider)
page15.csa: Delete nets MA_A<12>, stuff R7E6,R7E7 (MEM_A_VREF0 resistor divider)
page16.csa: Delete nets MB_A<12>, stuff R7R2,R7R4 (MEM_B_VREF1 resistor divider)
page17.csa: Delete nets MB_A<12>, stuff R7D4,R7D5 (MEM_B_VREF0 resistor divider)
page18.csa: Delete nets MC_A<12>, stuff R5U5,R5U4 (MEM_C_VREF1 resistor divider)
page19.csa: Delete nets MC_A<12>, stuff R5F1,R5F2 (MEM_C_VREF0 resistor divider)
page20.csa: Delete nets MD_A<12>, stuff R6U5,R6U4 (MEM_D_VREF1 resistor divider)
page21.csa: Delete nets MD_A<12>, stuff R6F1,R6F2 (MEM_D_VREF0 resistor divider)
page62.csa: Delete MEM_VREFS shunt circuitry
page63.csa: blank, deleted entire page
page64.csa: blank, deleted entire page

Waitsburg XDK to Retail changes
Start from Waitsburg Fab A XDK and replace the following files/pages:
page1.csa: Rename title page to RETAIL
page2.csa: Delete CPU_TE pulldown resistor (R4R20) and tie U5E1.E7 to ground
page3.csa: Disconnect U5E1 TB[0:7],TB[11:15] from CPU_DBG<0..15> bus, ground FTPs for TB[0:7], float FTPs for TB[11:15]
page5.csa: Diconnect U5E1.E3 (SROM_EN) from pullup resistor (R4R2)
page25.csa: Short U3D1.A6 (SMC_UART1_RXD) to V_3P3STBY
page38.csa: Delete V_12P0 shunt resistors (R7A1,R7A2), short V_12P0 to J7A1 pins 5-8. Remove short_parts (ST7A1,ST7A2).
page38.csa: Delete V_5P0STBY shunt resistor (R6B1), short V_5P0STBY to J7A1.10. Remove short_parts (ST6B1,ST6B2)
page47.csa: Delete V_5P0 shunt resistor (R2G1), short V_5P0 to L2F1.2. Remove short_parts (ST2G2,ST2G3)
page48.csa: Delete V_3P3 shunt resistor (R1F9), short V_3P3 to L1F1.2. Remove short_parts (ST1F1,ST1F3)
page49.csa: Delete V_CPUEDRAM shunt resistor (R4F1), short V_CPUEDRAM to L4F1.2. Remove short_parts (ST4F2,ST4F4)
page50.csa: Delete V_MEM shunt resistor (R7F1), short V_MEM to L7F1.1. Remove short_parts (ST7F2,ST7F3)
page51.csa: Delete V_CPUVCS shunt resistor (R5C2), short V_CPUVCS to L5C1.2. Remove short_parts (ST5C1,ST5C3)
page52.csa: Delete V_5P0 shunt resistor (R2G1), short V_5P0 to L2F1.2. Remove short_parts (ST2G2,ST2G3)
page53.csa: Delete V_1P2 shunt resistor (R4B14), short V_1P2 to U3B1.4. Remove short_parts (ST4B1,ST4B2). Remove PART_LINE_ITEM FOR R4B14
page53.csa: Delete V_GPUPCIE shunt resistor (R4E11), short V_GPUPCIE to U4E2 pin 2 and 4.
page53.csa: Delete V_1P8 shunt resistor (R4E12), short V_1P8 to U4E1.2. Delete V_CPUPLL shunt resistor (R4E4) replace with short_part
page53.csa: Delete V_EFUSE shunt resistor (R4P10), short V_EFUSE to U4P2.5.
page62.csa: Delete PCIe mid-bus probe (J3E1). Delete GPU_CLK probe (J4D2). Delete SMC_DBG_RXD filter (R2C2,C2C1). Float J2C3.3
page65.csa: Delete DBPADs to V_5P0, V_3P3, GND

Caveat:
On the Waitsburg Debug, page 38, there is a thermistor, RT3B1. This thermistor is the same thermistor on Waitsburg Retail, page 38, RT3A2.
During layout, ref des were renumbered on the Waitsburg Retail schematic page.
TO DO: Respin Waitsburg Retail schematic and fab and change RT3A2 to RT3B1
TO DO: Respin Waitsburg Retail schematic and fab and change RT3A2 to RT3B1
TO DO: Respin Waitsburg Retail schematic and fab and change RT3A2 to RT3B1

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